FINITE-SAMPLE METRIC NON-COLLAPSE FOR GEOMETRICALLY SUPERVISED LATENT WORLD MODELS IN CONTROL

ALAIN BENSOUSSAN, MINH-NHAT PHUNG, AND MINH-BINH TRAN

Abstract. We establish a finite-sample learning-to-control theory for geometrically supervised latent models of nonlinear deterministic systems. Geometric supervision is used only during training: simulator state, proprioception, or state estimates with independently validated metric and directional error bounds supply observable-state distances and tangent directions, while deployment remains observation- and action-conditioned. We introduce an encoder-only local–global metric hinge that enforces directional resolution and separated-state discrimination. Under regular observable-factor, coverage, finite-capacity approximation, and uniform $C^{1,1}$ hypotheses, a computable one-sided regularization regime has a strong selection property: with high probability, every approximate empirical minimizer is simultaneously pointwise co-Lipschitz and uniformly approximately semiconjugate to the controlled dynamics. Approximation, sampling, and optimization errors remain explicit and separate. Norm-constrained tensor-product B-spline classes constructively realize the approximation hypotheses, and the interpolation exponent converting mean residual control into a uniform bound is sharp. A modular deterministic corollary transfers the learned certificates to trajectory, finite-horizon cost, learned-cost-head, and optimizer guarantees, while a validated finite-net result enables sharper model-specific certification. Controlled experiments isolate collapse and folding, quantify the analytic certificate's reserve, and demonstrate the control benefit of restored metric resolution. The principal contribution is a complete finite-sample implication from approximate empirical optimization to metric faithfulness, uniform controlled dynamics, and reliable planning for the same learned model.

1. Introduction

A learned latent world model is useful for control only if it does more than fit one-step data on average. It must retain the state-space distinctions that matter under interventions, and its latent transition must remain uniformly compatible with the controlled dynamics. Otherwise a planner may exploit folds, overlaps, or poorly resolved regions of the representation even when the empirical prediction loss is small. The central question of this paper is therefore control-theoretic: under what conditions does an empirically trained latent model become a metrically faithful approximate realization of a nonlinear deterministic control system, with quantitative guarantees for downstream planning?

The model-based reinforcement-learning viewpoint that learned dynamics can support planning and policy improvement goes back to Dyna and underlies modern world-model methods for visual control [1, 2, 3, 4, 5]. Methods such as PlaNet and Dreamer encode high-dimensional observations into compact latent variables and learn the dynamics in latent space. Joint-embedding predictive architectures (JEPA) adopt a related but distinct principle: instead of reconstructing every detail of the next observation, the model predicts its representation [6]. This principle has led to I-JEPA for images [7], V-JEPA for video [8], and V-JEPA 2, which combines video pretraining with robot interaction data to construct an action-conditioned latent model for planning [9]. These developments motivate a precise mathematical problem: when does action-conditioned latent prediction produce both a non-collapsed representation and a latent transition that faithfully represents the underlying controlled system?

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We consider a deterministic controlled system

$$s_{t+1} = G(s_t, a_t), \qquad x_t = H(s_t),$$

where the deployed model receives the observation x_t. An action-conditioned latent model consists of an encoder Φ and a transition map F,

$$z_t = \Phi(H(s_t)), \qquad z_{t+1} \approx F(z_t, a_t),$$

with residual

$$R[\Phi, F](s, a) := \Phi(H(G(s, a))) - F(\Phi(H(s)), a).$$

A uniform bound on this residual is an approximate controlled-semiconjugacy estimate: evolving the physical system and then encoding nearly agrees with encoding first and evolving in latent space. The prediction loss is observation-based, but the metric regularizer studied here uses additional training information. Observable-factor coordinates, or equivalent state distances and tangent directions, are used to evaluate its local and separated-pair terms. The framework is formulated for settings where observable-state metric information is available, including simulator-state, proprioceptive, and state-estimation regimes.

Forward prediction alone cannot yield the representation property needed for deterministic control. The constant encoder $\Phi \equiv z_0$, together with $F \equiv z_0$, has zero prediction loss. More subtly, positive latent variance or covariance does not prevent folding: a representation may have substantial global spread while identifying distinct states or losing arbitrarily much local metric resolution. For the worst-case deterministic planning guarantees pursued here, the relevant non-collapse condition is not merely distributional. It is a pointwise lower metric estimate of the form

$$\|\Phi(H(s)) - \Phi(H(s'))\|_Z \geq c_* |s - s'|, \qquad s, s' \in S,$$

which makes the learned representation quantitatively injective on the observable state factor.

To enforce this geometry, we introduce an encoder-only local–global metric hinge. Its local directional component penalizes infinitesimal collapse of $D(\Phi \circ H)$, while its separated-pair component penalizes overlap of states that are a prescribed distance apart in the observable-state metric. The penalty is one-sided: it vanishes after the desired local and global margins are reached. It is therefore decoder-free, inverse-model-free, and independent of a reconstruction objective. The two components have antecedents in conditioning and lower co-Lipschitz regularization; the contribution here is the finite-sample control theorem obtained from their joint enforcement.

The theorem-level question is whether approximate optimization of the empirical training objective itself forces both a quantitatively faithful representation and a uniformly accurate controlled latent transition. The proof has the following parallel-output structure:

$$\boxed{\begin{tikzcd} & \text{empirical objective comparison and margin clearing} \arrow[dl] \arrow[dr] & \\ \begin{array}{c}\text{pointwise}\\ \text{co-Lipschitzness}\end{array} \arrow[dr] & & \begin{array}{c}\text{uniform approximate}\\ \text{controlled semiconjugacy}\end{array} \arrow[dl] \\ & \text{worst-case finite-horizon optimizer transfer} & \end{tikzcd}}$$

Thus the result is a finite-sample realization theorem rather than only an anti-collapse or planning-regret statement. The representation theorem supplies the metric and dynamics constants in parallel; the deterministic planning corollary then uses both, while approximation, statistical, training-optimization, cost-head, and planner errors remain explicit and separate.

Precise novelty statement. Under explicit observable-geometry, approximation, coverage, regularity, and optimization hypotheses, our principal theorem proves a finite-sample selection principle with quantitative geometric coercivity: throughout a computable one-sided regularization half-line, every approximate empirical minimizer of the fixed objective is pointwise co-Lipschitz and uniformly approximately semiconjugate to the controlled dynamics. This every-minimizer objective-to-certificate implication, followed by a modular deterministic planning transfer, is the paper's central contribution.

The principal contributions are:

(i) We identify collapse and folding obstructions for prediction and covariance control, and introduce an encoder-only local–global hinge aligned with pointwise metric faithfulness.

(ii) We prove the every-minimizer finite-sample co-Lipschitz and uniform controlled-semiconjugacy theorem, with approximation, statistical, and optimization errors separated and an explicit certified regularization regime.

(iii) We verify the finite-capacity hypotheses constructively for norm-constrained tensor-product B-splines, establish sharpness of the interpolation exponent, and transfer the learned certificates to deterministic planning with compatible or learned costs.

(iv) We provide controlled experiments that isolate the obstructions, quantify the analytic certificate's uniformity reserve, and demonstrate the control benefit of restored metric resolution.

The explicit lower bound on λ is the comparison scale at which the faithful zero-penalty comparator forces every approximate minimizer into the margin-clearing regime. It defines an a priori certified half-line and an error-budget map; after training, the finite-net proposition gives a sharper model-specific a posteriori certificate. The experiments use the same quantities diagnostically without identifying finite-set performance with the theorem's continuum guarantee.

Theorem 1.1 (Main representation and planning theorem, informal form)**.** *Suppose that the controlled system admits a finite-dimensional regular observable factor, that the training and metric-sampling distributions satisfy the stated coverage conditions, and that the finite-capacity model classes approximate a smooth, non-collapsed exact realization with uniform $C^{1,1}$ bounds. Then, for sufficiently large capacity W and sample size n, for every confidence level $\delta \in (0,1)$ and training tolerance $\varepsilon_{\mathrm{train}} \geq 0$, there exists an explicit regularization threshold such that, with probability at least $1-\delta$, every $\varepsilon_{\mathrm{train}}$-approximate empirical minimizer $(\Phi^\star, F^\star)$ satisfies*

$$\|\Phi^\star(H(s)) - \Phi^\star(H(s'))\|_Z \geq c_* |s - s'| \qquad \textit{for all } s, s' \in S,$$

for an explicit constant $c_* > 0$, and

$$\sup_{(s,a)\in S\times A} \|\Phi^\star(H(G(s,a))) - F^\star(\Phi^\star(H(s)), a)\|_Z \leq \eta_{W,n,\delta}(\varepsilon_{\mathrm{train}}).$$

The modulus $\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})$ tends to zero with the approximation, statistical, and training-optimization errors. Separately, if the physical stage and terminal costs are Lipschitz continuous in the state variable and $\widehat{\mathbf{a}}$ is a ξ_{plan}-approximate optimizer of the induced latent problem, then, for horizon T, there exists an explicit bound Δ_T such that

$$J_{\mathrm{true}}(s_0; \widehat{\mathbf{a}}) - J^\star(s_0) \leq 2\Delta_T(\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})) + \xi_{\mathrm{plan}},$$

where J_{true} denotes the total physical cost under the true dynamics and $J^\star$ denotes the corresponding optimal value.

Learned latent cost heads add $2(T\varepsilon_\ell + \varepsilon_g)$; exact constants and transfer bounds are in Theorem 7.11 and Corollary 8.3.

Section 2 positions the result; Sections 3–7 develop the model and prove the theorem; Sections 8–10 give planning transfer, evidence, scope, and outlook.

2. Relation to coercivity, metric embeddings, abstraction, and planning transfer

Coercivity and stability gaps. A recurring theme in inverse problems and statistical learning is that a data-fit residual becomes scientifically meaningful only after it is paired with a stability mechanism; see, for example, the regularization framework of Engl, Hanke, and Neubauer [10]. The present theory brings that perspective to latent control. Pure prediction exposes a geometric stability gap, and the local–global hinge supplies a conditional lower stability estimate from empirical objective value to pointwise latent geometry under explicit observable-factor, realizability, regularity, coverage, supervision, and optimization hypotheses.

Quantitative injectivity and metric regularization. Bi-Lipschitz and co-Lipschitz embeddings quantify injectivity through lower metric distortion, while pairwise and contrastive objectives shape representation geometry [11]. The local term controls directional singular values on the observable factor, and the separated-pair term resolves distant folds. The theorem then propagates sampled lower-margin control to a continuum co-Lipschitz constant under $C^{1,1}$ regularity and coverage, producing a quantitative injectivity certificate for every approximate empirical minimizer.

Observability, abstraction, bisimulation, and semiconjugacy. Observable quotients and state abstractions identify the distinctions that must be retained for dynamics and control. Approximation metrics and approximate bisimulation provide system-level notions of behavioral closeness [12, 13, 14, 15], with learned counterparts such as DeepMDP and DBC [16, 17]. Here the observable factor isolates the recoverable control state, and the theorem equips it simultaneously with a lower metric certificate and a uniform deterministic controlled-semiconjugacy estimate.

Empirical-to-uniform estimates. Uniform laws of large numbers and covering-number methods control deviations over function classes [18, 19]. We combine these population estimates with capacity-uniform Lipschitz budgets and lower-Ahlfors coverage to obtain pointwise continuum bounds. The sharp L^2-to-L^∞ exponent identifies the exact regularity-limited conversion rate and makes the dimension dependence of the guarantee explicit.

Simulation lemmas and planning transfer. Classical simulation lemmas and Lipschitz model-based bounds propagate one-step model error to finite-horizon value or planning error [20, 21]. Our deterministic sup-norm formulation is tailored to optimization over action sequences. The representation theorem supplies exactly the two constants required by the modular transfer argument: η controls the latent dynamics mismatch, and c_* controls the Lipschitz constants of compatible latent costs.

JEPA and learned world models. Modern world models and JEPA methods motivate latent action-conditioned prediction [1, 2, 3, 4, 6, 7, 8, 9]. Recent theory establishes complementary results on finite-sample JEPA generalization and planning regret, structured identifiability, planning-aligned geometry, reachability, action sensitivity, and prediction–control mismatch [22, 23, 24, 25, 26, 27, 28, 29, 30]. Auxiliary-task JEPA theory characterizes which distinctions an additional prediction target can preserve [31], while bi-Lipschitz autoencoder regularization targets injectivity and robustness in reconstruction models [32]. Controlled-world-model identifiability addresses latent recovery under structural hypotheses [24]. The present theorem addresses the subsequent objective-level implication: a fixed geometrically supervised finite-sample objective forces pointwise non-collapse and uniform semiconjugacy for all approximate minimizers in the certified regime, and those certificates feed directly into deterministic optimizer transfer.

3. Deterministic controlled systems and the observable quotient

In this section, we mathematically formalize the deterministic system and the observable quotient.

First, let the state space $S \subset \mathbb{R}^{d_s}$ be a compact convex Euclidean domain with nonempty interior, and let the action space $A \subset \mathbb{R}^{d_a}$ be a compact Euclidean domain. No smoothness of ∂S is imposed. Whenever derivatives up to order $C^{1,1}$ are used, the relevant maps are assumed to admit $C^{1,1}$ extensions to open Euclidean neighborhoods of their compact domains. The metric spaces (S, d_S) and (A, d_A) are Euclidean, with

$$d_S(s, s') := |s - s'|, \qquad d_A(a, a') := |a - a'|.$$

The state-action space $M = S \times A$ is equipped with the usual product metric

$$d_M((s, a), (s', a')) := d_S(s, s') + d_A(a, a').$$

The dimension of the state-action space is denoted by $d_m := d_s + d_a$.

The true deterministic dynamics and observation map are

$$s_{t+1} = G(s_t, a_t), \qquad x_t = H(s_t),$$

where

$$G : S \times A \to S, \qquad H : S \to X.$$

Here X is a separable Banach space, typically $\mathbb{R}^{d_x}$ with $d_x \gg d_s$. We assume throughout that G and H are Lipschitz. In the metric-regularized results, Lemma 7.3 and those that follow, the neighborhood extensions stipulated above are assumed to be of class $C^{1,1}$.

If H is not injective, no representation built from observations can recover more than the information contained in controlled observation histories. The correct object is therefore an observable quotient.

Definition 3.1 (Observational equivalence)**.** For $s, s' \in S$, write $s \sim s'$ if, for every $k \geq 0$ and every action sequence $(a_0, \ldots, a_{k-1}) \in A^k$,

$$H(G_{a_{0:k-1}}(s)) = H(G_{a_{0:k-1}}(s')),$$

where $G_{a_{0:k-1}}$ denotes the k-fold composition of G along the prescribed actions, that is

$$G_{a_{0:-1}}(s) = s, \quad G_{a_{0:0}}(s) = G(s, a_0), \quad G_{a_{0:k}}(s) = G(G_{a_{0:k-1}}(s), a_k).$$

This is the deterministic, controlled analogue of bisimulation and state abstraction [14, 33].

Definition 3.2 (Observable quotient)**.** For $s \in S$, denote its observational equivalence class by

$$[s] := \{r \in S : \ r \sim s\}.$$

The observable quotient is

$$S_{\text{obs}} := S/\!\sim := \{[s] : s \in S\}.$$

For two equivalence classes $[s], [s'] \in S_{\text{obs}}$, define the quotient chain pseudometric[1]

$$d_{\text{obs}}([s], [s']) := \inf_{N \geq 1} \inf \left\{ \sum_{j=0}^{N-1} d_S(u_j, v_j) : u_j, v_j \in S, \quad j = 0, \ldots, N-1, \right.$$
$$u_0 \in [s], \quad v_{N-1} \in [s'],$$
$$\left. v_j \sim u_{j+1}, \quad j = 0, \ldots, N-2 \right\}.$$

[1]It is necessary to have chains in the definition, the naive definition $\tilde{d}([s], [s']) = \inf\{d_S(u, v) : u \in [s], v \in [s']\}$ is not enough. For example, $S = \{0, 1, 2, 3\}$, which is cut into 3 equivalent classes $\{0\}, \{1, 2\}, \{3\}$; $\tilde{d}$ does not define a metric in such case.

The condition $v_j \sim u_{j+1}$ allows zero-cost jumps within observational equivalence classes. If d_{obs} is only a pseudometric, we quotient once more by the zero-distance relation. Equivalently, in the sequel we work on the metric quotient and write

$$d_{S_{\mathrm{obs}}} := d_{\mathrm{obs}}.$$

The dynamics descend to the quotient. Indeed, if $s \sim s'$, then for every $a \in A$ and every future action sequence, the observation sequences generated from $G(s,a)$ and $G(s',a)$ coincide. Hence

$$\bar{G}([s],a) := [G(s,a)]$$

is well-defined. The observation map descends to

$$\bar{H} : S_{\mathrm{obs}} \to X, \qquad \bar{H}([s]) = H(s),$$

because $s \sim s'$ implies $H(s) = H(s')$. However, this quotient construction alone need not make $\bar{H}$ bi-Lipschitz. Thus the following observability condition is a substantive structural assumption.

Assumption A1 (Regular observable Markov factor). *We assume that there exists a finite-dimensional observable factor*

$$q : S \to S_{\mathrm{o}} \subset \mathbb{R}^{d_o},$$

where S_{o} is a compact convex Euclidean domain with nonempty interior, such that

$$q(s) = q(s') \iff s \sim s'.$$

Moreover, the dynamics, observation map, and costs factor through q: there exist

$$G_{\mathrm{o}} : S_{\mathrm{o}} \times A \to S_{\mathrm{o}}, \qquad H_{\mathrm{o}} : S_{\mathrm{o}} \to X,$$

and

$$\ell_{\mathrm{o}} : S_{\mathrm{o}} \times A \to \mathbb{R}, \qquad g_{\mathrm{o}} : S_{\mathrm{o}} \to \mathbb{R}$$

such that

$$q(G(s,a)) = G_{\mathrm{o}}(q(s),a), \qquad H(s) = H_{\mathrm{o}}(q(s)),$$

and

$$\ell(s,a) = \ell_{\mathrm{o}}(q(s),a), \qquad g(s) = g_{\mathrm{o}}(q(s)).$$

The maps G_{o} and H_{o} admit $C^{1,1}$ extensions to open Euclidean neighborhoods of their compact domains, and H_{o} is bi-Lipschitz onto its image:

$$c_H|y-y'| \le \|H_{\mathrm{o}}(y) - H_{\mathrm{o}}(y')\|_X \le L_H|y-y'|, \qquad \text{for all } y, y' \in S_{\mathrm{o}}.$$

After this identification, we work on $S := S_{\mathrm{o}}$, $G := G_{\mathrm{o}}$, and $H := H_{\mathrm{o}}$. All derivatives below are taken with respect to the Euclidean observable coordinate $y \in S_{\mathrm{o}}$.

Under Assumption A1, the observable factor is identifiable from the current observation through the bi-Lipschitz map H_{o}. Controlled histories motivate the quotient construction, but they are not used by the deployed encoder in the theorem.

Remark 3.3. An arbitrary metric quotient space is not automatically a smooth Euclidean domain. Accordingly, the present Euclidean $C^{1,1}$ proof framework imposes the regular finite-dimensional factor structure of Assumption A1.

Classical indistinguishability theory in control has addressed the manifold structure of the quotient space and has provided a sufficient condition in [34, Theorem 3.5].

Under certain assumptions, the quotient metric d_{obs} endows a topology that agrees with the quotient topology in [35, Ex. 3.1.14].

Assumption A1 supplies a finite-dimensional Euclidean coordinate for the observable factor. The Euclidean factor metric used in all subsequent estimates need not coincide with the chain metric d_{obs}. If one wishes to translate the estimates back to the chain metric, an additional bi-Lipschitz comparison between the two metrics must be assumed.

For the remainder of the paper, the metric on the observable factor is defined by

$$\bar{d}_{\mathrm{obs}}([s],[s']) := |q(s) - q(s')|,$$

and after identifying S with S_{o} we write $d_S(s,s') := |s - s'|$. All lower co-Lipschitz constants below refer to this Euclidean observable-factor metric.

Assumption A2 (Observable-factor excitation / coverage). *Let μ be the training measure on $S \times A$, and define the pushforward measure*

$$\mu_{\mathrm{o}} := (q, \mathrm{Id}_A)_{\#}\mu$$

on $S_{\mathrm{o}} \times A$. We assume that μ_{o} is lower Ahlfors $(d_o + d_a)$-regular: there exist $m_0 > 0$ and $r_0 > 0$ such that

$$\mu_{\mathrm{o}}(B_r(s,a)) \geq m_0 r^{d_o + d_a} \qquad \textit{for all } (s,a) \in S_{\mathrm{o}} \times A, \quad 0 < r \leq r_0.$$

After the identification in Assumption A1, we write $\mu := \mu_{\mathrm{o}}$, $d_s := d_o$, and $d_m := d_o + d_a$.

In the common absolutely continuous case in which $\mathrm{d}\mu = \rho\, \mathrm{d}y$ on $M \subset \mathbb{R}^{d_m}$, a lower density bound $\rho \geq \rho_{\min} > 0$ implies Assumption A2 provided the domain also has a uniform interior-volume lower bound, namely $|M \cap B_r(y)| \geq c_M r^{d_m}$ for all $y \in M$ and all sufficiently small r. This geometric thickness condition holds, for example, for compact convex domains and for bounded Lipschitz domains after fixing the small-radius range.

4. Action-conditioned latent world models

In this section, we define the notation employed for the JEPA model.

Let $Z = \mathbb{R}^{d_z}$ be the latent space and let K_Z be a compact convex subset of Z with nonempty interior. A positive local directional margin requires $D\psi(s) : \mathbb{R}^{d_s} \to \mathbb{R}^{d_z}$ to be injective, and therefore necessarily $d_z \geq d_s$; we assume this dimension condition throughout the metric results. Because $S \subset \mathbb{R}^{d_s}$ is compact and K_Z has nonempty interior, a translated and scaled coordinate inclusion embeds S smoothly and bi-Lipschitzly into $\operatorname{int} K_Z$. Supplementary Section S1 gives this construction and a smooth range-enforcing map explicitly. An action-conditioned latent world model consists of an encoder

$$\Phi : X \to K_Z$$

and a latent transition map

$$F : K_Z \times A \to K_Z.$$

Only the restriction of Φ to the set $H(S)$ is utilized. Consequently, we define the induced latent state mapping

$$\psi := \Phi \circ H : S \to K_Z.$$

To analyze the learning properties of the JEPA model, we fix uniformly bounded Lipschitz classes of ambient maps

$$\mathcal{A}_\Phi \subset \big\{\Phi : X \to K_Z : \ \Phi|_{H(S)} \in C^{0,1}(H(S); K_Z)\big\}, \qquad \mathcal{A}_F \subset C^{0,1}(K_Z \times A; K_Z), \qquad \mathcal{A} := \mathcal{A}_\Phi \times \mathcal{A}_F.$$

Only the restrictions of the encoders to $H(S)$ enter the objective. For compactness of minimizers, we assume that these restrictions and the transition classes are closed in the relevant uniform topology.

For the metric results in Section 7, we take $X = \mathbb{R}^{d_x}$ and require each admissible encoder to be supplied with a specified $C^{1,1}$ ambient representative on an open neighborhood of $H(S)$ (the concrete spline encoders are global maps on X). For such a representative, we use

$$\|\Phi\|_{C^1(H(S))} := \sup_{x\in H(S)} \|\Phi(x)\|_Z + \sup_{x\in H(S)} \|D\Phi(x)\|_{\mathrm{op}}.$$

Thus every derivative of Φ appearing below is the derivative of its specified ambient representative. Equivalently, the geometric regularity may be expressed through the induced map $\psi = \Phi \circ H$ on the Euclidean observable factor.

For a horizon $T \in \mathbb{N}$, a stage cost $\ell : S \times A \to \mathbb{R}$, and a terminal cost $g : S \to \mathbb{R}$, we denote an action sequence by

$$a_{0:T-1} := (a_0, \dots, a_{T-1}) \in A^T.$$

When the horizon T is clear from the context, we also write $\mathbf{a} := a_{0:T-1}$. The deterministic control problem is

$$J^\star(s_0) = \inf_{\mathbf{a}\in A^T} J_{\mathrm{true}}(s_0; \mathbf{a}),$$

where

$$J_{\mathrm{true}}(s_0; \mathbf{a}) = \sum_{t=0}^{T-1} \ell(s_t^{\mathbf{a}}, a_t) + g(s_T^{\mathbf{a}}), \qquad s_{t+1}^{\mathbf{a}} = G(s_t^{\mathbf{a}}, a_t).$$

If G, ℓ, and g are continuous, compactness of A^T implies existence of an optimal control sequence.

The next lemma establishes the mathematical availability of exact compatible latent costs once metric non-collapse has been proved. Section 8 then extends this construction to learned cost heads through explicit compatibility errors.

Lemma 4.1 (Compatible latent costs induced by metric non-collapse)**.** *Let $\psi : S \to K_Z$ satisfy*

$$\|\psi(s) - \psi(s')\|_Z \geq c_* |s - s'| \qquad \textit{for all } s, s' \in S,$$

with $c_* > 0$. Assume

$$|\ell(s, a) - \ell(s', a')| \leq L_{\ell,s}|s - s'| + L_{\ell,a}|a - a'|, \qquad |g(s) - g(s')| \leq L_g|s - s'|. \tag{4.1}$$

Then

$$\ell_\psi(\psi(s), a) := \ell(s, a), \qquad g_\psi(\psi(s)) := g(s)$$

are well-defined. They admit extensions

$$\widetilde{\ell} : K_Z \times A \to \mathbb{R}, \qquad \widetilde{g} : K_Z \to \mathbb{R}$$

satisfying exact compatibility

$$\widetilde{\ell}(\psi(s), a) = \ell(s, a), \qquad \widetilde{g}(\psi(s)) = g(s),$$

and the separate Lipschitz estimate

$$|\widetilde{\ell}(z, a) - \widetilde{\ell}(z', a')| \leq \frac{L_{\ell,s}}{c_*}\|z - z'\|_Z + L_{\ell,a}|a - a'|, \qquad |\widetilde{g}(z) - \widetilde{g}(z')| \leq \frac{L_g}{c_*}\|z - z'\|_Z. \tag{4.2}$$

In particular,

$$L_{\widetilde{\ell},z} \leq \frac{L_{\ell,s}}{c_*}, \qquad L_{\widetilde{g}} \leq \frac{L_g}{c_*}.$$

Proof. The lower co-Lipschitz estimate implies injectivity and

$$|s-s'| \le c_*^{-1}\|\psi(s)-\psi(s')\|_Z.$$

Hence, ℓ_ψ is 1-Lipschitz on $\psi(S)\times A$ with respect to the weighted product distance

$$d_\ell((z,a),(z',a')) := \frac{L_{\ell,s}}{c_*}\|z-z'\|_Z + L_{\ell,a}|a-a'|.$$

When both weights are positive this is a metric; if one weight vanishes it is a pseudometric, and the same argument is applied after quotienting the corresponding zero-distance equivalence classes. Likewise, g_ψ is L_g/c_*-Lipschitz on $\psi(S)$. Therefore, applying McShane's extension theorem [36] to the resulting metric quotient (or directly when the weights are positive), we conclude that the functions

$$\widetilde{\ell}(z,a) := \inf_{(s',a')\in S\times A}\left\{\ell(s',a') + \frac{L_{\ell,s}}{c_*}\|z-\psi(s')\|_Z + L_{\ell,a}|a-a'|\right\}$$

and

$$\widetilde{g}(z) := \inf_{s'\in S}\left\{g(s') + \frac{L_g}{c_*}\|z-\psi(s')\|_Z\right\}$$

agree with the original functions on the encoded state set and satisfy (4.2). ■

For a representation (Φ,F) such that $\psi=\Phi\circ H$ is lower co-Lipschitz, and for Lipschitz costs ℓ and g as in Lemma 4.1, we obtain induced latent costs $\widetilde{\ell}$ and $\widetilde{g}$. The corresponding total latent cost is defined by

$$J_{\text{lat}}(\psi(s_0);\mathbf{a}) := \sum_{t=0}^{T-1}\widetilde{\ell}(z_t,a_t) + \widetilde{g}(z_T), \qquad z_0=\psi(s_0), \quad z_{t+1}=F(z_t,a_t), \quad t=0,\dots,T-1.$$

Let $Y\subset\mathbb{R}^d$ be a Euclidean domain and let E be a normed space. The Lipschitz constant of $f:Y\to E$ is denoted by $\mathrm{Lip}(f)$. A function class $\mathfrak{F}$ has Lipschitz budget L if every $f\in\mathfrak{F}$ is Lipschitz and $\mathrm{Lip}(f)\le L$. Furthermore, $\mathfrak{F}$ has $C^{1,1}$ budget Λ if every $f\in\mathfrak{F}$ has a specified extension that is $C^{1,1}$ on an open neighborhood of Y and

$$\|f\|_{C^0(Y)} + \|Df\|_{C^0(Y)} + \mathrm{Lip}(Df;Y) \le \Lambda.$$

Here, $Df: Y\to\mathcal{L}(\mathbb{R}^d,E)$ denotes the derivative of the specified extension, and

$$\|Df\|_{C^0(Y)} := \sup_{y\in Y}\|Df(y)\|_{\mathcal{L}(\mathbb{R}^d,E)}.$$

Definition 4.2 (JEPA residual and prediction loss)**.** For $(\Phi,F)\in\mathcal{A}$, define

$$R[\Phi,F](s,a) := \Phi(H(G(s,a))) - F(\Phi(H(s)),a) = \psi(G(s,a)) - F(\psi(s),a).$$

The population prediction loss is

$$\mathcal{E}_{\text{pred}}(\Phi,F) := \int_M \|R[\Phi,F](s,a)\|_Z^2\,\mathrm{d}\mu(s,a).$$

Given samples $(s_i,a_i)_{i=1}^n \sim \mu^{\otimes n}$, the empirical prediction loss is

$$\widehat{\mathcal{E}_{\text{pred}}}_n(\Phi,F) := \frac{1}{n}\sum_{i=1}^n \|R[\Phi,F](s_i,a_i)\|_Z^2.$$

The residual is uniformly Lipschitz on M whenever G,H,Φ,F have uniform Lipschitz constants. We denote a structural Lipschitz bound by L_R.

Proposition 4.3 (Existence of regularized empirical minimizers)**.** *Let τ be a topology on the admissible class $\mathcal{A} = \mathcal{A}_\Phi \times \mathcal{A}_F$. Assume that $\mathcal{A}$ is compact in τ, and that the empirical prediction loss and $\widehat{\mathcal{N}}_n : \mathcal{A}_\Phi \to [0, \infty]$ are τ-lower semicontinuous. Then, for every $\lambda > 0$, the empirical objective*

$$\widehat{\mathcal{E}}_{\lambda,n}(\Phi, F) := \widehat{\mathcal{E}_{\mathrm{pred}}}_n(\Phi, F) + \lambda \widehat{\mathcal{N}}_n(\Phi)$$

attains its infimum on $\mathcal{A}$.

For the Lipschitz-only setting one may take τ to be uniform convergence and use Arzelà–Ascoli. For the metric regularizer, which depends on $D(\Phi \circ H)$, one takes τ to be the C^1 topology. At each fixed capacity W, a class that is closed in C^1 and uniformly bounded in $C^{1,1}$ is compact in C^1; the empirical local and global hinge terms are then continuous.

Proof. Let $\{(\Phi_j, F_j)\}_{j\geq 1}$ be a minimizing sequence for the empirical objective. Since $\mathcal{A}$ is τ-compact, there exists a τ-convergent subsequence $\{(\Phi_{j_k}, F_{j_k})\}_{k\geq 1}$ with limit $(\Phi_0, F_0) \in \mathcal{A}$. Because the prediction term and the regularizer are τ-lower semicontinuous, we have

$$\widehat{\mathcal{E}}_{\lambda,n}(\Phi_0, F_0) \leq \liminf_{k\to\infty} \widehat{\mathcal{E}}_{\lambda,n}(\Phi_{j_k}, F_{j_k}).$$

Thus, (Φ_0, F_0) is a minimizer. ■

We refer to the parameter λ as *regularization strength*. The metric regularizer is presented in Section 6. The admissible range of λ for the uniform approximate controlled semiconjugacy estimate is discussed in Section 7.

5. Geometric obstructions for pure forward prediction and covariance/spectral spread

The following two propositions identify the obstruction that the metric regularizer will address. They show that forward prediction alone, and forward prediction plus purely second-moment spread penalties, do not imply a controlled semiconjugacy useful for planning.

Proposition 5.1 (Pure forward JEPA admits collapsed zero-loss minimizers)**.** *Assume $K_Z \neq \emptyset$. For any $z_0 \in K_Z$, the constant encoder and transition map*

$$\Phi(x) \equiv z_0, \qquad F(z, a) \equiv z_0$$

satisfy

$$R[\Phi, F](s, a) = 0 \qquad \text{for all } (s, a) \in M.$$

Hence pure forward prediction cannot by itself imply non-collapse.

Proof. For every (s, a), we have

$$\Phi(H(G(s, a))) = z_0 = F(\Phi(H(s)), a).$$

Thus, the residual is identically zero. On the other hand, we also observe that the induced latent map $\psi \equiv z_0$ has no positive lower co-Lipschitz constant unless S is a singleton. ■

To prevent a total collapse in pure forward JEPA, a natural attempt is to add covariance or spectral spread.

Definition 5.2 (State marginal of the training measure)**.** Let μ be the training measure on the state–action space $M = S \times A$. We denote by

$$\pi_S : M \to S, \qquad \pi_S(s, a) = s,$$

the canonical projection. The S-marginal of μ is the pushforward probability measure

$$\mu_S := (\pi_S)_\# \mu.$$

Equivalently, for every Borel set $B \subset S$,

$$\mu_S(B) = \mu(B \times A).$$

Equivalently, if $(\mathbf{S}, \mathbf{A})$ is an M-valued random variable with $\mathcal{L}(\mathbf{S}, \mathbf{A}) = \mu$, then

$$\mathcal{L}(\mathbf{S}) = \mu_S.$$

For a latent map $\psi : S \to Z = \mathbb{R}^{d_z}$, define its μ_S-mean by

$$\bar{\psi} := \int_S \psi(s) \mathrm{d}\mu_S(s) \in \mathbb{R}^{d_z}.$$

The covariance matrix of $\psi(s)$ under $s \sim \mu_S$ is

$$\Sigma_\psi := \mathrm{Cov}_{s\sim\mu_S}(\psi(s)) := \int_S \big(\psi(s) - \bar{\psi}\big)\big(\psi(s) - \bar{\psi}\big)^\top \mathrm{d}\mu_S(s) \in \mathbb{R}^{d_z \times d_z}.$$

Equivalently,

$$(\Sigma_\psi)_{ij} = \int_S \big(\psi_i(s) - \bar{\psi}_i\big)\big(\psi_j(s) - \bar{\psi}_j\big) \mathrm{d}\mu_S(s).$$

Variance-type regularizers penalize small coordinate variances, i.e. small diagonal entries $(\Sigma_\psi)_{jj}$. Spectral regularizers penalize small eigenvalues, for example through a term such as

$$\sum_{j=1}^{d_z} \big[\gamma - \lambda_j(\Sigma_\psi)\big]_+^2 \qquad \text{or} \qquad \big[\gamma - \lambda_{\min}(\Sigma_\psi)\big]_+^2.$$

Such penalties rule out total collapse in a global second-moment sense, but they do not imply injectivity or a pointwise lower co-Lipschitz bound.

Proposition 5.3 (Covariance/Spectral spread does not imply injectivity)**.** *There exists a deterministic control system and a non-injective zero-residual latent model whose latent code has positive variance. Consequently, covariance or spectral spread alone cannot imply the desired pointwise lower co-Lipschitz property without additional structural assumptions.*

Proof. We consider the example

$$S = [-1, 1], \qquad A = \{0\}, \qquad G(s, 0) = s, \qquad H(s) = s.$$

Let μ_S be the uniform probability measure on $[-1, 1]$. Here the singleton action set is only a notational shorthand for an uncontrolled special case. If one insists that the action domain have nonempty Euclidean interior, one may instead take any compact interval A and define $G(s, a) = s$ and $F(z, a) = z$ independently of a; the argument is unchanged. Let $Z = \mathbb{R}$. We choose a constant $C > 0$, and set

$$\psi(s) = Cs^2, \qquad F(z, 0) = z.$$

Then, we obtain the residual

$$R[\psi, F](s, 0) = \psi(G(s, 0)) - F(\psi(s), 0) = Cs^2 - Cs^2 = 0.$$

Moreover, we compute the variance

$$\mathrm{Var}_{\mu_S}(\psi(s)) = C^2 \,\mathrm{Var}_{\mu_S}(s^2) = C^2(\mathbb{E}_{\mu_S}[s^4] - (\mathbb{E}_{\mu_S}[s^2])^2) = \frac{4C^2}{45}.$$

Thus, $\psi(s)$ has positive variance for every fixed $C > 0$. For any prescribed variance threshold, one may choose a sufficiently large compact latent interval $K_Z = [0, C]$ and then choose C so that the threshold is cleared. No claim of unbounded variance is made within a single fixed compact latent range.

However, $\psi(s) = \psi(-s)$, so ψ is not injective and no inequality of the form

$$\|\psi(s) - \psi(s')\|_Z \geq c\, d_S(s, s')$$

can hold with $c > 0$. ■

These examples identify the precise role of geometric regularization. Pure forward JEPA leaves non-collapse unconstrained, and covariance or spectral spread leaves pointwise lower geometry unconstrained. The local–global objective supplies exactly the directional and separated-pair information used by the positive theorem to establish a uniform co-Lipschitz certificate.

6. The metric hinge regularizer

We now introduce a soft encoder-only metric regularizer. It requires neither a decoder nor an inverse model and imposes no hard architectural constraint. Instead, it directly promotes two complementary geometric properties of the latent representation: infinitesimal directional resolution and separation of well-separated states.

Throughout this section, $S \subset \mathbb{R}^{d_s}$ is a compact convex Euclidean domain and all maps are understood as restrictions of $C^{1,1}$ maps defined on open neighborhoods.

Training-information convention. Although the prediction residual is evaluated from observation–action transitions, the metric penalty is supervised by the observable-factor geometry. Its empirical implementation uses samples of the coordinate s (or an equivalent metric representation), state distances $|s - s'|$, and tangent directions v. This directly covers simulator-state and proprioceptive settings while preserving observation–action deployment. State-estimation or pixel-based geometry may be used when its distance and direction errors are independently bounded and absorbed into reduced effective margins before applying Assumptions A3–A4; the theorem is then applied to those certified effective quantities. No unquantified state-estimation claim is needed, and the deployed encoder–transition interface is unchanged.

Let

$$\mathcal{U} := S \times \mathbb{S}^{d_s - 1},$$

where $\mathbb{S}^{d_s-1}$ is the Euclidean unit sphere. We call $\mathcal{U}$ the unit-direction sampling set. It is equipped with the product probability measure

$$\omega := \mu_S \otimes \sigma,$$

where σ is normalized surface measure on $\mathbb{S}^{d_s-1}$. Fix a scale $0 < \rho < \operatorname{diam}(S)$. Let

$$P_\rho := \{(s, s') \in S \times S : \ |s - s'| \geq \rho\}.$$

Assuming $(\mu_S \otimes \mu_S)(P_\rho) > 0$, we define ν_ρ to be the normalized restriction of $\mu_S \otimes \mu_S$ to P_ρ, namely

$$\nu_\rho(E) := \frac{(\mu_S \otimes \mu_S)(E \cap P_\rho)}{(\mu_S \otimes \mu_S)(P_\rho)} \qquad \text{for every Borel set } E \subset S \times S.$$

Equivalently, ν_ρ is the conditional law of two independent μ_S-distributed states s, s', conditioned on the event $|s - s'| \geq \rho$.

The spaces M, $\mathcal{U}$ and P_ρ are endowed with the product metrics

$$d_M((s, a), (s', a')) := |s - s'| + |a - a'|,$$

$$d_U((s,v),(s',v')) := |s-s'| + |v-v'|,$$

and

$$d_P((s_1,s_2),(s_1',s_2')) := |s_1-s_1'| + |s_2-s_2'|.$$

Fix margins $\kappa > 0$ and $\alpha > 0$. For $\psi = \Phi \circ H$, we define the metric regularizers

$$\mathcal{N}_{\mathrm{loc}}(\Phi) := \int_{\mathcal{U}} \left[\kappa - \|D\psi(s)[v]\|_Z\right]_+^2 \mathrm{d}\omega(s,v),$$

$$\mathcal{N}_{\mathrm{glob}}(\Phi) := \int_{P_\rho} \left[\alpha - \|\psi(s)-\psi(s')\|_Z\right]_+^2 \mathrm{d}\nu_\rho(s,s'),$$

and

$$\mathcal{N}_{\mathrm{met}}(\Phi) := \mathcal{N}_{\mathrm{loc}}(\Phi) + \mathcal{N}_{\mathrm{glob}}(\Phi).$$

The local term is an averaged hinge version of the pointwise infinitesimal condition

$$\sigma_{\min}(D\psi(s)) \geq \kappa, \qquad \sigma_{\min}(D\psi(s)) = \inf_{|v|=1} \|D\psi(s)[v]\|_Z.$$

In the same perspective, the global term is an averaged hinge version of the metric condition

$$\|\psi(s)-\psi(s')\|_Z \geq \alpha, \quad \text{when } |s-s'| \geq \rho.$$

We now define the empirical version of the metric regularizer. Let

$$\{(s_i,a_i)\}_{i=1}^{n_M}$$

be independent samples from the training measure μ on $M = S \times A$. Let

$$\{(\xi_j,v_j)\}_{j=1}^{n_U}$$

be independent samples from the unit-direction measure

$$\omega = \mu_S \otimes \sigma \qquad \text{on} \qquad \mathcal{U} = S \times \mathbb{S}^{d_s-1},$$

where σ is normalized surface measure on $\mathbb{S}^{d_s-1}$. Finally, let

$$\{(\zeta_k,\zeta_k')\}_{k=1}^{n_P}$$

be independent samples from the separated-pair measure ν_ρ on

$$P_\rho = \{(s,s') \in S \times S : \ |s-s'| \geq \rho\}.$$

We assume that these three sample families are mutually independent.

We denote by $\mathcal{A}(W)$ an admissible class of encoder–latent-transition pairs depending on some trainable parameter W. For $(\Phi,F) \in \mathcal{A}(W)$, with $\psi = \Phi \circ H$, we define the empirical prediction loss by

$$\widehat{\mathcal{E}_{\mathrm{pred}}}_{n_M}(\Phi,F) := \frac{1}{n_M}\sum_{i=1}^{n_M} \|\psi(G(s_i,a_i)) - F(\psi(s_i),a_i)\|_Z^2.$$

We define the empirical local metric penalty by

$$\widehat{\mathcal{N}_{\mathrm{loc}}}_{n_U}(\Phi) := \frac{1}{n_U}\sum_{j=1}^{n_U} \left[\kappa - \|D\psi(\xi_j)[v_j]\|_Z\right]_+^2,$$

and define the empirical separated-pair metric penalty by

$$\widehat{\mathcal{N}_{\mathrm{glob}}}_{n_P}(\Phi) := \frac{1}{n_P}\sum_{k=1}^{n_P} \left[\alpha - \|\psi(\zeta_k) - \psi(\zeta_k')\|_Z\right]_+^2.$$

The empirical metric regularizer is then

$$\widehat{\mathcal{N}_{\mathrm{met}}}_{n_U,n_P}(\Phi) := \widehat{\mathcal{N}_{\mathrm{loc}}}_{n_U}(\Phi) + \widehat{\mathcal{N}_{\mathrm{glob}}}_{n_P}(\Phi).$$

Consequently, for a regularization parameter $\lambda > 0$, the metric-regularized empirical JEPA objective is

$$\widehat{\mathcal{E}}^{\mathrm{met}}_{\lambda,n_M,n_U,n_P}(\Phi, F) := \widehat{\mathcal{E}_{\mathrm{pred}}}_{n_M}(\Phi, F) + \lambda\widehat{\mathcal{N}_{\mathrm{met}}}_{n_U,n_P}(\Phi).$$

When no distinction between the three sample sizes is needed, we take

$$n_M = n_U = n_P = n$$

and write simply

$$\widehat{\mathcal{E}}^{\mathrm{met}}_{\lambda,n}(\Phi, F) := \widehat{\mathcal{E}_{\mathrm{pred}}}_n(\Phi, F) + \lambda\widehat{\mathcal{N}_{\mathrm{met}}}_n(\Phi), \qquad \widehat{\mathcal{N}_{\mathrm{met}}}_n(\Phi) := \widehat{\mathcal{N}_{\mathrm{loc}}}_n(\Phi) + \widehat{\mathcal{N}_{\mathrm{glob}}}_n(\Phi).$$

Assumption A3 (Metric sampling geometry). *The metric sampling spaces $(\mathcal{U}, d_U, \omega)$ and (P_ρ, d_P, ν_ρ) are lower Ahlfors regular. That is, there exist constants $m_U, m_P, r_U, r_P > 0$ and dimensions*

$$q_U = 2d_s - 1, \qquad q_P = 2d_s,$$

such that

$$\omega(B_r(s, v)) \geq m_U r^{q_U} \qquad \text{for all } (s, v) \in \mathcal{U}, \quad 0 < r \leq r_U,$$

$$\nu_\rho(B_{\bar r}(s, s')) \geq m_P \bar r^{q_P} \qquad \text{for all } (s, s') \in P_\rho, \quad 0 < \bar r \leq r_P.$$

The lower Ahlfors conditions in Assumptions A2 and A3 yield the following $L^2 \to L^\infty$ interpolation property.

Lemma 6.1 (Lipschitz–Ahlfors L^2-to-L^∞ interpolation)**.** *Let (Y, d, ν) be a compact metric probability space. Assume that there exist constants $m > 0$, $r_0 > 0$, and an exponent $q > 0$ such that*

$$\nu(B_r(y)) \geq m r^q \qquad \text{for all } y \in Y, \quad 0 < r \leq r_0.$$

If $u : Y \to [0, \infty)$ is 0-Lipschitz, then it is constant and

$$\|u\|^2_{L^\infty(Y)} = \int_Y u^2 \,\mathrm{d}\nu.$$

Thus the constant case is immediate. In the nonconstant case, let u be L-Lipschitz with $L > 0$, and suppose

$$\int_Y u^2 \mathrm{d}\nu \leq \varepsilon.$$

Then

$$\|u\|_{L^\infty(Y)} \leq \Theta_Y(\varepsilon),$$

where

$$\Theta_Y(\varepsilon) := 2 \max\left\{ \left(\frac{L^q \varepsilon}{m}\right)^{1/(q+2)}, \quad \left(\frac{\varepsilon}{m r_0^q}\right)^{1/2} \right\}.$$

In particular, $\Theta_Y(\varepsilon) \to 0$ as $\varepsilon \to 0$. Furthermore, we set

$$\Theta_Y^{-1}(\tau) := \min\left\{ \frac{m}{L^q}\left(\frac{\tau}{2}\right)^{q+2}, m r_0^q \left(\frac{\tau}{2}\right)^2 \right\},$$

which satisfies $\Theta_Y(\Theta_Y^{-1}(\tau)) = \tau$ for $\tau > 0$. In the case $L = 0$, we set $1/L^q = +\infty$ by convention.

Hypothesis	**Exact role in the proof**
A1: regular observable Markov factor	Supplies a finite-dimensional Euclidean observable state factor, a bi-Lipschitz observation embedding, and descended dynamics and costs.
Smooth realizability hypotheses of Lemma 7.3	Supply a $C^{1,1}$ bi-Lipschitz latent embedding, a smooth range-enforcing map, and hence the exact non-collapsed comparator used in the objective comparison.
A2: lower-Ahlfors dynamics coverage	Converts the population L^2 prediction residual into a uniform semiconjugacy defect through Lemma 6.1.
A3: tangent and separated-pair coverage	Converts population local and global hinge defects into uniform directional and separated-pair margins.
A4: finite-capacity approximation and uniform budgets	Provides a margin-clearing comparator, compactness/equi-Lipschitz control, and the second-order remainder estimate for short chords.
Approximate empirical global minimization	Compares the learned objective with the zero-metric-penalty comparator and turns the regularization threshold into population defect bounds.

Table 1. Assumption-to-conclusion map. The hypotheses provide geometry, regularity, coverage, and optimization control; the theorem proves stability of all approximate empirical minimizers within that regime.

Proof. We set $h = \|u\|_{L^\infty(Y)}$, and choose $y_0 \in Y$ such that $u(y_0) = h$. We suppose first that $h/(2L) \le r_0$, then the Lipschitz bound implies

$$u(y) \ge \frac{h}{2} \qquad \text{for all } y \in B_{h/(2L)}(y_0).$$

Consequently, using the hypothesis, we obtain

$$\varepsilon \ge \int_Y u^2 \mathrm{d}\nu \ge \frac{h^2}{4}\nu(B_{h/(2L)}(y_0)) \ge \frac{h^2}{4} m \left(\frac{h}{2L}\right)^q = \frac{m}{2^{q+2}L^q} h^{q+2}.$$

Hence, in the first case, we arrive at

$$h \le 2\left(\frac{L^q}{m}\varepsilon\right)^{1/(q+2)}.$$

In a similar way, we suppose now that $h/(2L) > r_0$, then $u \ge h/2$ on $B_{r_0}(y_0)$, and therefore

$$\varepsilon \ge \frac{h^2}{4}\nu(B_{r_0}(y_0)) \ge \frac{h^2}{4} m r_0^q.$$

Thus, in the second case, we obtain

$$h \le 2\left(\frac{\varepsilon}{m r_0^q}\right)^{1/2}.$$

Combining the two cases gives the stated estimate. ∎

7. Finite-sample geometric coercivity and controlled semiconjugacy

This section proves the principal finite-sample implication. The main text retains the comparator, margin-clearing, empirical-process, and representation arguments that drive the theorem. Detailed spline localization, coefficient stability, and architecture-specific covering-number calculations are provided in Supplementary Sections S1–S2.

7.1. **A smooth non-collapsed comparator and a concrete finite-capacity class.** The representation theorem requires a faithful comparator and finite-capacity classes that approximate it without losing regularity. Supplementary Section S1 contains the affine latent embedding, local inverse charts, extension and range-enforcement arguments, tensor-product spline construction, compactness proof, coefficient counts, margin-clearing proof, and nonempty-parameter calculation; Supplementary Section S2 contains the architecture-specific covering estimates.

Lemma 7.1 (Exact non-collapsed Lipschitz realization). *Under Assumption A1 and the standing conditions $d_z \ge d_s$ and $\operatorname{int} K_Z \neq \varnothing$, there are a smooth affine bi-Lipschitz embedding $\psi_0 : S \to \operatorname{int} K_Z$, a $C^{1,1}$ range-enforcing map $\Pi_Z : Z \to K_Z$ that is the identity near $\psi_0(S)$, and Lipschitz maps Φ_0, F_0 such that $\Phi_0 \circ H = \psi_0$ and $F_0(\psi_0(s), a) = \psi_0(G(s, a))$.*

Lemma 7.2 (Quantitative inverse regularity). *Every $C^{1,1}$ map on a neighborhood of a compact convex Euclidean domain that is quantitatively co-Lipschitz on the domain admits finitely many $C^{1,1}$ inverse charts near its compact image, with constants determined by the co-Lipschitz constant and the fixed-neighborhood $C^{1,1}$ budget.*

Lemma 7.3 ($C^{1,1}$ exact representatives). *Assume A1 and $X = \mathbb{R}^{d_x}$, and use the affine embedding and smooth range-enforcing map furnished by Lemma 7.1. Then the exact realization may be chosen with $\Phi_0 \in C^{1,1}(X; K_Z)$ and $F_0 \in C^{1,1}(K_Z \times A; K_Z)$, while retaining the exact identities on $H(S)$ and $\psi_0(S) \times A$.*

Assumption A4 (Finite-capacity C^1 approximation with uniform $C^{1,1}$ budgets). *The exact pair is approximated by closed classes $\mathcal{A}_\Phi(W) \times \mathcal{A}_F(W)$ mapping into K_Z, with capacity-independent Lipschitz and $C^{1,1}$ budgets and*

$$\|\Phi_W - \Phi_0\|_{C^1(H(S))} + \|F_W - F_0\|_{C^1(\psi_0(S)\times A)} \le \delta(W), \qquad \delta(W) \to 0.$$

Proposition 7.4 (Concrete finite-capacity realization). *For $X = \mathbb{R}^{d_x}$ and the exact representatives and range-enforcing map of Lemma 7.3, norm-constrained tensor-product cubic B-spline classes on fixed rectangular neighborhoods of $H(S)$ and $K_Z \times A$ satisfy Assumption A4. They may be chosen so that*

$$\delta(W) \le C_\Phi W^{-1/d_x} + C_F W^{-1/(d_z+d_a)},$$

their Lipschitz and $C^{1,1}$ budgets are independent of W, and each class is compact in C^1.

Proposition 7.5 (Margin-clearing competitors). *Choose $0 < \kappa < \inf_{\mathcal{U}} \|D\psi_0(s)[v]\|_Z$ and $0 < \alpha < \inf_{P_\rho} \|\psi_0(s) - \psi_0(s')\|_Z$. For all sufficiently large W, the approximants clear both margins, have zero population and empirical metric penalty, and satisfy*

$$\mathcal{E}_{\text{pred}}(\Phi_W, F_W) \le \beta(W) := C_\sharp \delta(W)^2 \to 0.$$

Lemma 7.6 (Nonempty parameter regime). *Let $\kappa_0 := \inf_{(s,v)\in\mathcal{U}} \|D\psi_0(s)[v]\|_Z > 0$ and let B_{loc} be the capacity-uniform second-order budget. Every $\kappa \in (0, \kappa_0)$ admits*

$$0 < \rho < \min\left\{\operatorname{diam}(S), \frac{\kappa}{4B_{\text{loc}}}\right\}, \qquad 0 < \alpha < \inf_{P_\rho} \|\psi_0(s) - \psi_0(s')\|_Z,$$

with the usual $+\infty$ convention when $B_{\text{loc}} = 0$. Hence the geometric margin regime required by Theorem 7.11 is nonempty.

These statements supply the only construction input used below: a zero-penalty comparator with prediction error $\beta(W)$ and uniform regularity. Their complete statements and proofs are in Supplementary Section S1; the architecture-specific entropy estimates are in Supplementary Section S2.

7.2. **Uniform statistical control.** Fix $\kappa, \alpha > 0$ and define

$$\mathcal{G}_M(W) = \{(s,a) \mapsto \|R[\Phi, F](s,a)\|_Z^2 : (\Phi, F) \in \mathcal{A}(W)\},$$

$$\mathcal{G}_U(W) = \{(s,v) \mapsto [\kappa - \|D(\Phi \circ H)(s)[v]\|_Z]_+^2 : \Phi \in \mathcal{A}_\Phi(W)\},$$

and

$$\mathcal{G}_P(W) = \{(s,s') \mapsto [\alpha - \|(\Phi \circ H)(s) - (\Phi \circ H)(s')\|_Z]_+^2 : \Phi \in \mathcal{A}_\Phi(W)\}.$$

Let $N_M(r;W), N_U(r;W), N_P(r;W)$ denote their C^0 covering numbers. The generic deviation argument is retained here; the spline coefficient and parameter-Lipschitz verification is in Supplementary Section S2.

Lemma 7.7 (Equi-Lipschitz integrand classes)**.** *Adopt the uniform $C^{1,1}$ budgets of Assumption A4 and the standing Lipschitz regularity of G and H. By Assumption A4, there are constants $C_\psi, L_F^{\mathrm{bud}} < \infty$, independent of W, such that for every $\Phi \in \mathcal{A}_\Phi(W)$, writing $\psi := \Phi \circ H$, we have*

$$\|\psi\|_{C^0(S)} + \|D\psi\|_{C^0(S)} + \mathrm{Lip}(D\psi; S) \le C_\psi,$$

and for every $F \in \mathcal{A}_F(W)$,

$$F : K_Z \times A \to K_Z, \qquad \mathrm{Lip}(F) \le L_F^{\mathrm{bud}}.$$

Then $\mathcal{G}_M(W), \mathcal{G}_U(W), \mathcal{G}_P(W)$ are uniformly bounded and equi-Lipschitz on their respective compact domains: there exist constants

$$B_M, B_U, B_P < \infty \quad \textit{and} \quad \widetilde{L}_M, \widetilde{L}_U, \widetilde{L}_P < \infty,$$

independent of W, such that every $g \in \mathcal{G}_Y(W)$ satisfies

$$0 \le g \le B_Y, \qquad \mathrm{Lip}(g) \le \widetilde{L}_Y, \qquad Y \in \{M, U, P\}.$$

Consequently, $N_Y(r;W)$ is finite for every $r > 0$ and $Y \in \{M, U, P\}$.

Furthermore, there exist constants $\overline{L}_M, \overline{L}_U, \overline{L}_P < \infty$ such that

$$\mathrm{Lip}(\sqrt{g}) \le \overline{L}_Y, \qquad \textit{for all } g \in \mathcal{G}_Y(W),\ Y \in \{M, U, P\}.$$

Proof. We first treat the case $Y = M$.

Since $K_Z \subset Z$ is compact, we set

$$D_Z := \mathrm{diam}\,(K_Z) < \infty.$$

For every admissible encoder Φ, the induced map

$$\psi = \Phi \circ H : S \to K_Z$$

takes values in K_Z. Hence

$$\|\psi(s) - \psi(s')\|_Z \le D_Z \qquad \text{for all } s, s' \in S.$$

The uniform $C^{1,1}$ budget gives constants, independent of W, such that

$$\mathrm{Lip}(\psi) \le L_\psi, \qquad \|D\psi\|_{C^0(S)} \le B_D, \qquad \mathrm{Lip}(D\psi) \le L_D.$$

We may take these constants to be bounded by C_ψ.

Let

$$g_M(s,a) := \|R[\Phi, F](s,a)\|_Z^2,$$

where

$$R[\Phi, F](s,a) = \psi(G(s,a)) - F(\psi(s), a).$$

Since both $\psi(G(s,a))$ and $F(\psi(s),a)$ lie in K_Z, we have

$$\|R[\Phi,F](s,a)\|_Z \le D_Z \qquad \text{for all } (s,a)\in M.$$

Thus, we obtain the bound

$$0 \le g_M(s,a) \le D_Z^2.$$

We may therefore take

$$B_M := D_Z^2.$$

We next prove a uniform Lipschitz bound for $R[\Phi,F]$. Let

$$p=(s,a), \qquad p'=(s',a').$$

Then, by the triangle inequality, we estimate

$$\|R[\Phi,F](p)-R[\Phi,F](p')\|_Z \le \|\psi(G(s,a))-\psi(G(s',a'))\|_Z + \|F(\psi(s),a)-F(\psi(s'),a')\|_Z.$$

For the first term, using the Lipschitz bounds for ψ and G, we have

$$\begin{aligned}\|\psi(G(s,a))-\psi(G(s',a'))\|_Z &\le L_\psi\, d_S(G(s,a),G(s',a'))\\ &\le L_\psi \operatorname{Lip}(G)\, d_M(p,p').\end{aligned}$$

For the second term, using the Lipschitz budget of F, we have

$$\begin{aligned}\|F(\psi(s),a)-F(\psi(s'),a')\|_Z &\le L_F^{\mathrm{bud}}\big(\|\psi(s)-\psi(s')\|_Z + d_A(a,a')\big)\\ &\le L_F^{\mathrm{bud}}\big(L_\psi|s-s'| + d_A(a,a')\big)\\ &\le L_F^{\mathrm{bud}}\max\{L_\psi,1\} d_M(p,p').\end{aligned}$$

Therefore, we arrive at

$$\operatorname{Lip}(R[\Phi,F]) \le L_R,$$

where

$$L_R := L_\psi \operatorname{Lip}(G) + L_F^{\mathrm{bud}}\max\{L_\psi,1\}.$$

The constant L_R is independent of W.

Now, we use the elementary inequality

$$\big|\|x\|_Z^2 - \|y\|_Z^2\big| \le (\|x\|_Z + \|y\|_Z)\|x-y\|_Z.$$

Since $\|R[\Phi,F]\| \le D_Z$, we obtain

$$\begin{aligned}|g_M(p)-g_M(p')| &= \big|\|R[\Phi,F](p)\|_Z^2 - \|R[\Phi,F](p')\|_Z^2\big|\\ &\le 2D_Z\|R[\Phi,F](p)-R[\Phi,F](p')\|_Z\\ &\le 2D_Z L_R\, d_M(p,p').\end{aligned}$$

By the reverse triangle inequality, we have

$$|\sqrt{g_M(p)}-\sqrt{g_M(p')}| = \big|\|R[\Phi,F](p)\|_Z - \|R[\Phi,F](p')\|_Z\big| \le \|R[\Phi,F](p)-R[\Phi,F](p')\|_Z \le L_R d_M(p,p').$$

Hence every element of $\mathcal{G}_M(W)$ satisfies

$$\operatorname{Lip}(g_M) \le \widetilde{L}_M, \qquad \widetilde{L}_M := 2(\operatorname{diam} K_Z)L_R, \qquad \overline{L}_M := L_R.$$

The cases $Y = U, P$ are analogous, using that $t \mapsto [c-t]_+^2$ is $2c$-Lipschitz on $[0,\infty)$ and that $(s,v) \mapsto \|D\psi(s)[v]\|_Z$ and $(s,s') \mapsto \|\psi(s)-\psi(s')\|_Z$ are Lipschitz with constants $\max\{B_D, L_D\}$ and L_ψ, respectively. The uniform bounds and the Lipschitz budgets are

$$B_U = \kappa^2, \quad \widetilde{L}_U = 2\kappa \max\{B_D, L_D\}, \quad \overline{L}_U = \max\{B_D, L_D\},$$

$$B_P = \alpha^2, \quad \widetilde{L}_P = 2\alpha L_\psi, \quad \overline{L}_P = L_\psi.$$

Finally, we show that the C^0-covering numbers $N_Y(r;W)$ are finite for every $r>0$ and each $Y \in \{M,U,P\}$.

Indeed, for any $Y \in \{M,U,P\}$, the class $\mathcal{G}_Y(W)$ is uniformly bounded and equi-Lipschitz on the corresponding compact domain, with constants independent of W. By the Arzelà–Ascoli theorem, any such class is relatively compact in $C^0(Y)$. In particular, it is totally bounded in the C^0-norm. Therefore, for every $r>0$, there exists a finite r-net in $C^0(Y)$, and hence $N_Y(r;W) < \infty$. ■

Lemma 7.8 (Uniform statistical deviation)**.** *Assume the hypotheses of Lemma 7.7. For each $Y \in \{M,U,P\}$, choose in advance a deterministic radius*

$$r_Y = r_Y(n,\delta,W) \in (0, B_Y),$$

and define

$$\varepsilon_Y(n,\delta;W) := 2r_Y + B_Y\sqrt{\frac{\log(6N_Y(r_Y;W)/\delta)}{2n}}.$$

Set

$$\varepsilon_{\mathrm{stat},W}(n,\delta) := \max\left\{\varepsilon_M(n,\delta;W), \varepsilon_U(n,\delta;W) + \varepsilon_P(n,\delta;W)\right\}.$$

Then, with probability at least $1-\delta$, simultaneously for all $(\Phi,F) \in \mathcal{A}(W)$,

$$\left|\widehat{\mathcal{E}_{\mathrm{pred}}}_n(\Phi,F) - \mathcal{E}_{\mathrm{pred}}(\Phi,F)\right| \le \varepsilon_{\mathrm{stat},W}(n,\delta),$$

and

$$\left|\widehat{\mathcal{N}_{\mathrm{met}}}_n(\Phi) - \mathcal{N}_{\mathrm{met}}(\Phi)\right| \le \varepsilon_{\mathrm{stat},W}(n,\delta).$$

For fixed W and δ, one may choose deterministic radii $r_Y(n) \downarrow 0$ slowly enough that

$$\frac{\log N_Y(r_Y(n);W)}{n} \to 0,$$

in which case $\varepsilon_{\mathrm{stat},W}(n,\delta) \to 0$.

Proof. Fix $Y \in \{M,U,P\}$ and let $\{g_1,\dots,g_N\}$ be an r_Y-net of $\mathcal{G}_Y(W)$ in $C^0(Y)$, where $N = N_Y(r_Y;W)$. If $\|g-g_j\|_{C^0(Y)} \le r_Y$, then

$$|(\mu_{Y,n}-\mu_Y)g| \le |(\mu_{Y,n}-\mu_Y)g_j| + 2r_Y.$$

Since $0 \le g_j \le B_Y$, Hoeffding's inequality and a union bound give

$$\mathbb{P}\left(\max_{1\le j\le N} |(\mu_{Y,n}-\mu_Y)g_j| > t\right) \le 2N\exp\left(-\frac{2nt^2}{B_Y^2}\right).$$

Taking

$$t = B_Y\sqrt{\frac{\log(6N_Y(r_Y;W)/\delta)}{2n}}$$

shows that the stated bound for class Y fails with probability at most $\delta/3$. A union bound over $Y = M, U, P$ yields the simultaneous event. The prediction loss is the integral of an element of $\mathcal{G}_M(W)$, while the two metric penalties are integrals of elements of $\mathcal{G}_U(W)$ and $\mathcal{G}_P(W)$; the displayed estimates follow by addition.

For fixed W, total boundedness gives $N_Y(r; W) < \infty$ for every $r > 0$. A diagonal choice of deterministic radii can therefore be made so that $r_Y(n) \downarrow 0$ and $\log N_Y(r_Y(n); W) = o(n)$, proving convergence. ∎

Corollary 7.9 (Finite-parametric estimate). *Assume the hypotheses of Lemma 7.8. Since the three classes $\mathcal{G}_M(W)$, $\mathcal{G}_U(W)$, $\mathcal{G}_P(W)$ are all determined by the parameters of the pair $(\Phi, F) \in \mathcal{A}(W)$, they share a common parametrization*

$$\theta \in \Theta_W \subset [-R_W, R_W]^{p_W}.$$

Suppose that, for each $Y \in \{M, U, P\}$, there is a constant $L_{\theta,Y,W} > 0$ with

$$\|g_\theta - g_{\theta'}\|_{C^0(Y)} \le L_{\theta,Y,W} \|\theta - \theta'\|_{\ell^\infty} \qquad \text{for all } \theta, \theta' \in \Theta_W.$$

Then, for every $r > 0$,

$$N_Y(r; W) \le \left(1 + \tfrac{2R_W L_{\theta,Y,W}}{r}\right)^{p_W},$$

and hence, for every deterministic choice $r_Y \in (0, B_Y)$,

$$\varepsilon_Y(n, \delta; W) \le 2r_Y + B_Y \sqrt{\frac{p_W \log\left(1 + \frac{2R_W L_{\theta,Y,W}}{r_Y}\right) + \log(6/\delta)}{2n}}.$$

In particular, taking $r_Y = B_Y n^{-1/2}$ gives, for $n > 1$,

$$\varepsilon_Y(n, \delta; W) \le \frac{2B_Y}{\sqrt{n}} + B_Y \sqrt{\frac{p_W \log\left(1 + \frac{2R_W L_{\theta,Y,W}\sqrt{n}}{B_Y}\right) + \log(6/\delta)}{2n}}.$$

If $R_W L_{\theta,Y,W} \le C_Y W^{q_Y}$ for constants C_Y, q_Y independent of n, then

$$\varepsilon_Y(n, \delta; W) = O\left(\sqrt{\frac{p_W(\log n + \log W) + \log(1/\delta)}{n}}\right).$$

The same rate holds for $\varepsilon_{\mathrm{stat},W}(n, \delta) = \max\{\varepsilon_M,\ \varepsilon_U + \varepsilon_P\}$:

$$\varepsilon_{\mathrm{stat},W}(n, \delta) = O\left(\sqrt{\frac{p_W(\log n + \log W) + \log(1/\delta)}{n}}\right).$$

In particular, if W is fixed, or if the capacity schedule satisfies $W = W(n) \le Cn^q$ for some $C, q > 0$, then $\log W = O(\log n)$ and

$$\varepsilon_{\mathrm{stat},W}(n, \delta) = O\left(\sqrt{\frac{p_W \log n + \log(1/\delta)}{n}}\right).$$

Proof. Fix $Y \in \{M, U, P\}$; we bound the covering number of $\mathcal{G}_Y(W)$ in the $C^0(Y)$-norm.

We partition each coordinate interval $[-R_W, R_W]$ into at most

$$1 + \frac{2R_W L_{\theta,Y,W}}{r}$$

subintervals of length at most $r/L_{\theta,Y,W}$. This partitions the parameter box into at most

$$N := \left(1 + \frac{2R_W L_{\theta,Y,W}}{r}\right)^{p_W}$$

cells. For every cell that intersects Θ_W, choose one representative $\theta_j \in \Theta_W$ from that intersection. Any two points in the same cell are at ℓ^∞-distance at most $r/L_{\theta,Y,W}$. Hence, for each $\theta \in \Theta_W$, the representative of its cell satisfies

$$\|g_\theta - g_{\theta_j}\|_{C^0(Y)} \le L_{\theta,Y,W}\|\theta - \theta_j\|_{\ell^\infty} \le r.$$

Therefore the selected g_{θ_j} form an r-net of $\mathcal{G}_Y(W)$ in $C^0(Y)$, so

$$N_Y(r; W) \le N \le \left(1 + \frac{2R_W L_{\theta,Y,W}}{r}\right)^{p_W}.$$

Substituting this bound into Lemma 7.8 gives the stated bound on $\varepsilon_Y(n, \delta; W)$; the remaining estimates follow by the elementary computations in the statement. ■

7.3. **Principal finite-sample representation theorem.** We now fix the three interpolation moduli used in the estimates. As in Subsection 7.2, we consider the function classes $\mathcal{G}_M(W)$, $\mathcal{G}_U(W)$, and $\mathcal{G}_P(W)$ with fixed $\kappa, \alpha > 0$. By Lemma 7.7, let $\overline{L}_M$, $\overline{L}_U$, and $\overline{L}_P$ denote the uniform Lipschitz bounds for the square-root integrands

$$\|R[\Phi, F]\|_Z, \qquad [\kappa - \|D(\Phi \circ H)(s)[v]\|_Z]_+, \qquad [\alpha - \|(\Phi \circ H)(s) - (\Phi \circ H)(s')\|_Z]_+,$$

respectively. These are the functions to which the Lipschitz–Ahlfors L^2-to-L^∞ lemma is applied.

The moduli $\Theta_M, \Theta_{\mathcal{U}}, \Theta_{P_\rho}$ are therefore the maps furnished by Lemma 6.1 with data

$$(d_m, m_0, r_0, \overline{L}_M), \qquad (q_U, m_U, r_U, \overline{L}_U), \qquad (q_P, m_P, r_P, \overline{L}_P),$$

respectively.

Definition 7.10 (Approximate empirical minimizer)**.** For $\lambda > 0$, define

$$\widehat{\mathcal{J}}_{\lambda,n}(\Phi, F) := \widehat{\mathcal{E}_{\mathrm{pred}}}_n(\Phi, F) + \lambda \widehat{\mathcal{N}_{\mathrm{met}}}_n(\Phi).$$

Given $\varepsilon_{\mathrm{train}} \ge 0$, a pair $(\Phi^\star, F^\star) \in \mathcal{A}(W)$ is called an $\varepsilon_{\mathrm{train}}$-approximate empirical minimizer if

$$\widehat{\mathcal{J}}_{\lambda,n}(\Phi^\star, F^\star) \le \inf_{(\Phi,F)\in\mathcal{A}(W)} \widehat{\mathcal{J}}_{\lambda,n}(\Phi, F) + \varepsilon_{\mathrm{train}}.$$

The case $\varepsilon_{\mathrm{train}} = 0$ is the exact global-minimizer case.

Theorem 7.11 (Finite-sample metric non-collapse and controlled semiconjugacy)**.** *Assume A1, A2, A4, the hypotheses of Lemma 7.3, and the statistical setup of Lemmas 7.7–7.8 at capacity W. Fix (κ, ρ, α) as in Lemma 7.6, and assume A3 for this ρ. In particular,*

$$0 < \rho < \mathrm{diam}(S), \qquad B_{\mathrm{loc}}\rho \le \frac{\kappa}{4}, \quad B_{\mathrm{loc}} := \sup_W \sup_{\Phi\in\mathcal{A}_\Phi(W)} \mathrm{Lip}(D(\Phi \circ H)).$$

Let $(\Phi_\sharp, F_\sharp)$ be the margin-clearing competitor of Proposition 7.5, with

$$\mathcal{E}_{\mathrm{pred}}(\Phi_\sharp, F_\sharp) \le \beta(W).$$

Let $\varepsilon_{\mathrm{stat},W}(n, \delta)$ be the error in Lemma 7.8, let $\varepsilon_{\mathrm{train}} \ge 0$, and set

$$P^* := \min\{\Theta_{\mathcal{U}}^{-1}(\kappa/2), \Theta_{P_\rho}^{-1}(\alpha/2)\}.$$

Suppose $\varepsilon_{\mathrm{stat},W}(n,\delta) < P^*$ *and choose*

$$\lambda \geq \lambda_{\min}(W,n,\delta;\varepsilon_{\mathrm{train}}) := \frac{\beta(W)+\varepsilon_{\mathrm{stat},W}(n,\delta)+\varepsilon_{\mathrm{train}}}{P^*-\varepsilon_{\mathrm{stat},W}(n,\delta)}. \tag{7.1}$$

Then, with probability at least $1-\delta$*, every* $\varepsilon_{\mathrm{train}}$*-approximate empirical minimizer* $(\Phi^\star, F^\star)$ *satisfies*

$$\|\Phi^\star(H(s)) - \Phi^\star(H(s'))\|_Z \geq c_* d_S(s,s') \qquad (s,s' \in S), \tag{7.2}$$

where

$$c_* := \min\left\{\frac{\kappa}{4}, \frac{\alpha}{2\,\mathrm{diam}(S)}\right\} > 0,$$

and

$$\sup_{(s,a)\in M} \|\Phi^\star(H(G(s,a))) - F^\star(\Phi^\star(H(s)),a)\|_Z \leq \eta_{W,n,\delta}(\varepsilon_{\mathrm{train}}), \tag{7.3}$$

with

$$\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}}) := \Theta_M(\beta(W) + 2\varepsilon_{\mathrm{stat},W}(n,\delta) + \varepsilon_{\mathrm{train}}). \tag{7.4}$$

If $W = W(n)$ *and the approximation, statistical, and training errors vanish, then* $\eta_{W(n),n,\delta}(\varepsilon_{\mathrm{train}}(n)) \to 0$*. In the finite-parametric setting of Corollary 7.9, it is enough that*

$$\beta(W(n)) \to 0, \qquad \varepsilon_{\mathrm{train}}(n) \to 0, \qquad \frac{p_{W(n)}(\log n + \log W(n))}{n} \to 0.$$

Proof. Write $\varepsilon := \varepsilon_{\mathrm{stat},W}(n,\delta)$. On the event in Lemma 7.8, the prediction and metric empirical/population deviations are at most ε uniformly over $\mathcal{A}(W)$. Approximate optimality against the margin-clearing comparator gives

$$\mathcal{E}_{\mathrm{pred}}(\Phi^\star, F^\star) \leq \beta(W) + 2\varepsilon + \varepsilon_{\mathrm{train}}, \qquad \mathcal{N}_{\mathrm{met}}(\Phi^\star) \leq \frac{\beta(W)+\varepsilon+\varepsilon_{\mathrm{train}}}{\lambda} + \varepsilon. \tag{7.5}$$

Equation (7.1) makes the second bound at most P^*.

Set $\psi^\star := \Phi^\star \circ H$. The local defect

$$q^\star_{\mathrm{loc}}(s,v) := [\kappa - \|D\psi^\star(s)[v]\|_Z]_+$$

is uniformly Lipschitz and has squared L^2 norm at most P^*. Lemma 6.1 and the definition of P^* yield

$$\|D\psi^\star(s)[v]\|_Z \geq \frac{\kappa}{2} \qquad ((s,v) \in \mathcal{U}). \tag{7.6}$$

The same argument for the separated-pair defect gives

$$\|\psi^\star(s) - \psi^\star(s')\|_Z \geq \frac{\alpha}{2} \qquad \text{when } d_S(s,s') \geq \rho. \tag{7.7}$$

For pairs at distance at least ρ, this implies the second term in c_*. For $h := |s'-s| < \rho$, let $e = (s'-s)/h$ and $\gamma(t) = s + t(s'-s)$. Convexity, the fundamental theorem of calculus, (7.6), and the second-order budget give

$$\begin{aligned}\|\psi^\star(s') - \psi^\star(s)\|_Z &\geq h\|D\psi^\star(s)[e]\|_Z - h\int_0^1 \|(D\psi^\star(\gamma(t)) - D\psi^\star(s))[e]\|_Z\,\mathrm{d}t \\ &\geq \frac{\kappa}{2}h - \frac{B_{\mathrm{loc}}}{2}h^2 \geq \frac{3\kappa}{8}h \geq c_* h.\end{aligned}$$

This proves (7.2).

Finally, the residual norm is uniformly Lipschitz on M, and (7.5) bounds its squared population L^2 norm. Assumption A2 and Lemma 6.1 therefore give (7.3). The convergence statements follow from $\Theta_M(r) \to 0$ as $r \to 0$ and Corollary 7.9. ■

Corollary 7.12 (Explicit admissible regularization range)**.** *Under the hypotheses of Theorem 7.11, the admissible strengths form the nonempty half-line*

$$[\lambda_{\min}(W, n, \delta; \varepsilon_{\text{train}}), \infty).$$

There is no finite upper endpoint. For fixed W and $\varepsilon_{\text{train}}$,

$$\lambda_{\min}(W, n, \delta; \varepsilon_{\text{train}}) \longrightarrow \frac{\beta(W) + \varepsilon_{\text{train}}}{P^*} \qquad (n \to \infty).$$

If all three errors vanish along $W = W(n)$, then $\lambda_{\min} \to 0$, so every fixed $\lambda > 0$ is eventually admissible.

Proof. The comparison uses only the lower bound on λ, while the comparator has zero empirical metric penalty. The limits follow directly from the formula. ■

7.4. **Sharpness of the interpolation exponent.** The exponent supplied by Lipschitz–Ahlfors interpolation cannot be improved under the stated regularity alone.

Proposition 7.13 (Sharpness of the $1/(q+2)$ exponent)**.** *For $Y = [0,1]^q$ and fixed $L > 0$, there are nonnegative L-Lipschitz functions u_h with $\|u_h\|_\infty = h$ and $\int_Y u_h^2 \le C_q L^{-q} h^{q+2}$. Consequently no uniform estimate with exponent larger than $1/(q+2)$ holds on this class.*

The truncated-cone construction and calculation are given in Supplementary Section S3.

7.5. **A finite-net a posteriori certificate and use of the threshold.** After training, the theorem's a priori design guarantee is complemented by a model-specific route that certifies the uniform defects directly.

Proposition 7.14 (Validated finite-net certificate)**.** *Let $\psi = \Phi \circ H$ and F satisfy the same compact-range and second-order budget as in Theorem 7.11. Define*

$$\mathfrak{g}_U(s, v) := [\kappa - \|D\psi(s)[v]\|_Z]_+, \quad \mathfrak{g}_P(s, s') := [\alpha - \|\psi(s) - \psi(s')\|_Z]_+,$$

and $\mathfrak{g}_M(s, a) := \|\psi(G(s, a)) - F(\psi(s), a)\|_Z$. Assume

$$\operatorname{Lip}(\mathfrak{g}_U) \le L_U, \qquad \operatorname{Lip}(\mathfrak{g}_P) \le L_P, \qquad \operatorname{Lip}(\mathfrak{g}_M) \le L_M.$$

Let $\mathcal{Q}_U, \mathcal{Q}_P, \mathcal{Q}_M$ be finite nets with fill distances h_U, h_P, h_M. If

$$\max_{\mathcal{Q}_U} \mathfrak{g}_U + L_U h_U \le \frac{\kappa}{2}, \qquad \max_{\mathcal{Q}_P} \mathfrak{g}_P + L_P h_P \le \frac{\alpha}{2}, \qquad B_{\text{loc}}\rho \le \frac{\kappa}{4},$$

then

$$\|\psi(s) - \psi(s')\|_Z \ge \min\left\{\frac{\kappa}{4}, \frac{\alpha}{2\,\operatorname{diam}(S)}\right\} d_S(s, s') \quad (s, s' \in S).$$

Moreover,

$$\sup_{(s,a)\in M} \mathfrak{g}_M(s, a) \le \max_{\mathcal{Q}_M} \mathfrak{g}_M + L_M h_M.$$

Consequently, the deterministic planning bounds of Corollary 8.3 apply with these certified constants.

Proof. Choose a nearest net point in each compact domain. Lipschitz continuity bounds each continuum defect by its net maximum plus the corresponding fill-distance remainder. The first two inequalities give the uniform margins used in the short-chord/long-chord proof of Theorem 7.11; the third defect bound is the uniform semiconjugacy estimate. ∎

The analytic threshold and Proposition 7.14 give complementary certification routes. The class-uniform formula for $\lambda_{\min}$ proves an a priori one-sided design half-line for every approximate empirical minimizer and displays the dependence on coverage, regularity, approximation, sampling, and optimization. The finite-net proposition is model-specific and is typically sharper: validated interval bounds, exact spline or finite-element estimates, or controlled fill-distance remainders convert measured local, separated-pair, residual, and Lipschitz quantities into the constants used by the same deterministic planning corollary.

For calibration, one may sweep λ upward and retain the smallest value for which held-out directional and separated-pair margins are stable across samples and restarts while prediction residual and upper distortion remain controlled. This is a certificate-aligned diagnostic, not a monotonicity claim for nonconvex optimization. It becomes a rigorous a posteriori certificate only after the Lipschitz and fill-distance remainders in Proposition 7.14 have been validated.

8. Deterministic planning as a downstream corollary

Theorem 7.11 supplies the representation and dynamics certificates. Adding Lipschitz costs, a finite horizon, and a planner converts those certificates into deterministic trajectory, cost, and optimizer-transfer guarantees. Isolating this modular step makes the reusable role of the learned geometric and semiconjugacy constants explicit.

For $T \in \mathbb{N}$ and $L \geq 0$, define

$$C(T, L) := \sum_{j=0}^{T-1} L^j, \qquad D(T, L) := \sum_{t=0}^{T-1} \sum_{j=0}^{t-1} L^j, \tag{8.1}$$

where the inner sum is zero at $t = 0$. Put

$$\Delta_{T,L}(\eta) := \eta[L_{\widetilde{\ell},z} D(T, L) + L_{\widetilde{g}} C(T, L)], \quad \widehat{\Delta}_{T,L}(\eta) := \eta[L_{\widehat{\ell},z} D(T, L) + L_{\widehat{g}} C(T, L)].$$

Lemma 8.1 (Deterministic simulation estimate)**.** *Let $\psi : S \to K_Z$ and $F : K_Z \times A \to K_Z$ satisfy*

$$\sup_{(s,a)} \|\psi(G(s, a)) - F(\psi(s), a)\|_Z \leq \eta,$$

and suppose F is L_F-Lipschitz in its latent-state argument. For a common action sequence, let $s_{t+1} = G(s_t, a_t)$, $z_{t+1} = F(z_t, a_t)$, and $z_0 = \psi(s_0)$. Then

$$\|\psi(s_t) - z_t\|_Z \leq \eta C(t, L_F).$$

If $\widetilde{\ell}$ and $\widetilde{g}$ are Lipschitz in the latent variable, the true-image and latent-rollout costs differ by at most $\Delta_{T,L_F}(\eta)$.

Proof. The recurrence $e_{t+1} \leq \eta + L_F e_t$, $e_0 = 0$, gives $e_t \leq \eta C(t, L_F)$. Summing the stage-cost discrepancies and adding the terminal discrepancy gives the stated bound. ∎

Corollary 8.2 (Uniform finite-horizon planning transfer)**.** *Assume ψ is c-co-Lipschitz, the physical costs are Lipschitz in state, and let $\widetilde{\ell}, \widetilde{g}$ be the compatible latent extensions of Lemma 4.1. If $\widehat{\mathbf{a}}$ is*

ξ_{plan}*-optimal for the latent horizon-T problem, then*

$$J_{\mathrm{true}}(s_0;\widehat{\mathbf{a}}) - J^\star(s_0) \le 2\Delta_{T,L_F}(\eta) + \xi_{\mathrm{plan}},$$

with $L_{\widetilde{\ell},z} \le L_{\ell,s}/c$ *and* $L_{\widetilde{g}} \le L_g/c$. *For learned cost heads satisfying the compatibility bounds*

$$\sup_{s,a} |\widehat{\ell}(\psi(s),a) - \ell(s,a)| \le \varepsilon_\ell, \qquad \sup_s |\widehat{g}(\psi(s)) - g(s)| \le \varepsilon_g,$$

and the latent-state Lipschitz estimates

$$|\widehat{\ell}(z,a) - \widehat{\ell}(z',a)| \le L_{\widehat{\ell},z}\|z-z'\|_Z, \qquad |\widehat{g}(z) - \widehat{g}(z')| \le L_{\widehat{g}}\|z-z'\|_Z,$$

the right-hand side becomes

$$2[\widehat{\Delta}_{T,L_F}(\eta) + T\varepsilon_\ell + \varepsilon_g] + \xi_{\mathrm{plan}}.$$

Proof. Lemma 8.1 bounds the true/latent objective discrepancy uniformly over the common action space. Add and subtract the true and latent costs of a latent optimizer and a true optimizer. Approximate latent optimization adds ξ_{plan}; learned-head compatibility contributes $T\varepsilon_\ell + \varepsilon_g$ before the two-sided optimizer comparison. ∎

Corollary 8.3 (Finite-sample planning consequence)**.** *Under Theorem 7.11, assume the physical stage and terminal costs satisfy*

$$|\ell(s,a) - \ell(s',a')| \le L_{\ell,s} d_S(s,s') + L_{\ell,a} d_A(a,a'), \qquad |g(s) - g(s')| \le L_g d_S(s,s').$$

For every fixed horizon T and every ξ_{plan}*-optimal latent action sequence,*

$$J_{\mathrm{true}}(s_0;\widehat{\mathbf{a}}) - J^\star(s_0) \le 2\Delta_{T,L_F^{\mathrm{bud}}}(\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})) + \xi_{\mathrm{plan}}. \tag{8.2}$$

For learned heads satisfying

$$\sup_{s,a} |\widehat{\ell}(\psi^\star(s),a) - \ell(s,a)| \le \varepsilon_\ell, \qquad \sup_s |\widehat{g}(\psi^\star(s)) - g(s)| \le \varepsilon_g,$$

and

$$|\widehat{\ell}(z,a) - \widehat{\ell}(z',a)| \le L_{\widehat{\ell},z}\|z-z'\|_Z, \qquad |\widehat{g}(z) - \widehat{g}(z')| \le L_{\widehat{g}}\|z-z'\|_Z$$

on the relevant compact latent domain, one instead has

$$J_{\mathrm{true}}(s_0;\widehat{\mathbf{a}}) - J^\star(s_0) \le 2[\widehat{\Delta}_{T,L_F^{\mathrm{bud}}}(\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})) + T\varepsilon_\ell + \varepsilon_g] + \xi_{\mathrm{plan}}. \tag{8.3}$$

Proof. Combine Theorem 7.11, Lemma 4.1, and Corollary 8.2. ∎

Horizon dependence. If $L_F^{\mathrm{bud}} = 1$, then $C(T,1) = T$ and $D(T,1) = T(T-1)/2$, so the running-cost contribution is generally $O(\eta T^2/c_*)$. If $L_F^{\mathrm{bud}} < 1$, rollout error is bounded by $\eta/(1-L_F^{\mathrm{bud}})$; if $L_F^{\mathrm{bud}} > 1$, the worst-case envelope grows geometrically. These formulas furnish certified horizon-dependent envelopes; trajectory-local or contractive structure can produce substantially tighter realized errors.

Mean-to-uniform residual control. The simulation lemma requires a supremum residual, not only a population mean. A localized Lipschitz bump can have fixed pointwise height while its L^2 norm tends to zero with its support radius, and a deterministic optimizer may visit that high-error region. Theorem 7.11 closes this gap by combining regularity with coverage. The exact bump calculation, an adversarial-planner diagnostic, and additional Lyapunov and reachability consequences are in Supplementary Section S3.

experiment	R_Z	upper control	$n_M/n_U/n_P$	runs per objective	architecture	params	preprocessing time (s)
A3 folding	2.0	bounded range; upper hinge in objective (vi)	512/512/256	12 seeds × 2 regimes	MLP 1-128-128-1	–	–
B spline (main grid)	2.0	coefficient box ±3.0	$n/n/n$, $n \le 2048$	2 seeds × 3 restarts	cubic B-spline	–	–
B spline (capacity sweep)	2.0	coefficient box	512/512/512	1 seed(s)	cubic B-spline	–	–
B bounded MLP	2.0	approx. PI spectral target 1.6	$n/n/n$	2 seeds	MLP width 64	–	–
C pendulum + planning	2.0	approx. PI spectral target 1.5	2048/1024/512	3 seeds	MLP width 64	13896	20.64 [14.87, 50.40]

TABLE 2. Protocol alignment. Every trained encoder in a given experiment maps into the same compact latent box $K_Z = [-R_Z, R_Z]^{d_z}$, uses the same fixed D_M, D_U, D_P arrays, and receives the same number of updates. Run counts are read from the authors' retained result records, which will be provided upon request. The timing column is the median [min, max] over individual representation preprocessing runs, including representation training, cost-head fitting, and diagnostics but excluding MPC planning; it is not a total experiment wall-clock.

9. NUMERICAL EVIDENCE

The experiments provide three complementary forms of evidence for the mathematical mechanism: exact collapse and folding obstructions, a controlled finite-capacity study of threshold ingredients and margin formation, and a nonlinear latent-MPC benchmark linking representation geometry to true-system control. The high-probability result is established analytically; the numerical study connects its principal quantities to observable finite-model behavior. All trained suprema are evaluated on declared finite sets with adversarial refinement. Supplementary Section S4 provides the protocol, summarized tables, threshold ingredients, and additional figures; complete per-seed outputs, objective and gradient histories, evaluation arrays, and other supporting numerical documents are retained by the authors and will be provided upon request.

Every trained encoder uses the same compact output range, and all objectives share fixed metric arrays, matched initializations, and update counts. Lower geometric diagnostics are reported together with upper chord and derivative diagnostics, so improved resolution is separated cleanly from scale inflation. Multi-restart objective values, gradient diagnostics, and held-out geometric margins provide a common certificate-aligned view of optimization stability.

9.1. **Exact and trained folding obstructions.** For $S = [-1, 1]$, identity dynamics, and $\psi_C(s) = Cs^2$, the JEPA residual is zero and the latent variance is $4C^2/45$, yet $\psi_C(s) = \psi_C(-s)$. In two dimensions,

$$\psi_{C,D}(s_1, s_2) = (Cs_1^2, Ds_2)$$

has full-rank covariance $\mathrm{diag}(4C^2/45, D^2/3)$ while remaining non-injective. These exact examples establish the geometric obstruction that motivates the local–global penalty.

The bounded trained stress test compares pure prediction, covariance spread, local-only, global-only, the local–global hinge, the hinge plus an upper directional penalty, and a two-sided baseline. Under plain initialization the combined hinge is injective in all 12 runs. Under strongly symmetry-biased even pretraining, adding upper control reduces the number of non-injective outcomes from 9 to 3 out of 12, demonstrating how the lower and upper diagnostics can guide optimization and architectural refinement even from a deliberately challenging initialization.

9.2. **Finite-capacity threshold diagnostic.** Experiment B uses a norm-constrained coefficient-box spline proxy with fixed nested samples and repeated deterministic restarts, providing a controlled finite-dimensional laboratory for the threshold ingredients. The interpolation threshold is $P^* = 4.81\times10^{-6}$, while, for the medium-capacity family, the class-uniform statistical deviations range from 5.19 to 18.46; their separation from the sufficient sample-size margin quantifies the uniformity reserve carried by the class-wide guarantee. At the same time, the trained models exhibit a clear intermediate regime in which

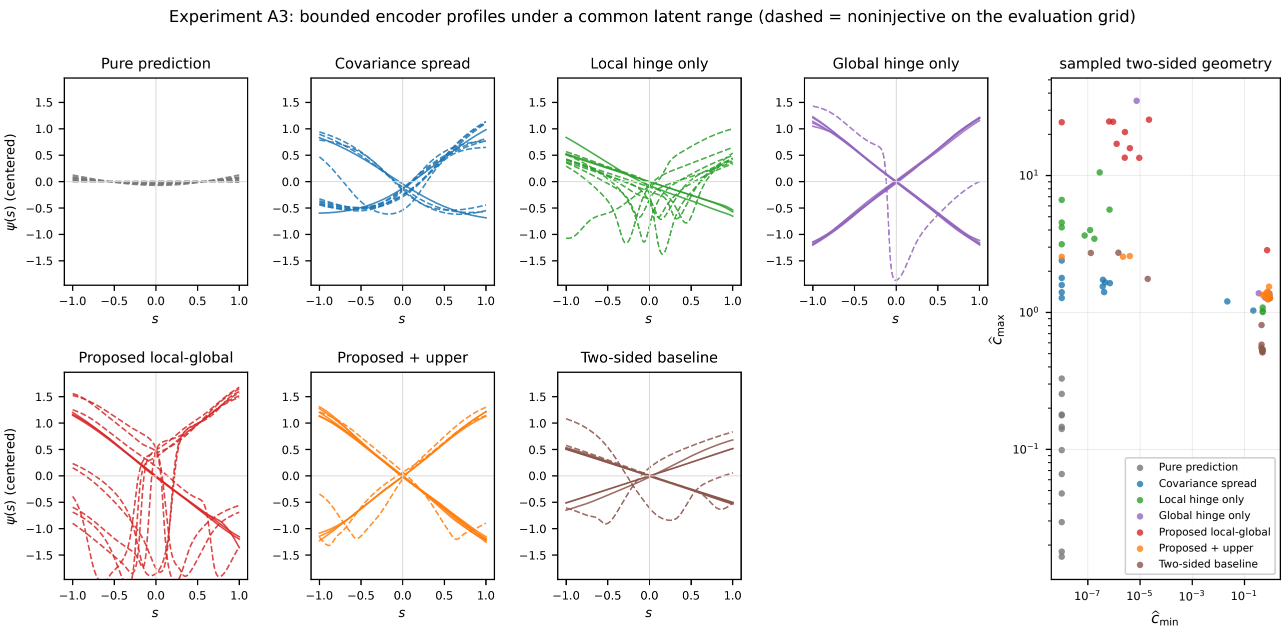

FIGURE 1. Bounded encoder profiles and sampled two-sided geometry in the folding stress test. Dashed profiles are non-injective on the evaluation grid. Reporting lower and upper geometry together enables direct comparison of resolution, folding, and scale across objectives.

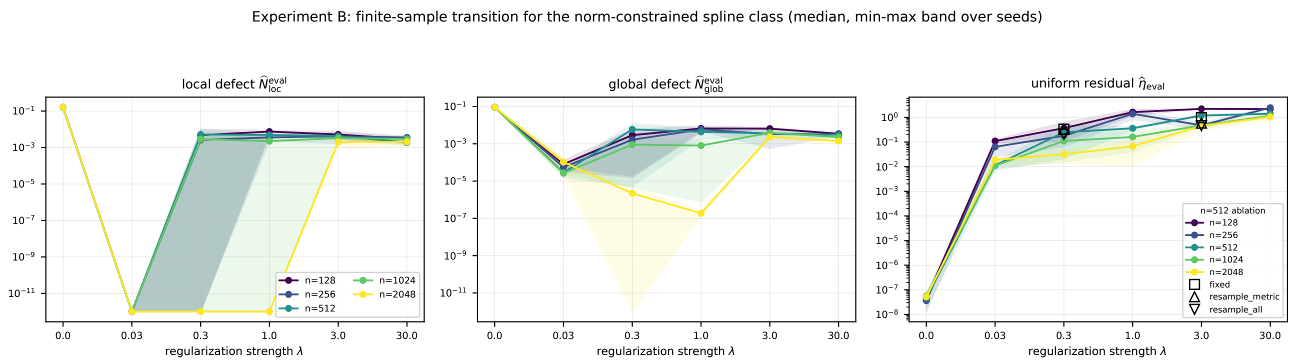

FIGURE 2. Finite-capacity spline proxy. Local and global empirical margins strengthen in an intermediate regularization regime, while the residual tradeoff identifies a certificate-aligned operating range. The analytic sufficient bound retains a substantial uniformity reserve across the full class at the tested sample sizes.

sampled local and global margins strengthen while the uniform evaluation residual remains controlled. This comparison turns the threshold formula into a diagnostic map and identifies the covering step as a concrete target for sharper data-dependent calibration.

9.3. **Nonlinear latent-MPC benchmark.** A smoothly saturated controlled pendulum with a nonlinear differentiable observation embedding is used to compare exact-model MPC and latent MPC. All learned objectives share architecture, bounded latent range, data, optimization budget, learned cost-head form, and planning budget. The exact-model reference is the best of 6 independent restarts and receives more total optimization effort than a learned arm, providing a stringent common baseline for the paired cost comparisons.

Pure prediction attains the smallest evaluated residual, $\widehat{\eta}_{\mathrm{eval}} = 0.01$, while the geometry-aware objectives raise the median chord constant from $\widehat{c}_{\min} = 1.75 \times 10^{-4}$ to a control-relevant range. The corresponding median paired closed-loop cost difference for pure prediction is 1.41 at $T = 5$ and 1.55

Objective	$\widehat{\eta}_{\rm eval}$	$\widehat{c}_{\rm min}$	median ΔJ, $T = 5$	median ΔJ, $T = 20$
Pure prediction	0.01	1.75×10^{-4}	1.41	1.55
Covariance spread	0.13	0.19	0.10	0.23
Local hinge only	0.08	0.12	0.08	0.38
Global hinge only	0.04	0.06	0.08	0.20
Local–global hinge	0.09	0.04	0.20	0.32
Two-sided baseline	0.06	0.14	0.09	0.18

Table 3. Selected Experiment C medians. Geometry-aware training raises the lower metric constant and substantially reduces paired cost differences relative to the residual-only reference. The regularized objectives achieve comparable control performance, and the local–global hinge pairs that competitiveness with the pointwise geometric quantities used by the theorem.

at $T = 20$, between 4.03 and 25.79 times the medians of the regularized arms. This is the central empirical finding: restoring the representation geometry singled out by the theorem converts accurate latent self-prediction into substantially more reliable control.

Pooled over methods, seeds, and horizons, the Spearman correlation between worst-case paired cost difference and $\widehat{\eta}_{\rm eval}/\widehat{c}_{\rm min}$ is 0.62 with cluster-bootstrap interval $[0.30, 0.80]$, whereas the correlation with $\widehat{\eta}_{\rm eval}$ alone is -0.34. Within the compact group of geometry-aware models, the corresponding correlation is 0.42 with interval $[0.04, 0.69]$. The ratio therefore provides a strong diagnostic gate between collapsed and non-collapsed models and a secondary, horizon-dependent diagnostic within the already reliable regime.

Covariance spread, local–global, and two-sided regularization have overlapping per-seed ranges and all achieve competitive performance on this controlled nonlinear benchmark. Their shared success demonstrates that geometry-aware training robustly restores control-relevant representations. The local–global hinge is mathematically distinguished because its directional and separated-pair terms are exactly the quantities for which the paper proves the complete finite-sample pointwise certificate and deterministic transfer theorem. The evaluation-set theorem proxy carries a substantial certification reserve—its median envelope factor is 81.95 and its maximum is 1.18×10^5—and provides a transparent decomposition of the regularity and horizon-amplification terms available for trajectory-local and model-specific calibration.

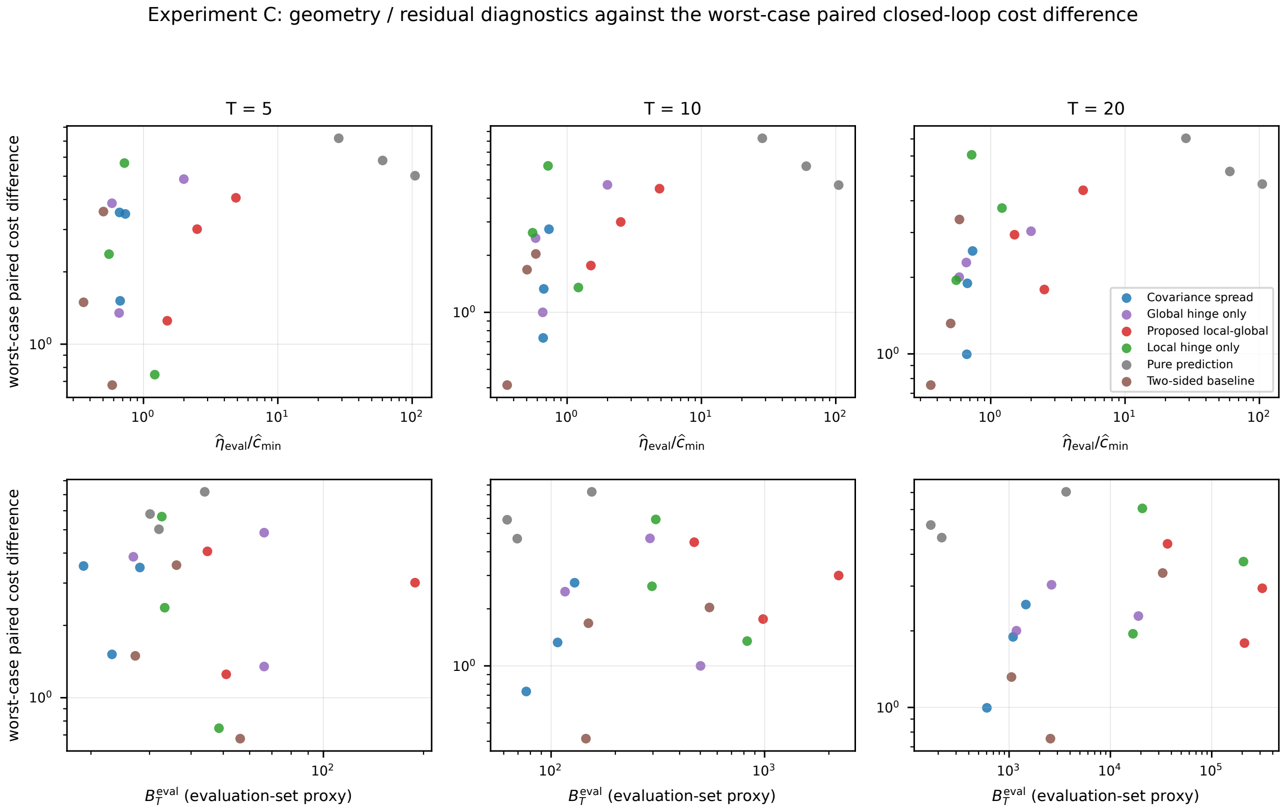

FIGURE 3. Worst-case paired closed-loop cost difference against the geometry/residual ratio (top) and the evaluation-set theorem proxy (bottom). The collapsed pure-prediction arm is clearly separated, while the regularized arms occupy a common control-relevant regime.

10. Discussion: scope, certification, and outlook

The scope is deliberately concrete: observable-state distances and tangent directions are available during training from simulator state, proprioception, or certified state-estimator outputs whose distance and tangent errors leave positive effective margins, while deployment remains observation- and action-conditioned. Independently validated pixel-based geometry estimators enter the same framework only after their errors have been absorbed into those margins. The spline construction shows that the regularity and approximation hypotheses are non-vacuous, and the coverage and optimization assumptions translate into actionable sampling and margin-monitoring requirements.

The analytic threshold is a class-uniform sufficient condition, not a claim about the smallest empirically useful regularization weight. Its reserve is visible in Experiment B and motivates sharper class-specific or finite-net certification. Experiment C supports the broader geometric mechanism: several regularized objectives restore non-collapse and achieve competitive control, while the proposed local–global hinge is distinguished by exact alignment with the pointwise theorem. Extending the same objective-to-certificate principle to partially observed stochastic systems and independently certified visual metrics is a natural next direction.

11. Conclusion

For geometrically supervised latent control, the local–global hinge provides the missing pointwise stability mechanism between empirical prediction and deterministic planning. The principal theorem proves that every approximate empirical minimizer in an explicit certified regime is co-Lipschitz and uniformly approximately semiconjugate to the controlled dynamics; the modular planning result transfers those certificates to trajectory, cost, learned-cost-head, and optimizer bounds.

The constructive approximation theory, sharp interpolation result, finite-net certificate, and controlled experiments complete the learning-to-control chain. Together they show how an empirically optimized objective can select metrically faithful dynamics models and how the resulting constants support reliable finite-horizon planning without weakening the distinction between analytic certification and finite-model evidence.

Declaration on the use of AI tools

AI tools were used for language editing, code review, and manuscript checking. The authors assume responsibility for all content.

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(Alain Bensoussan) Naveen Jindal School of Management, University of Texas at Dallas, Richardson, TX, 75080 USA

Email address: `alain.bensoussan@utdallas.edu`

(Minh-Nhat Phung) Department of Mathematics, Texas A&M University, College Station, TX, 77843 USA

Email address, Minh-Nhat Phung: `pmnt1114@tamu.edu`

(Minh-Binh Tran) Department of Mathematics, Texas A&M University, College Station, TX, 77843 USA

Email address, Minh-Binh Tran: `minhbinh@tamu.edu`

SUPPLEMENTARY MATERIAL FOR
FINITE-SAMPLE METRIC NON-COLLAPSE FOR GEOMETRICALLY SUPERVISED LATENT WORLD MODELS IN CONTROL

ALAIN BENSOUSSAN, MINH-NHAT PHUNG, AND MINH-BINH TRAN

Contents

S1. Exact realizers and detailed finite-capacity spline construction

This section contains the complete construction material used by the main theorem. It first constructs an exact non-collapsed latent realization, quantitative inverse charts, and global $C^{1,1}$ representatives. It then verifies Assumption A4 for norm-constrained tensor-product splines, including range enforcement, compactness, coefficient counts, margin clearing, and a nonempty geometric parameter regime. All arguments relocated from the main article are retained here in full.

S1.1. **Exact and smooth non-collapsed realizers.**

Lemma S1.1 (Exact non-collapsed Lipschitz realization). *Assume Assumption A1, $d_z \geq d_s$, and let $K_Z \subset \mathbb{R}^{d_z}$ be compact and convex with nonempty interior. Then there exist a smooth affine bi-Lipschitz embedding $\psi_0 : S \to \operatorname{int} K_Z$ with constants $0 < c_{\psi_0} \leq L_{\psi_0}$ and a $C^{1,1}$ map $\Pi_Z : Z \to K_Z$ that equals the identity on an open neighborhood of $\psi_0(S)$. Moreover, there exist Lipschitz maps*

$$\Phi_0 : X \to K_Z, \qquad F_0 : K_Z \times A \to K_Z,$$

such that

$$\Phi_0(H(s)) = \psi_0(s), \qquad F_0(\psi_0(s), a) = \psi_0(G(s,a)), \qquad R[\Phi_0, F_0] \equiv 0.$$

They may be chosen with bounds

$$\operatorname{Lip}(\Phi_0) \leq \sqrt{d_z}\,\frac{L_{\psi_0}}{c_H}, \qquad \operatorname{Lip}(F_0) \leq \sqrt{d_z}\, L_{\psi_0} L_G \max\{c_{\psi_0}^{-1}, 1\}.$$

Consequently, any admissible hypothesis classes that contain these maps admit an exact non-collapsed realization. In the finite-capacity theorem, richness is supplied by the explicit approximation property, while the Lipschitz budget provides the uniform regularity needed for empirical-to-continuum control.

Proof. Choose $z_c \in \operatorname{int} K_Z$ and $r > 0$ such that

$$\overline{B}_{2r}(z_c) \subset \operatorname{int} K_Z.$$

Let $\iota : \mathbb{R}^{d_s} \to \mathbb{R}^{d_z}$ be the coordinate inclusion, fix $s_c \in S$, and set $D_S := \max_{s \in S} |s - s_c| > 0$. With $\gamma := r/(2D_S)$, define

$$\psi_0(s) := z_c + \gamma\, \iota(s - s_c).$$

Then $\psi_0(S) \subset B_r(z_c)$ and

$$\|\psi_0(s) - \psi_0(s')\|_Z = \gamma |s - s'|,$$

so ψ_0 is a smooth affine bi-Lipschitz embedding with $c_{\psi_0} = L_{\psi_0} = \gamma$. Choose a smooth scalar saturation $\vartheta : [0,\infty) \to [0,\infty)$ such that $\vartheta(t) = 1$ for $t \leq r^2$ and $\vartheta(t)\sqrt{t} \leq 2r$ for all $t \geq 0$, and define

$$\Pi_Z(z) := z_c + \vartheta(|z - z_c|^2)(z - z_c).$$

This map is $C^{1,1}$ (indeed smooth), takes values in $\overline{B}_{2r}(z_c) \subset K_Z$, and is the identity on $B_r(z_c)$, an open neighborhood of $\psi_0(S)$.

Since H is bi-Lipschitz, H is injective. We define the map $\widehat{\Phi}_0 : H(S) \to Z$ by $\widehat{\Phi}_0(x) := \psi_0(H^{-1}(x))$. Then, we get that $\widehat{\Phi}_0$ is Lipschitz with $\operatorname{Lip}(\widehat{\Phi}_0) \leq L_{\psi_0} c_H^{-1}$, where c_H is the lower co-Lipschitz constant of H. We extend each scalar coordinate of $\widehat{\Phi}_0$ to X by the McShane extension theorem [1]. Combining these coordinatewise extensions, we obtain the vector-valued extension $\widetilde{\Phi}_0$ with

$$\operatorname{Lip}(\widetilde{\Phi}_0) \leq \sqrt{d_z}\,\frac{L_{\psi_0}}{c_H}.$$

Since K_Z is closed and convex, its metric projection Π_{K_Z} is 1-Lipschitz. As the values of $\widehat{\Phi}_0$ lie in K_Z on $H(S)$, by setting $\Phi_0 := \Pi_{K_Z} \circ \widetilde{\Phi}_0$, we obtain the encoder satisfying

$$\mathrm{Lip}(\Phi_0) \le \sqrt{d_z}\,\frac{L_{\psi_0}}{c_H} \qquad \text{and} \qquad \Phi_0(H(s)) = \psi_0(s).$$

Next, on $\psi_0(S) \times A$, we define $\widehat{F}(\zeta, a) := \psi_0\big(G(\psi_0^{-1}(\zeta), a)\big)$. With respect to the product metric

$$d_F((\zeta, a), (\zeta', a')) := \|\zeta - \zeta'\|_Z + |a - a'|,$$

the map $\widehat{F}$ is Lipschitz with constant at most $L_{\psi_0} L_G \max\{c_{\psi_0}^{-1}, 1\}$. As in the construction of Φ_0, applying coordinatewise McShane extension and then composing with Π_{K_Z}, we obtain F_0 such that

$$\mathrm{Lip}(F_0) \le \sqrt{d_z}\, L_{\psi_0} L_G \max\{c_{\psi_0}^{-1}, 1\} \qquad \text{and} \qquad F_0(\psi_0(s), a) = \psi_0(G(s, a)).$$

Therefore, we have the identity

$$F_0(\Phi_0(H(s)), a) = \psi_0(G(s, a)) = \Phi_0(H(G(s, a))).$$

Thus, the residual vanishes identically. The lower co-Lipschitz estimate is the assumed inequality $\|\psi_0(s) - \psi_0(s')\|_Z \ge c_{\psi_0} d_S(s, s')$. ■

Lemma S1.2 (Quantitative inverse regularity for Euclidean observable embeddings)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded convex open set, let $U \subset \mathbb{R}^d$ be an open neighborhood of $\overline{\Omega}$, and let*

$$f : U \to \mathbb{R}^m, \qquad m \ge d,$$

be of class $C^{1,1}$. Suppose that $f|_{\overline{\Omega}}$ is quantitatively co-Lipschitz: there exists $c_f > 0$ such that

$$\|f(s) - f(s')\|_{\mathbb{R}^m} \ge c_f |s - s'| \qquad \textit{for all } s, s' \in \overline{\Omega}.$$

Then $f|_{\overline{\Omega}}$ is injective and

$$\sigma_{\min}(Df(s)) := \inf_{|v|=1} \|Df(s)v\|_{\mathbb{R}^m} \ge c_f \qquad \textit{for every } s \in \overline{\Omega}.$$

Choose $\delta > 0$ such that the closed neighborhood

$$\overline{\Omega}_\delta := \{x \in \mathbb{R}^d : \mathrm{dist}(x, \overline{\Omega}) \le \delta\}$$

is contained in U, and set

$$L_f := \mathrm{Lip}(Df; \overline{\Omega}_\delta), \qquad r_* := \min\left\{\delta, \frac{c_f}{2L_f}\right\},$$

with the convention $c_f/(2L_f) = +\infty$ when $L_f = 0$. For every $s_0 \in \overline{\Omega}$, define

$$A_0 := Df(s_0), \qquad P_{s_0} := (A_0^\top A_0)^{-1} A_0^\top, \qquad \mathcal{F}_{s_0} := P_{s_0} \circ f, \qquad V_{s_0} := B_{r_*}(s_0).$$

Then

$$P_{s_0} A_0 = I_{\mathbb{R}^d}, \qquad \|P_{s_0}\|_{\mathrm{op}} \le c_f^{-1},$$

and, for all $s, s' \in V_{s_0}$,

$$\frac{1}{2}|s - s'| \le |\mathcal{F}_{s_0}(s) - \mathcal{F}_{s_0}(s')| \le \frac{3}{2}|s - s'|.$$

Consequently, $O_{s_0} := \mathcal{F}_{s_0}(V_{s_0})$ is open, $\mathcal{F}_{s_0} : V_{s_0} \to O_{s_0}$ is a $C^{1,1}$ diffeomorphism, and its inverse $\Gamma_{s_0} : O_{s_0} \to V_{s_0}$ satisfies the quantitative estimates

$$\mathrm{Lip}(\Gamma_{s_0}; O_{s_0}) \le 2, \qquad \sup_{y \in O_{s_0}} \|D\Gamma_{s_0}(y)\|_{\mathrm{op}} \le 2, \qquad \mathrm{Lip}(D\Gamma_{s_0}; O_{s_0}) \le \frac{8L_f}{c_f}.$$

Moreover,

$$(f|_{\overline{\Omega}})^{-1}(y) = \Gamma_{s_0}(P_{s_0} y) \qquad \textit{for every } y \in f(\overline{\Omega} \cap V_{s_0}).$$

Thus the local inverse charts may be chosen with a uniform radius and with constants depending only on c_f, the fixed-neighborhood $C^{1,1}$ budget of f, and the finite covering of $\overline{\Omega}$ by these charts.

Proof. The co-Lipschitz estimate immediately implies injectivity on $\overline{\Omega}$. Let $s \in \Omega$ and $v \in \mathbb{S}^{d-1}$. For all sufficiently small $t \neq 0$, convexity gives $s + tv \in \Omega$, and hence we have

$$\frac{\|f(s+tv) - f(s)\|_{\mathbb{R}^m}}{|t|} \ge c_f.$$

Letting $t \to 0$, we get $\|Df(s)v\| \ge c_f$. The same conclusion holds on $\partial\Omega$ by approximation from the interior and continuity of Df. Therefore, we arrive at $\sigma_{\min}(Df(s)) \ge c_f$ on $\overline{\Omega}$.

Fix $s_0 \in \overline{\Omega}$. Since $A_0 = Df(s_0)$ has full column rank, $A_0^\top A_0$ is invertible and P_{s_0} is the Moore–Penrose left inverse of A_0. Thus, we have

$$P_{s_0} A_0 = I, \qquad \|P_{s_0}\|_{\mathrm{op}} = \sigma_{\min}(A_0)^{-1} \le c_f^{-1}.$$

For $s \in V_{s_0}$, by the definitions of r_* and V_{s_0}, we estimate that

$$\|D\mathcal{F}_{s_0}(s) - I\|_{\mathrm{op}} = \|P_{s_0}(Df(s) - Df(s_0))\|_{\mathrm{op}} \le \frac{L_f}{c_f}|s - s_0| < \frac{1}{2}.$$

In particular, $D\mathcal{F}_{s_0}(s)$ is invertible and

$$\|D\mathcal{F}_{s_0}(s)^{-1}\|_{\mathrm{op}} \le 2.$$

Because V_{s_0} is convex, for $s, s' \in V_{s_0}$ we may integrate along the segment joining them:

$$\mathcal{F}_{s_0}(s) - \mathcal{F}_{s_0}(s') - (s - s') = \int_0^1 \big(D\mathcal{F}_{s_0}(s' + t(s - s')) - I\big)(s - s')\,\mathrm{d}t.$$

The preceding derivative estimate therefore gives

$$\big|\mathcal{F}_{s_0}(s) - \mathcal{F}_{s_0}(s') - (s - s')\big| \le \frac{1}{2}|s - s'|,$$

which proves the two-sided estimate

$$\frac{1}{2}|s - s'| \le |\mathcal{F}_{s_0}(s) - \mathcal{F}_{s_0}(s')| \le \frac{3}{2}|s - s'|.$$

Hence $\mathcal{F}_{s_0}$ is injective on V_{s_0}. Since its derivative is invertible everywhere, the inverse function theorem shows that it is a local diffeomorphism and an open map. It follows that $O_{s_0} = \mathcal{F}_{s_0}(V_{s_0})$ is open and that $\mathcal{F}_{s_0} : V_{s_0} \to O_{s_0}$ is a global C^1 diffeomorphism. The lower estimate above implies

$$\mathrm{Lip}(\Gamma_{s_0}; O_{s_0}) \le 2,$$

and the inverse derivative formula gives

$$D\Gamma_{s_0}(y) = D\mathcal{F}_{s_0}(\Gamma_{s_0}(y))^{-1}, \qquad \sup_{y \in O_{s_0}} \|D\Gamma_{s_0}(y)\|_{\mathrm{op}} \le 2.$$

For $y_i = \mathcal{F}_{s_0}(s_i) \in O_{s_0}$, the matrix identity $A^{-1} - B^{-1} = A^{-1}(B-A)B^{-1}$ yields

$$\begin{aligned}\|D\Gamma_{s_0}(y_1) - D\Gamma_{s_0}(y_2)\|_{\mathrm{op}} &\le 4\|D\mathcal{F}_{s_0}(s_1) - D\mathcal{F}_{s_0}(s_2)\|_{\mathrm{op}} \\ &\le 4\|P_{s_0}\|_{\mathrm{op}} L_f |s_1 - s_2| \\ &\le \frac{8L_f}{c_f}|y_1 - y_2|.\end{aligned}$$

Thus $\Gamma_{s_0} \in C^{1,1}(O_{s_0})$ with the claimed quantitative bounds. Finally, if $y = f(s)$ with $s \in \overline{\Omega} \cap V_{s_0}$, then $P_{s_0} y = \mathcal{F}_{s_0}(s)$, and therefore $\Gamma_{s_0}(P_{s_0} y) = s = (f|_{\overline{\Omega}})^{-1}(y)$. Compactness of $\overline{\Omega}$ supplies a finite subcover by the neighborhoods V_{s_0}, while the radius and all displayed constants are uniform. ■

Lemma S1.3 ($C^{1,1}$ exact representatives)**.** *Suppose that A1 holds and let $X = \mathbb{R}^{d_x}$. Let $\psi_0 : S \to \operatorname{int} K_Z$ and $\Pi_Z : Z \to K_Z$ be the affine bi-Lipschitz embedding and the smooth range-enforcing map constructed in Lemma S1.1; in particular, Π_Z is the identity on an open neighborhood of $\psi_0(S)$. Then the exact realization admits representatives*

$$\Phi_0 \in C^{1,1}(X; K_Z), \qquad F_0 \in C^{1,1}(K_Z \times A; K_Z),$$

with

$$\Phi_0(H(s)) = \psi_0(s), \qquad F_0(\psi_0(s), a) = \psi_0(G(s,a)), \qquad R[\Phi_0, F_0] \equiv 0.$$

Proof. We write

$$\psi_0 : S \to Z = \mathbb{R}^{d_z}$$

for the affine $C^{1,1}$ bi-Lipschitz embedding. By Assumption A1 and the construction in Lemma S1.1, H, G, and ψ_0 are restrictions of $C^{1,1}$ maps defined on open Euclidean neighborhoods of their domains. After possibly shrinking these neighborhoods, we use the same notation for the corresponding extensions.

Let

$$\Pi_Z : Z \to K_Z$$

be the constructed $C^{1,1}$ range-enforcing map, and let

$$\mathcal{O}_\psi \subset Z$$

be an open neighborhood of $\psi_0(S)$ on which

$$\Pi_Z(\zeta) = \zeta.$$

We first construct the encoder Φ_0. By Lemma $S1.2$, for each $s_i \in S$, there exist an open neighborhood $V_i^H \subset \mathbb{R}^{d_s}$ of s_i, a linear map

$$P_i^H : \mathbb{R}^{d_x} \to \mathbb{R}^{d_s},$$

an open set $O_i^H \subset \mathbb{R}^{d_s}$, and a $C^{1,1}$ map

$$\Gamma_i^H : O_i^H \to V_i^H$$

such that

$$P_i^H \circ H : V_i^H \to O_i^H$$

is a $C^{1,1}$ diffeomorphism with inverse Γ_i^H. In particular, for $s \in S \cap V_i^H$, we have

$$\Gamma_i^H(P_i^H H(s)) = s.$$

Since S is compact, we may pass to a finite subcover and assume that

$$S \subset \bigcup_{i=1}^{N_H} V_i^H.$$

For each i, we choose an open ambient neighborhood $\mathcal{N}_i^H \subset \mathbb{R}^{d_x}$ of $H(S \cap V_i^H)$ such that

$$P_i^H(\mathcal{N}_i^H) \subset O_i^H \quad \text{and} \quad \mathcal{N}_i^H \cap H(S) \subset H(S \cap V_i^H).$$

This is possible after shrinking $\mathcal{N}_i^H$, because $H(S \cap V_i^H)$ is relatively open in $H(S)$ and $H(S)$ is compact. On $\mathcal{N}_i^H$, we define

$$\Phi_i^H(x) := \psi_0\big(\Gamma_i^H(P_i^H x)\big).$$

Then $\Phi_i^H \in C^{1,1}(\mathcal{N}_i^H; Z)$, and whenever $x = H(s) \in \mathcal{N}_i^H \cap H(S)$, we have

$$\Phi_i^H(H(s)) = \psi_0\big(\Gamma_i^H(P_i^H H(s))\big) = \psi_0(s).$$

We now choose smooth functions

$$\chi_i^H \in C_c^\infty(\mathcal{N}_i^H), \qquad i = 1, \dots, N_H,$$

such that

$$\sum_{i=1}^{N_H} \chi_i^H(x) = 1 \qquad \text{for all } x \text{ in a neighborhood of } H(S).$$

Such a finite smooth partition of unity exists because the sets $\mathcal{N}_i^H$ cover the compact set $H(S)$.

Let $z_* \in Z$ be fixed. We define on all of $X = \mathbb{R}^{d_x}$

$$\widetilde{\Phi}_0(x) := \sum_{i=1}^{N_H} \chi_i^H(x)\Phi_i^H(x) + \left(1 - \sum_{i=1}^{N_H} \chi_i^H(x)\right) z_*,$$

where each product $\chi_i^H \Phi_i^H$ is extended by zero outside $\mathcal{N}_i^H$. Since this is a finite sum of products of smooth functions and $C^{1,1}$ maps,

$$\widetilde{\Phi}_0 \in C^{1,1}(X; Z).$$

Using the range-enforcing map, we set

$$\Phi_0 := \Pi_Z \circ \widetilde{\Phi}_0.$$

This encoder satisfies $\Phi_0(H(s)) = \psi_0(s)$ and $\Phi_0 \in C^{1,1}(X; K_Z)$.

We next construct F_0. Apply Lemma S1.2 to the embedding ψ_0. For finitely many points $s_j \in S$, there exist open neighborhoods $V_j^\psi \subset \mathbb{R}^{d_s}$, linear maps

$$P_j^\psi : Z \to \mathbb{R}^{d_s},$$

open sets $O_j^\psi \subset \mathbb{R}^{d_s}$, and local inverse maps

$$\Gamma_j^\psi : O_j^\psi \to V_j^\psi$$

such that $P_j^\psi \circ \psi_0 : V_j^\psi \to O_j^\psi$ is a $C^{1,1}$ diffeomorphism and the sets V_j^ψ cover S. There also exist ambient neighborhoods $\mathcal{N}_j^\psi \subset Z$ of $\psi_0(S \cap V_j^\psi)$ satisfying

$$P_j^\psi(\mathcal{N}_j^\psi) \subset O_j^\psi, \qquad \mathcal{N}_j^\psi \cap \psi_0(S) \subset \psi_0(S \cap V_j^\psi).$$

We choose $\chi_j^\psi \in C_c^\infty(\mathcal{N}_j^\psi)$ such that $\sum_j \chi_j^\psi = 1$ on a neighborhood of $\psi_0(S)$. Then, we define

$$\widetilde{F}_0(\zeta, a) := \sum_j \chi_j^\psi(\zeta)\psi_0\big(G(\Gamma_j^\psi(P_j^\psi\zeta), a)\big) + \left(1 - \sum_j \chi_j^\psi(\zeta)\right) z_*.$$

Applying the range-enforcing map to $\widetilde{F}_0$, that is $F_0 := \Pi_Z \circ \widetilde{F}_0$, we obtain $F_0 \in C^{1,1}(K_Z \times A; K_Z)$ with

$$F_0(\psi_0(s), a) = \psi_0(G(s, a)).$$

It immediately follows that Φ_0, F_0 are the desired representatives. ■

The next assumption makes explicit the finite-capacity and capacity-uniform regularity requirement. It provides a concrete architecture-level verification target, and the spline construction below demonstrates that this target is attained by nontrivial finite-capacity classes with quantitative approximation rates.

Assumption A4 (Finite-capacity C^1-approximation with uniform $C^{1,1}$ budgets). *Let (Φ_0, F_0) be the $C^{1,1}$ exact representatives from Lemma S1.3. There exist classes $\mathcal{A}_\Phi(W), \mathcal{A}_F(W)$, indexed by a capacity parameter W and closed in the C^1 topology, whose elements map into K_Z. Their Lipschitz and $C^{1,1}$ budgets depend only on S, A, Z, K_Z, Π_Z, and (Φ_0, F_0); in particular, they are independent of W. For each sufficiently large W, there exist $(\Phi_W, F_W) \in \mathcal{A}_\Phi(W) \times \mathcal{A}_F(W)$ with*

$$\|\Phi_W - \Phi_0\|_{C^1(H(S))} + \|F_W - F_0\|_{C^1(\psi_0(S)\times A)} \le \delta(W), \qquad \delta(W) \to 0 \quad \text{as } W \to \infty.$$

S1.2. Finite-capacity spline approximation.

Lemma S1.4 (Range-constrained spline approximation on a box)**.** *Let $\Omega \subset \mathbb{R}^d$ be a compact rectangular box, let $f_0 : \Omega \to \mathbb{R}^m$ be the restriction of a $C^{1,1}$ map defined on an open neighborhood of Ω, and let $K \subset \mathbb{R}^m$ be a compact convex set with $f_0(\Omega) \subset \operatorname{int} K$. Assume there is a $C^{1,1}$ map $\Pi_K : \mathbb{R}^m \to K$ that is the identity on a fixed open neighborhood of $f_0(\Omega)$.*

For a mesh width $h > 0$, let $\mathcal{S}_h^m(\Omega)$ be the vector-valued tensor-product cubic B-spline space on a uniform grid of mesh size h over Ω; its dimension is mN_h with $N_h \asymp h^{-d}$ as $h \downarrow 0$. Writing $W_h := mN_h \asymp h^{-d}$ for the number of scalar coefficients, we have $h \asymp W_h^{-1/d}$. For $B > 0$, we define the range-constrained, norm-constrained class

$$\mathcal{A}_h^K(B; \Omega, m) := \Big\{\Pi_K \circ u : \ u \in \mathcal{S}_h^m(\Omega), \ \|u\|_{C^0(\Omega)} + \|Du\|_{C^0(\Omega)} + \operatorname{Lip}(Du; \Omega) \le B\Big\}.$$

Every element of $\mathcal{A}_h^K(B; \Omega, m)$ is $C^{1,1}$, with a $C^{1,1}$ bound depending only on B and Π_K, uniformly in h.

Fix a rectangular neighborhood Ω^+ with $\Omega \Subset \Omega^+$ on which a $C^{1,1}$ extension of f_0 is defined. Then there exist constants $B, C > 0$ and $h_0 > 0$, depending only on Ω, Ω^+, m, K, Π_K, and the $C^{1,1}(\Omega^+)$-norm of this fixed extension, and functions $\widetilde{u}_h \in \mathcal{A}_h^K(B; \Omega, m)$ for $0 < h < h_0$, such that

$$\|\widetilde{u}_h - f_0\|_{C^1(\Omega)} \le C\, h \operatorname{Lip}(Df_0; \Omega^+), \qquad \sup_{0<h<h_0} \|\widetilde{u}_h\|_{C^{1,1}(\Omega)} \le C\|f_0\|_{C^{1,1}(\Omega^+)}.$$

Equivalently, in terms of the number W_h of scalar coefficients,

$$\|\widetilde{u}_h - f_0\|_{C^1(\Omega)} \le C\, W_h^{-1/d}.$$

Proof. We use a standard local tensor-product cubic B-spline quasi-interpolant Q_h on the fixed box Ω^+, restricted afterward to Ω. The uniform grid is extended from Ω to Ω^+, so this restriction lies in the same mesh- h spline space $\mathcal{S}_h^m(\Omega)$. We choose the usual local construction that reproduces affine polynomials and whose coefficient functionals are uniformly bounded under scaling; see [2, Chapter 12] for the construction of local spline quasi-interpolants. For completeness, we derive the derivative estimates required below directly from locality, scaling, affine reproduction, and the $C^{1,1}$ Taylor remainder.

Let K_h be a mesh cell meeting Ω, and let $\omega_{K_h} \subset \Omega^+$ be the union of the fixed finite number of neighboring cells on which the local coefficient functionals contributing to Q_h over K_h depend. Because only a uniformly bounded number of fixed-degree tensor-product B-splines are active on a cell, the local coefficient functionals are uniformly bounded, and a derivative of order j scales as h^{-j}, there is a constant independent of h such that, for $j = 0, 1, 2$,

$$\|D^j Q_h v\|_{L^\infty(K_h)} \leq C h^{-j} \|v\|_{L^\infty(\omega_{K_h})}.$$

We choose $x_{K_h} \in K_h$ and let $p_{K_h}(y) = f_0(x_{K_h}) + Df_0(x_{K_h})(y - x_{K_h})$. Since Df_0 is Lipschitz on Ω^+ and ω_{K_h} has diameter $O(h)$, Taylor's formula with Lipschitz derivative gives

$$\|f_0 - p_{K_h}\|_{L^\infty(\omega_{K_h})} \leq Ch^2 \operatorname{Lip}(Df_0; \Omega^+), \qquad \|D(f_0 - p_{K_h})\|_{L^\infty(\omega_{K_h})} \leq Ch \operatorname{Lip}(Df_0; \Omega^+).$$

Affine reproduction gives $Q_h p_{K_h} = p_{K_h}$. Applying the local stability estimate to $f_0 - p_{K_h}$, and combining it with the Taylor remainder, yields on every cell meeting Ω

$$\|Q_h f_0 - f_0\|_{L^\infty(K_h)} \leq Ch^2 \operatorname{Lip}(Df_0; \Omega^+), \qquad \|D(Q_h f_0 - f_0)\|_{L^\infty(K_h)} \leq Ch \operatorname{Lip}(Df_0; \Omega^+),$$

and, because $D^2 p_{K_h} = 0$, we find

$$\|D^2 Q_h f_0\|_{L^\infty(K_h)} = \|D^2 Q_h(f_0 - p_{K_h})\|_{L^\infty(K_h)} \leq C \operatorname{Lip}(Df_0; \Omega^+).$$

Taking the maximum over the finitely overlapping cells, we obtain the global estimates

$$\|Q_h f_0 - f_0\|_{C^0(\Omega)} \leq Ch^2 \operatorname{Lip}(Df_0; \Omega^+), \qquad \|D(Q_h f_0 - f_0)\|_{C^0(\Omega)} \leq Ch \operatorname{Lip}(Df_0; \Omega^+),$$

together with a mesh-independent bound on $\operatorname{Lip}(DQ_h f_0; \Omega)$. In particular, after increasing C and taking $h \leq 1$, we have

$$\|Q_h f_0\|_{C^0(\Omega)} + \|D(Q_h f_0)\|_{C^0(\Omega)} + \operatorname{Lip}(D(Q_h f_0); \Omega) \leq C \|f_0\|_{C^{1,1}(\Omega^+)}.$$

Setting $u_h := Q_h f_0$ and choosing $B \geq 2C\|f_0\|_{C^{1,1}(\Omega^+)}$, we obtain $\widetilde{u}_h := \Pi_K \circ u_h \in \mathcal{A}_h^K(B; \Omega, m)$.

By the hypothesis, Π_K coincides with the identity on an open neighborhood $\mathcal{O}_K$ of $f_0(\Omega)$. For sufficiently small $h_0 > 0$, we have $u_h(\Omega) \subset \mathcal{O}_K$ for every h with $0 < h < h_0$. Consequently, $\widetilde{u}_h$ agrees with u_h on Ω for $0 < h < h_0$. Therefore, we have

$$\|\widetilde{u}_h - f_0\|_{C^1(\Omega)} \leq Ch \operatorname{Lip}(Df_0; \Omega^+), \qquad \|\widetilde{u}_h\|_{C^{1,1}(\Omega)} \leq C\|f_0\|_{C^{1,1}(\Omega^+)}.$$

To complete the proof, we establish a $C^{1,1}$ budget for $\mathcal{A}_h^K(B; \Omega, m)$.

Let $w = \Pi_K \circ u \in \mathcal{A}_h^K(B; \Omega, m)$, where

$$u \in \mathcal{S}_h^m(\Omega), \quad \|u\|_{C^0(\Omega)} + \|Du\|_{C^0(\Omega)} + \operatorname{Lip}(Du; \Omega) \leq B.$$

Since $\|u\|_{C^0(\Omega)} \leq B$, we have $u(\Omega) \subset \overline{B}_B(0)$. We define the constants

$$M_0 := \|\Pi_K\|_{C^0(\overline{B}_B(0))}, \quad M_1 := \|D\Pi_K\|_{C^0(\overline{B}_B(0))}, \quad M_2 := \operatorname{Lip}(D\Pi_K; \overline{B}_B(0)).$$

Then, it follows that

$$\|w\|_{C^0(\Omega)} \leq M_0.$$

By the chain rule, we get

$$Dw(x) = D\Pi_K(u(x))Du(x).$$

Hence, we further get

$$\|Dw\|_{C^0(\Omega)} \leq M_1\|Du\|_{C^0(\Omega)} \leq M_1 B.$$

For $x, y \in \Omega$, we estimate

$$\begin{aligned}
\|Dw(x) - Dw(y)\| &\leq \|D\Pi_K(u(x))\|\|Du(x) - Du(y)\| \\
&\quad + \|D\Pi_K(u(x)) - D\Pi_K(u(y))\|\|Du(y)\| \\
&\leq M_1 \operatorname{Lip}(Du;\Omega)|x-y| + M_2\|u(x) - u(y)\|\|Du\|_{C^0(\Omega)} \\
&\leq M_1 B|x-y| + M_2 B^2|x-y|.
\end{aligned}$$

Therefore, we arrive at

$$\operatorname{Lip}(Dw;\Omega) \leq M_1 B + M_2 B^2.$$

Thus every element of $\mathcal{A}_h^K(B;\Omega,m)$ has a $C^{1,1}$ norm bounded by a constant depending only on B and Π_K. ■

Lemma S1.5 (Localized range-constrained spline approximation). *Let $\Omega \subset \mathbb{R}^d$ be a compact rectangular box, let $E \subset \Omega$ be compact, and let $f_0 : \Omega \to \mathbb{R}^m$ be the restriction of a $C^{1,1}$ map defined on an open neighborhood of Ω. Let $K \subset \mathbb{R}^m$ be compact and convex, and suppose that*

$$\Pi_K : \mathbb{R}^m \to K$$

is of class $C^{1,1}$. Assume that there exists an open set $\mathcal{O} \subset \mathbb{R}^m$ such that

$$f_0(E) \Subset \mathcal{O}, \qquad \Pi_K(y) = y \quad \text{for every } y \in \mathcal{O}.$$

Fix a rectangular neighborhood Ω^+ with $\Omega \Subset \Omega^+$ and a fixed $C^{1,1}$ extension of f_0 to Ω^+. Let $Q_h f_0$ denote the restriction to Ω of the same local tensor-product cubic B-spline quasi-interpolant construction used in Lemma S1.4, on the extended uniform mesh, and set

$$u_h := Q_h f_0, \qquad \widetilde{u}_h := \Pi_K \circ u_h.$$

Then there exist constants $C > 0$ and $h_0 > 0$, depending only on Ω, Ω^+, m, Π_K, $\mathcal{O}$, and the $C^{1,1}(\Omega^+)$-norm of the fixed extension, such that, for $0 < h < h_0$,

$$\|\widetilde{u}_h - f_0\|_{C^1(E)} \leq Ch \operatorname{Lip}(Df_0;\Omega^+).$$

Here

$$\|w\|_{C^1(E)} := \sup_{x\in E}\|w(x)\| + \sup_{x\in E}\|Dw(x)\|.$$

Moreover,

$$\|u_h\|_{C^0(\Omega)} + \|Du_h\|_{C^0(\Omega)} + \operatorname{Lip}(Du_h;\Omega) \leq C\|f_0\|_{C^{1,1}(\Omega^+)},$$

uniformly in h, and

$$\sup_{0<h<h_0}\|\widetilde{u}_h\|_{C^{1,1}(\Omega)} \leq C.$$

Proof. The cellwise affine-reproduction argument in Lemma S1.4, applied on the fixed outer box, gives constants independent of h such that

$$\|Q_h f_0 - f_0\|_{C^0(\Omega)} \le C_0 h^2 \operatorname{Lip}(Df_0;\Omega^+), \qquad \|D(Q_h f_0 - f_0)\|_{C^0(\Omega)} \le C_0 h \operatorname{Lip}(Df_0;\Omega^+),$$

and

$$\|Q_h f_0\|_{C^0(\Omega)} + \|DQ_h f_0\|_{C^0(\Omega)} + \operatorname{Lip}(DQ_h f_0;\Omega) \le C_1 \|f_0\|_{C^{1,1}(\Omega^+)}.$$

These are precisely the mesh-independent $C^{1,1}$ stability and C^1 approximation estimates needed in the localized argument.

Because $f_0(E) \Subset \mathcal{O}$, we observe that

$$d_{\mathcal{O}} := \operatorname{dist}\big(f_0(E), \mathbb{R}^m \setminus \mathcal{O}\big) > 0.$$

Thus, we can choose $h_0 > 0$ such that

$$C_0 h^2 \operatorname{Lip}(Df_0;\Omega^+) < \frac{d_{\mathcal{O}}}{2} \qquad \text{for all } 0 < h < h_0.$$

Then $u_h(E) \subset \mathcal{O}$. Since Π_K equals the identity on the open set $\mathcal{O}$, we get

$$\Pi_K(u_h(x)) = u_h(x), \qquad D\Pi_K(u_h(x)) = I_{\mathbb{R}^m} \qquad \text{for every } x \in E.$$

Thus, we further get

$$\widetilde{u}_h(x) = u_h(x), \qquad D\widetilde{u}_h(x) = Du_h(x) \qquad \text{for every } x \in E.$$

Hence, after increasing the constant and taking $h \le 1$, we obtain

$$\begin{aligned}
\|\widetilde{u}_h - f_0\|_{C^1(E)} &= \|u_h - f_0\|_{C^1(E)} \\
&\le \|u_h - f_0\|_{C^0(\Omega)} + \|D(u_h - f_0)\|_{C^0(\Omega)} \\
&\le Ch \operatorname{Lip}(Df_0;\Omega^+).
\end{aligned}$$

It remains to establish the uniform $C^{1,1}$ bound after composition with Π_K. The stability estimate implies that all sets $u_h(\Omega)$ are contained in a fixed compact ball. On this ball, let

$$M_1 := \|D\Pi_K\|_{C^0}, \qquad M_2 := \operatorname{Lip}(D\Pi_K).$$

The chain rule gives

$$D\widetilde{u}_h(x) = D\Pi_K(u_h(x)) Du_h(x),$$

so we have

$$\|D\widetilde{u}_h\|_{C^0(\Omega)} \le M_1 \|Du_h\|_{C^0(\Omega)}.$$

For $x, y \in \Omega$, we estimate

$$\begin{aligned}
\|D\widetilde{u}_h(x) - D\widetilde{u}_h(y)\| \le M_1 &\|Du_h(x) - Du_h(y)\| \\
&+ M_2 \|u_h(x) - u_h(y)\| \, \|Du_h(y)\|.
\end{aligned}$$

Therefore, we obtain the bound

$$\operatorname{Lip}(D\widetilde{u}_h;\Omega) \le M_1 \operatorname{Lip}(Du_h;\Omega) + M_2 \|Du_h\|^2_{C^0(\Omega)}.$$

Together with the compactness of K, which controls the C^0-norm of $\widetilde{u}_h$, the uniform spline stability estimate proves the claimed $C^{1,1}$ bound. ∎

The next proposition records one concrete way in which Assumption A4 of the main article can be satisfied. Thus the assumption is not merely conditional on an abstract approximation axiom: there are finite-dimensional, norm-constrained architectures for which the required uniform $C^{1,1}$ budget, the output constraint into K_Z, and the C^1-approximation property hold simultaneously.

Proposition S1.6 (A norm-constrained spline class satisfying Assumption A4)**.** *Let (Φ_0, F_0) be the $C^{1,1}$ exact representatives from Lemma S1.3, and let*

$$\Pi_Z : Z \to K_Z, \qquad \mathcal{O}_Z \subset Z$$

be the range-enforcing map and identity neighborhood from that lemma:

$$\psi_0(S) \Subset \mathcal{O}_Z, \qquad \Pi_Z(z) = z \quad \text{for every } z \in \mathcal{O}_Z.$$

Choose compact rectangular boxes

$$\Omega_\Phi \subset \mathbb{R}^{d_x}, \qquad \Omega_F \subset \mathbb{R}^{d_z+d_a},$$

such that

$$H(S) \Subset \operatorname{int} \Omega_\Phi, \qquad K_Z \times A \Subset \operatorname{int} \Omega_F.$$

Choose a fixed $C^{1,1}$ map

$$\rho_\Phi : \mathbb{R}^{d_x} \to \Omega_\Phi$$

that equals the identity on an open neighborhood of $H(S)$. Such a map may be obtained by applying a fixed smooth coordinatewise saturation to the rectangular box.

For mesh widths $h_\Phi, h_F > 0$, define

$$\mathcal{A}_\Phi(h_\Phi) := \Big\{ \Phi(x) = \Pi_Z(u(\rho_\Phi(x))) : u \in \mathcal{S}^{d_z}_{h_\Phi}(\Omega_\Phi), \\ \|u\|_{C^0(\Omega_\Phi)} + \|Du\|_{C^0(\Omega_\Phi)} + \mathrm{Lip}(Du; \Omega_\Phi) \le B_\Phi \Big\},$$

and

$$\mathcal{A}_F(h_F) := \Big\{ F(z,a) = \Pi_Z(v(z,a)) : v \in \mathcal{S}^{d_z}_{h_F}(\Omega_F), \\ \|v\|_{C^0(\Omega_F)} + \|Dv\|_{C^0(\Omega_F)} + \mathrm{Lip}(Dv; \Omega_F) \le B_F \Big\},$$

where F is restricted to $K_Z \times A$.

For each sufficiently large integer capacity level W, choose admissible uniform mesh widths $h_\Phi(W)$ and $h_F(W)$ such that

$$h_\Phi(W) \asymp W^{-1/d_x}, \qquad h_F(W) \asymp W^{-1/(d_z+d_a)},$$

with comparison constants independent of W. For example, one may choose the nearest admissible tensor-grid resolutions to the displayed powers. Then, for sufficiently large fixed constants B_Φ, B_F, the classes

$$\mathcal{A}_\Phi(W) := \mathcal{A}_\Phi(h_\Phi(W)), \qquad \mathcal{A}_F(W) := \mathcal{A}_F(h_F(W))$$

satisfy Assumption A4 of the main article. More precisely, there exist

$$(\Phi_W, F_W) \in \mathcal{A}_\Phi(W) \times \mathcal{A}_F(W)$$

such that

$$\|\Phi_W - \Phi_0\|_{C^1(H(S))} + \|F_W - F_0\|_{C^1(\psi_0(S)\times A)} \le \delta(W),$$

where

$$\delta(W) \le C_\Phi W^{-1/d_x} + C_F W^{-1/(d_z+d_a)} \longrightarrow 0.$$

The Lipschitz and $C^{1,1}$ budgets of the two classes are independent of W, and each class is compact, hence closed, in the C^1 topology. Each vector-valued class has $O(d_z W)$ scalar coefficients.

Proof. We divide the argument into four steps.

Step 1: $C^{1,1}$ extensions on the coordinate boxes. The encoder Φ_0 is defined on all of $X = \mathbb{R}^{d_x}$, so we set

$$\overline{\Phi}_0 := \Phi_0|_{\Omega_\Phi}.$$

By the definition of $C^{1,1}(K_Z \times A; K_Z)$, the transition map F_0 admits a $C^{1,1}$ extension F_0^{ext} to an open neighborhood U_F of $K_Z \times A$. We choose a smooth cut-off

$$\chi_F \in C_c^\infty(U_F), \qquad 0 \le \chi_F \le 1,$$

satisfying $\chi_F = 1$ on an open neighborhood of $K_Z \times A$. Fix $z_* \in Z$, we define

$$\overline{F}_0 := z_* + \chi_F(F_0^{\text{ext}} - z_*)$$

on U_F, extending the compactly supported second term by zero to $\mathbb{R}^{d_z+d_a}$. Then

$$\overline{F}_0 \in C^{1,1}(\mathbb{R}^{d_z+d_a}; Z)$$

and $\overline{F}_0 = F_0^{\text{ext}}$ on an open neighborhood of $K_Z \times A$; in particular, $\overline{F}_0 = F_0$ on $K_Z \times A$.

We now define

$$E_\Phi := H(S), \qquad E_F := \psi_0(S) \times A.$$

The exact-realization identities give

$$\overline{\Phi}_0(E_\Phi) = \psi_0(S)$$

and

$$\overline{F}_0(E_F) = \{\psi_0(G(s,a)) : (s,a) \in S \times A\} \subset \psi_0(S).$$

Thus both target images are compactly contained in $\mathcal{O}_Z$.

Step 2: localized spline approximation. We apply Lemma S1.5 to

$$(\Omega, E, f_0) = (\Omega_\Phi, E_\Phi, \overline{\Phi}_0)$$

and to

$$(\Omega, E, f_0) = (\Omega_F, E_F, \overline{F}_0),$$

with $K = K_Z$, $\Pi_K = \Pi_Z$, and $\mathcal{O} = \mathcal{O}_Z$. We obtain raw spline quasi-interpolants

$$u_{h_\Phi} \in \mathcal{S}_{h_\Phi}^{d_z}(\Omega_\Phi), \qquad v_{h_F} \in \mathcal{S}_{h_F}^{d_z}(\Omega_F),$$

whose $C^{1,1}$ norms are bounded independently of the mesh widths and such that

$$\|\Pi_Z \circ u_{h_\Phi} - \Phi_0\|_{C^1(H(S))} \le C_\Phi h_\Phi,$$

$$\|\Pi_Z \circ v_{h_F} - F_0\|_{C^1(\psi_0(S)\times A)} \le C_F h_F.$$

By Lemma S1.4, we can choose B_Φ, B_F such that these constants are larger than the corresponding uniform raw-spline bounds and are independent of h_Φ, h_F, and hence of W.

We define

$$\Phi_W(x) := \Pi_Z\big(u_{h_\Phi(W)}(\rho_\Phi(x))\big),$$

$$F_W(z,a) := \Pi_Z\big(v_{h_F(W)}(z,a)\big).$$

Then, we get

$$\Phi_W \in \mathcal{A}_\Phi(W), \qquad F_W \in \mathcal{A}_F(W).$$

Because ρ_Φ equals the identity on a neighborhood of $H(S)$, we have

$$\rho_\Phi(x) = x, \qquad D\rho_\Phi(x) = I \quad \text{for } x \in H(S).$$

Therefore, we arrive at

$$\|\Phi_W - \Phi_0\|_{C^1(H(S))} \le C_\Phi h_\Phi(W),$$

$$\|F_W - F_0\|_{C^1(\psi_0(S)\times A)} \le C_F h_F(W).$$

The stated estimate for $\delta(W)$ follows.

Step 3: uniform $C^{1,1}$ budgets. We consider the encoder

$$\Phi = \Pi_Z \circ u \circ \rho_\Phi \in \mathcal{A}_\Phi(W),$$

and set

$$R_1 := \|D\rho_\Phi\|_{C^0}, \qquad R_2 := \mathrm{Lip}(D\rho_\Phi).$$

The raw-spline constraint gives

$$\|D(u \circ \rho_\Phi)\|_{C^0} \le B_\Phi R_1,$$

$$\mathrm{Lip}(D(u \circ \rho_\Phi)) \le B_\Phi(R_1^2 + R_2).$$

All values of $u \circ \rho_\Phi$ lie in the fixed ball $\overline{B}_{B_\Phi}(0)$. On this ball, we define

$$P_1 := \|D\Pi_Z\|_{C^0}, \qquad P_2 := \mathrm{Lip}(D\Pi_Z).$$

The chain rule yields

$$\|D\Phi\|_{C^0} \le P_1 B_\Phi R_1$$

and

$$\mathrm{Lip}(D\Phi) \le P_1 B_\Phi(R_1^2 + R_2) + P_2 B_\Phi^2 R_1^2.$$

Since Φ takes values in the compact set K_Z, this gives a $C^{1,1}$ budget independent of W.

For $F = \Pi_Z \circ v \in \mathcal{A}_F(W)$, the same argument gives

$$\|DF\|_{C^0} \le \widetilde{P}_1 B_F, \qquad \mathrm{Lip}(DF) \le \widetilde{P}_1 B_F + \widetilde{P}_2 B_F^2,$$

where $\widetilde{P}_1, \widetilde{P}_2$ are bounds for $D\Pi_Z$ and $\mathrm{Lip}(D\Pi_Z)$ on the fixed ball containing the raw spline values. Thus the transition classes also have uniform $C^{1,1}$ budgets, and both classes map into K_Z by construction.

Step 4: finite capacity and C^1-closedness. At fixed mesh width, each spline space is finite-dimensional. The raw coefficient sets determined by

$$\|u\|_{C^0} + \|Du\|_{C^0} + \mathrm{Lip}(Du) \le B_\Phi$$

and

$$\|v\|_{C^0} + \|Dv\|_{C^0} + \mathrm{Lip}(Dv) \le B_F$$

are closed and bounded in finite-dimensional spaces. By equivalence of norms, they are compact.

The maps

$$u \longmapsto \Pi_Z \circ u \circ \rho_\Phi$$

and

$$v \longmapsto (\Pi_Z \circ v)|_{K_Z \times A}$$

are continuous into the C^1 topology. Hence $\mathcal{A}_\Phi(W)$ and $\mathcal{A}_F(W)$ are continuous images of compact sets and are therefore compact, in particular closed, in C^1.

Finally, a tensor-product spline space in input dimension d has $O(h^{-d})$ scalar basis functions per output coordinate. The chosen mesh widths satisfy

$$h_\Phi(W)^{-d_x} \asymp W, \qquad h_F(W)^{-(d_z+d_a)} \asymp W.$$

Thus each vector-valued class has $O(d_z W)$ scalar coefficients. All requirements of Assumption A4 of the main article are satisfied. ■

S1.3. **Margin-clearing competitors and a nonempty parameter regime.** The margin-clearing comparator is the bridge between realizability and the empirical objective.

Proposition S1.7 (Margin-clearing competitors)**.** *Adopt the hypotheses of Lemma S1.3 and Assumption A4. We define*

$$\kappa_0 := \inf_{(s,v)\in\mathcal{U}} \|D\psi_0(s)[v]\|_Z, \qquad \alpha_0(\rho) := \inf_{(s,s')\in P_\rho} \|\psi_0(s) - \psi_0(s')\|_Z.$$

By the bi-Lipschitz property of ψ_0, $\kappa_0 > 0$ and $\alpha_0(\rho) > 0$, we choose margins

$$0 < \kappa < \kappa_0, \qquad 0 < \alpha < \alpha_0(\rho).$$

Then, for every W sufficiently large, the pair $(\Phi_W, F_W) \in \mathcal{A}_\Phi(W) \times \mathcal{A}_F(W)$ from Assumption A4 is a margin-clearing competitor. More precisely, writing $\psi_W := \Phi_W \circ H$, we get

$$\|D\psi_W(s)[v]\|_Z \geq \kappa \quad \text{for all } (s,v) \in \mathcal{U}, \qquad \|\psi_W(s) - \psi_W(s')\|_Z \geq \alpha \quad \text{for all } (s,s') \in P_\rho.$$

Consequently,

$$\mathcal{N}_{\text{loc}}(\Phi_W) = \mathcal{N}_{\text{glob}}(\Phi_W) = \mathcal{N}_{\text{met}}(\Phi_W) = 0,$$

and the same identities hold for the empirical penalties, for every sample:

$$\widehat{\mathcal{N}_{\text{loc}n}}(\Phi_W) = \widehat{\mathcal{N}_{\text{glob}_n}}(\Phi_W) = \widehat{\mathcal{N}_{\text{met}n}}(\Phi_W) = 0.$$

Moreover,

$$\mathcal{E}_{\text{pred}}(\Phi_W, F_W) \leq \beta(W), \qquad \beta(W) := C_\sharp\, \delta(W)^2 \to 0,$$

where $C_\sharp$ depends only on H, G, and the uniform Lipschitz and $C^{1,1}$ budgets.

Proof. We have the identities

$$\psi_W := \Phi_W \circ H, \quad \psi_0 := \Phi_0 \circ H,$$

and by the chain rule,

$$D\psi_W(s) - D\psi_0(s) = \big(D\Phi_W(H(s)) - D\Phi_0(H(s))\big) DH(s).$$

Consequently, we estimate

$$\|\psi_W - \psi_0\|_{C^1(S)} \leq \|\Phi_W - \Phi_0\|_{C^0(H(S))} + \|D\Phi_W - D\Phi_0\|_{C^0(H(S))} \|DH\|_{C^0(S)} \leq C_H^\sharp \delta(W),$$

where $C_H^\sharp := 1 + \|DH\|_{C^0(S)}$.

For every $(s,v) \in \mathcal{U} = S \times \mathbb{S}^{d_s-1}$, we have

$$\begin{aligned} \|D\psi_W(s)[v]\|_Z &\geq \|D\psi_0(s)[v]\|_Z - \|(D\psi_W(s) - D\psi_0(s))[v]\|_Z \\ &\geq \kappa_0 - \|D\psi_W - D\psi_0\|_{C^0(S)} \\ &\geq \kappa_0 - C_H^\sharp \delta(W). \end{aligned}$$

Since $0 < \kappa < \kappa_0$ and $\delta(W) \to 0$, we may choose W sufficiently large so that

$$C_H^\sharp \delta(W) \le \kappa_0 - \kappa.$$

Then, it follows that

$$\|D\psi_W(s)[v]\|_Z \ge \kappa \qquad \text{for all } (s, v) \in \mathcal{U}.$$

Therefore, we have proved

$$\mathcal{N}_{\mathrm{loc}}(\Phi_W) = 0 \quad \text{and} \quad \widehat{\mathcal{N}_{\mathrm{loc}n}}(\Phi_W) = 0.$$

We next show the separated-point margin. For every $(s, s') \in P_\rho$, we estimate

$$\begin{aligned} \|\psi_W(s) - \psi_W(s')\|_Z &\ge \|\psi_0(s) - \psi_0(s')\|_Z \\ &\qquad - \|\psi_W(s) - \psi_0(s)\|_Z - \|\psi_W(s') - \psi_0(s')\|_Z \\ &\ge \alpha_0(\rho) - 2\|\psi_W - \psi_0\|_{C^0(S)} \\ &\ge \alpha_0(\rho) - 2C_H^\sharp \delta(W). \end{aligned}$$

Since $0 < \alpha < \alpha_0(\rho)$, we may choose W sufficiently large so that

$$2C_H^\sharp \delta(W) \le \alpha_0(\rho) - \alpha.$$

Then, it follows that

$$\|\psi_W(s) - \psi_W(s')\|_Z \ge \alpha \qquad \text{for all } (s, s') \in P_\rho.$$

Therefore, we get

$$\mathcal{N}_{\mathrm{glob}}(\Phi_W) = 0 \quad \text{and} \quad \widehat{\mathcal{N}_{\mathrm{glob}}}_n(\Phi_W) = 0.$$

Consequently, we obtain

$$\mathcal{N}_{\mathrm{met}}(\Phi_W) = 0 \quad \text{and} \quad \widehat{\mathcal{N}_{\mathrm{met}n}}(\Phi_W) = 0.$$

It remains to estimate the prediction loss. Since (Φ_0, F_0) is the smooth exact realizer given by Lemma S1.3, we have

$$\psi_0(G(s, a)) = F_0(\psi_0(s), a) \qquad \text{for all } (s, a) \in S \times A.$$

Using the exact identity for (ψ_0, F_0), we obtain, for $(s, a) \in M$,

$$\begin{aligned} \|R[\Phi_W, F_W](s, a)\|_Z &= \|\psi_W(G(s, a)) - F_W(\psi_W(s), a)\|_Z \\ &\le \|\psi_W(G(s, a)) - \psi_0(G(s, a))\|_Z \\ &\quad + \|F_0(\psi_0(s), a) - F_W(\psi_W(s), a)\|_Z. \end{aligned}$$

Since $G(s, a) \in S$, the first term is bounded by

$$\|\psi_W(G(s, a)) - \psi_0(G(s, a))\|_Z \le \|\psi_W - \psi_0\|_{C^0(S)} \le C_H^\sharp \delta(W).$$

For the second term, we add and subtract $F_W(\psi_0(s), a)$, which gives

$$\begin{aligned} \|F_0(\psi_0(s), a) - F_W(\psi_W(s), a)\|_Z &\le \|F_0(\psi_0(s), a) - F_W(\psi_0(s), a)\|_Z \\ &\quad + \|F_W(\psi_0(s), a) - F_W(\psi_W(s), a)\|_Z. \end{aligned}$$

By Assumption A4, we estimate

$$\|F_0(\psi_0(s), a) - F_W(\psi_0(s), a)\|_Z \le \|F_W - F_0\|_{C^0(\psi_0(S) \times A)} \le \delta(W).$$

For the remaining term, we use the uniform Lipschitz budget of the admissible transition class. Namely, there exists a constant L_F^{bud}, independent of W, such that

$$\|F_W(z,a) - F_W(z',a)\|_Z \le L_F^{\mathrm{bud}} \|z - z'\|_Z \qquad \text{for all } z, z' \in K_Z, \text{ and } a \in A.$$

Therefore, we estimate

$$\begin{aligned}\|F_W(\psi_0(s),a) - F_W(\psi_W(s),a)\|_Z &\le L_F^{\mathrm{bud}} \|\psi_0(s) - \psi_W(s)\|_Z \\ &\le L_F^{\mathrm{bud}} C_H^{\sharp} \delta(W).\end{aligned}$$

Combining the preceding estimates yields

$$\|R[\Phi_W, F_W](s,a)\|_Z \le C_R \delta(W) \qquad \text{for all } (s,a) \in M,$$

where

$$C_R := 1 + (1 + L_F^{\mathrm{bud}}) C_H^{\sharp}.$$

Since μ is a probability measure on M, we obtain

$$\begin{aligned}\mathcal{E}_{\mathrm{pred}}(\Phi_W, F_W) &= \int_M \|R[\Phi_W, F_W](s,a)\|_Z^2 \mathrm{d}\mu(s,a) \\ &\le \left(\sup_{(s,a)\in M} \|R[\Phi_W, F_W](s,a)\|_Z \right)^2 \\ &\le C_R^2 \delta(W)^2.\end{aligned}$$

Thus the desired estimate holds with

$$C_\sharp := C_R^2, \qquad \beta(W) := C_\sharp \delta(W)^2.$$

Since $\delta(W) \to 0$, we have $\beta(W) \to 0$. ■

Lemma S1.8 (Nonempty parameter regime)**.** *Let $S \subset \mathbb{R}^{d_s}$ be compact with $\operatorname{diam}(S) > 0$, and let $\psi_0 : S \to Z$ be the $C^{1,1}$ bi-Lipschitz embedding from Lemma S1.3, with co-Lipschitz constant $c_{\psi_0} > 0$. Assume*

$$\kappa_0 := \inf_{(s,v)\in\mathcal{U}} \|D\psi_0(s)[v]\|_Z > 0,$$

and fix κ with $0 < \kappa < \kappa_0$. Let

$$B_{\mathrm{loc}} := \sup_W \sup_{\Phi \in \mathcal{A}_\Phi(W)} \operatorname{Lip}\big(D(\Phi \circ H)\big),$$

which is finite by Assumption A4 together with the $C^{1,1}$ regularity of H. We use the convention

$$\frac{\kappa}{4B_{\mathrm{loc}}} := +\infty \qquad \text{when } B_{\mathrm{loc}} = 0,$$

and choose

$$0 < \rho < \min\Big\{\operatorname{diam}(S),\ \frac{\kappa}{4B_{\mathrm{loc}}}\Big\}, \qquad \text{so that} \qquad B_{\mathrm{loc}}\, \rho \le \frac{\kappa}{4}.$$

Since $0 < \rho < \operatorname{diam}(S)$, the compact set

$$P_\rho := \{(s,s') \in S \times S :\ |s - s'| \ge \rho\}$$

is nonempty and disjoint from the diagonal. By the co-Lipschitz constant c_{ψ_0}, we have

$$\alpha_0(\rho) := \inf_{(s,s')\in P_\rho} \|\psi_0(s) - \psi_0(s')\|_Z \ge c_{\psi_0}\, \rho > 0,$$

so any $\alpha \in (0, \alpha_0(\rho))$ is a valid margin. In particular, the set of parameter tuples (κ, ρ, α) satisfying

$$0 < \kappa < \kappa_0, \qquad B_{\mathrm{loc}}\,\rho \le \frac{\kappa}{4}, \qquad 0 < \alpha < \alpha_0(\rho)$$

is nonempty.

Proof. By the uniform budgets in Assumption A4, the $C^{1,1}$ regularity of H, and the chain rule, the quantity

$$B_{\mathrm{loc}} = \sup_W \sup_{\Phi \in \mathcal{A}_\Phi(W)} \mathrm{Lip}\big(D(\Phi \circ H)\big) < \infty.$$

By the convention $\kappa/(4B_{\mathrm{loc}}) = +\infty$ when $B_{\mathrm{loc}} = 0$, and since $\kappa > 0$ and $\mathrm{diam}(S) > 0$, the set $\big(0, \min\{\mathrm{diam}(S), \kappa/(4B_{\mathrm{loc}})\}\big)$ is nonempty. We may therefore choose ρ in it. Then,

$$0 < \rho < \mathrm{diam}(S) \quad \text{and} \quad B_{\mathrm{loc}}\rho \le \frac{\kappa}{4}.$$

We next verify the properties of P_ρ. We observe that the map $(s, s') \mapsto |s - s'|$ is continuous on the compact set $S \times S$, so

$$P_\rho = \{(s, s') \in S \times S : \ |s - s'| \ge \rho\}$$

is compact, and the diameter is attained: there exist $s_1, s_2 \in S$ with $|s_1 - s_2| = \mathrm{diam}(S)$.

Since $\rho < \mathrm{diam}(S)$, we have $(s_1, s_2) \in P_\rho$, so P_ρ is nonempty. Moreover, $(s, s') \in P_\rho$ implies $|s - s'| \ge \rho > 0$, hence $s \neq s'$. Thus, P_ρ is disjoint from the diagonal.

Finally, we show the positivity of $\alpha_0(\rho)$. By the co-Lipschitz bound for ψ_0,

$$\|\psi_0(s) - \psi_0(s')\|_Z \ge c_{\psi_0}|s - s'| \qquad \text{for all } s, s' \in S.$$

This implies, for every $(s, s') \in P_\rho$,

$$\|\psi_0(s) - \psi_0(s')\|_Z \ge c_{\psi_0}|s - s'| \ge c_{\psi_0}\rho.$$

Taking the infimum, we obtain

$$\alpha_0(\rho) \ge c_{\psi_0}\rho > 0.$$

We may therefore choose $0 < \alpha < \alpha_0(\rho)$. Together with the already fixed choice $0 < \kappa < \kappa_0$ and the choice of ρ satisfying $B_{\mathrm{loc}}\rho \le \kappa/4$, we obtain parameters satisfying

$$B_{\mathrm{loc}}\rho \le \frac{\kappa}{4}, \qquad 0 < \kappa < \kappa_0, \qquad 0 < \alpha < \alpha_0(\rho).$$

Therefore the admissible parameter regime is nonempty. ■

S2. Architecture-specific covering estimates

The main article proves the abstract uniform deviation bound and the finite-parametric estimate. Here we verify the required coefficient stability and parameter-Lipschitz bounds for the concrete spline dictionaries of Supplementary Section S1.

Let us fix the parameters $\kappa, \alpha > 0$. Define the three function classes

$$\mathcal{G}_M(W) := \left\{(s, a) \mapsto \|R[\Phi, F](s, a)\|_Z^2 : (\Phi, F) \in \mathcal{A}(W)\right\},$$

$$\mathcal{G}_U(W) := \left\{(s, v) \mapsto \big[\kappa - \|D(\Phi \circ H)(s)[v]\|_Z\big]_+^2 : \Phi \in \mathcal{A}_\Phi(W)\right\},$$

$$\mathcal{G}_P(W) := \left\{(s, s') \mapsto \big[\alpha - \|(\Phi \circ H)(s) - (\Phi \circ H)(s')\|_Z\big]_+^2 : \Phi \in \mathcal{A}_\Phi(W)\right\}.$$

Let μ_M, μ_U, μ_P be the population measures on $M, \mathcal{U}, P_\rho$, respectively, and let $\mu_{M,n}, \mu_{U,n}, \mu_{P,n}$ be the corresponding empirical measures from independent samples of size n.

We denote by $N_M(r;W), N_U(r;W), N_P(r;W)$ the r-covering numbers of $\mathcal{G}_M(W), \mathcal{G}_U(W), \mathcal{G}_P(W)$ in the C^0 norm.

Lemma S2.1 (Uniform coefficient stability for the spline dictionaries)**.** *Fix the spline degree and a compact rectangular box $\Omega \subset \mathbb{R}^d$. For the normalized tensor-product B-spline bases $\{B_j^h\}_j$ on the admissible uniform meshes used above, there exists a constant $C_{\mathrm{stab}} \geq 1$, independent of h, such that every scalar spline*

$$s_c = \sum_j c_j B_j^h$$

satisfies

$$C_{\mathrm{stab}}^{-1} \|c\|_{\ell^\infty} \leq \|s_c\|_{L^\infty(\Omega)} \leq C_{\mathrm{stab}} \|c\|_{\ell^\infty}.$$

The same estimate holds componentwise for vector-valued splines.

Proof. The upper bound is immediate from nonnegativity and the partition-of-unity property of the normalized B-spline basis. Indeed, for every $x \in \Omega$, we have

$$|s_c(x)| = \left| \sum_j c_j B_j^h(x) \right| \leq \|c\|_{\ell^\infty} \sum_j B_j^h(x) = \|c\|_{\ell^\infty}.$$

Thus, we get

$$\|s_c\|_{L^\infty(\Omega)} \leq \|c\|_{\ell^\infty}.$$

It remains to prove the inverse estimate. Let $\mathcal{T}_h$ denote the underlying uniform tensor-product mesh, and for each cell $K \in \mathcal{T}_h$ let

$$I(K) := \{j : B_j^h|_K \not\equiv 0\}$$

be the set of indices of B-splines active on K. Since the spline degree is fixed, the cardinality of $I(K)$ is uniformly bounded, independently of h.

Let $A_K : \widehat{K} \to K$ be the affine map from the reference cube $\widehat{K} = [0,1]^d$ to K. For $j \in I(K)$, we define the rescaled restriction

$$\widehat{B}_{j,K} := B_j^h \circ A_K.$$

By uniformity of the mesh and by the fixed degree assumption, the collection of possible families

$$\{\widehat{B}_{j,K} : j \in I(K)\}$$

belongs to a finite set of reference patterns. This finite set includes the interior pattern and the finitely many boundary knot patterns.

For each such reference pattern $P = \{\beta_1, \ldots, \beta_m\}$, we consider the linear map

$$T_P : \mathbb{R}^m \to C(\widehat{K}), \qquad T_P b := \sum_{r=1}^m b_r \beta_r.$$

The functions $\beta_1, \ldots, \beta_m$ are locally linearly independent. Hence T_P is injective. Since $\mathbb{R}^m$ is finite-dimensional, the quantity

$$C_P := \inf_{\|b\|_{\ell^\infty}=1} \|T_P b\|_{L^\infty(\widehat{K})} > 0.$$

Because only finitely many patterns P can occur, we may define

$$C_{\min} := \min_P C_P > 0.$$

Therefore, for every cell K and every coefficient vector $b = (b_j)_{j\in I(K)}$, we have

$$\|b\|_{\ell^\infty} \le C_{\min}^{-1} \left\| \sum_{j\in I(K)} b_j B_j^h \right\|_{L^\infty(K)}.$$

Now, we fix an arbitrary coefficient index j and choose a mesh cell K_j on which B_j^h is active. Applying the preceding local estimate to the coefficient vector $(c_i)_{i\in I(K_j)}$, we obtain

$$|c_j| \le \|(c_i)_{i\in I(K_j)}\|_{\ell^\infty} \le C_{\min}^{-1}\|s_c\|_{L^\infty(K_j)} \le C_{\min}^{-1}\|s_c\|_{L^\infty(\Omega)}.$$

Taking the supremum over j gives

$$\|c\|_{\ell^\infty} \le C_{\min}^{-1}\|s_c\|_{L^\infty(\Omega)}.$$

Equivalently,

$$C_{\min}\|c\|_{\ell^\infty} \le \|s_c\|_{L^\infty(\Omega)}.$$

Combining this lower bound with the upper bound, the desired estimate holds with, for example,

$$C_{\text{stab}} := \max\{1, C_{\min}^{-1}\}.$$

The constant depends only on the spline degree, the dimension, and the admissible boundary knot patterns, and is independent of h.

For vector-valued splines, the same argument applies to each component separately, which gives the stated componentwise estimate. ∎

The following proposition shows that the spline architecture satisfies the finite-parametric hypothesis of the finite-parametric deviation corollary in the main article.

Proposition S2.2 (Spline parameter-Lipschitz bound)**.** *Consider the range-constrained tensor-product cubic B-spline classes of Proposition S1.6, parametrized by the vector θ of spline coefficients of (Φ, F), constrained to a box $[-R_W, R_W]^{p_W}$. Then the hypothesis of the finite-parametric deviation corollary in the main article holds for each $Y \in \{M, U, P\}$: each integrand class $\mathcal{G}_Y(W)$ satisfies*

$$\|g_\theta - g_{\theta'}\|_{C^0(Y)} \le L_{\theta,Y,W}\|\theta - \theta'\|_{\ell^\infty},$$

and there are constants C_Y, q_Y, independent of n, with

$$R_W L_{\theta,Y,W} \le C_Y W^{q_Y}.$$

Consequently, the finite-parametric estimate of the finite-parametric deviation corollary in the main article applies to the norm-constrained spline architecture of Proposition S1.6.

Proof. We write the proof for one fixed capacity level W.

Let $\Omega_\Phi \subset X$ and $\Omega_F \subset Z \times \mathbb{R}^{d_a}$ be the compact coordinate boxes containing $H(S)$ and $K_Z \times A$, respectively, as in Proposition S1.6.

We recall the construction of the encoder and transition spline classes $\mathcal{A}_\Phi(W)$ and $\mathcal{A}_F(W)$ in Proposition S1.6. They have different mesh widths, which we denote by h_Φ and h_F.

The encoder dictionary is a tensor-product cubic B-spline dictionary on Ω_Φ, and the transition dictionary is a tensor-product cubic B-spline dictionary on Ω_F.

We first record the basic coefficient-stability estimates. A vector-valued spline map on a d-dimensional coordinate box has the form

$$u_c(x) = \sum_{j=1}^{p} c_j B_j^h(x),$$

where B_j^h are scalar tensor-product cubic B-splines, $c_j \in \mathbb{R}^m$ are the output coefficients, and $p \asymp h^{-d}$ is the number of basis functions. Applied to the two classes, the encoder splines on $\Omega_\Phi \subset \mathbb{R}^{d_x}$ have $p_\Phi \asymp h_\Phi^{-d_x}$ basis functions and the transition splines on $\Omega_F \subset \mathbb{R}^{d_z+d_a}$ have $p_F \asymp h_F^{-(d_z+d_a)}$. Since both maps are $\mathbb{R}^{d_z}$-valued, the combined parameter vector $\theta = (c_\Phi, c_F)$ has dimension

$$p_W = d_z\,(p_\Phi + p_F) \asymp d_z\big(h_\Phi^{-d_x} + h_F^{-(d_z+d_a)}\big),$$

which is the ambient dimension of the box $[-R_W, R_W]^{p_W}$ in the finite-parametric deviation corollary in the main article.

Since the spline degree is fixed and the grid is uniform, the basis has uniformly bounded overlap. Therefore there exist constants $C_0, C_1 > 0$, independent of h, such that for every two coefficient vectors c, c',

$$\|u_c - u_{c'}\|_{C^0} \le C_0 \|c - c'\|_{\ell^\infty},$$

and

$$\|D(u_c - u_{c'})\|_{C^0} \le C_1 h^{-1} \|c - c'\|_{\ell^\infty}.$$

Here the constants depend only on the coordinate box, the dimension, and the fixed spline degree, but not on h.

We now incorporate the output range-enforcing map. For the encoder and transition maps, we use the range-constrained form $\widetilde{u}_c := \Pi_Z \circ u_c$, where $\Pi_Z : Z \to K_Z$ is the fixed $C^{1,1}$ range-enforcing map. Since the norm-constrained spline class satisfies

$$\|u_c\|_{C^0} + \|Du_c\|_{C^0} + \mathrm{Lip}(Du_c) \le B,$$

all spline maps u_c take values in a fixed compact subset of Z. Hence, $\|D\Pi_{K_Z}\|_{C^0}$ and $\mathrm{Lip}(D\Pi_{K_Z})$ are bounded on that compact set by constants independent of h and W.

For the values, the mean value theorem gives

$$\begin{aligned}
\|\widetilde{u}_c - \widetilde{u}_{c'}\|_{C^0} &= \|\Pi_{K_Z}(u_c) - \Pi_{K_Z}(u_{c'})\|_{C^0} \\
&\le \|D\Pi_{K_Z}\|_{C^0} \|u_c - u_{c'}\|_{C^0} \\
&\le C\|c - c'\|_{\ell^\infty}.
\end{aligned}$$

For the derivatives, by the chain rule,

$$D\widetilde{u}_c(x) = D\Pi_{K_Z}(u_c(x)) Du_c(x).$$

Thus, we have the identity

$$\begin{aligned}
&D\widetilde{u}_c(x) - D\widetilde{u}_{c'}(x) \\
&\quad = \big[D\Pi_{K_Z}(u_c(x)) - D\Pi_{K_Z}(u_{c'}(x))\big] Du_c(x) + D\Pi_{K_Z}(u_{c'}(x))\big[Du_c(x) - Du_{c'}(x)\big].
\end{aligned}$$

Using the uniform C^1-bound on u_c, the Lipschitz bound on $D\Pi_{K_Z}$, and the spline stability estimates above, we obtain

$$\begin{aligned}
\|D\widetilde{u}_c - D\widetilde{u}_{c'}\|_{C^0} &\le C\|u_c - u_{c'}\|_{C^0} + C\|D(u_c - u_{c'})\|_{C^0} \\
&\le C\left(1 + h^{-1}\right)\|c - c'\|_{\ell^\infty}.
\end{aligned}$$

For $0 < h \le 1$, this gives

$$\|D\widetilde{u}_c - D\widetilde{u}_{c'}\|_{C^0} \le Ch^{-1}\|c - c'\|_{\ell^\infty}.$$

By Lemma S2.1 and the raw norm constraints,

$$\|c_\Phi\|_{\ell^\infty} \le C_{\mathrm{stab}} B_\Phi, \qquad \|c_F\|_{\ell^\infty} \le C_{\mathrm{stab}} B_F.$$

Thus the combined coefficient set is contained in a cube $[-R, R]^{p_W}$ with a radius R independent of W; in the finite-parametric deviation corollary in the main article one may therefore take $R_W = R$.

We apply these estimates separately to the transition maps and to the encoder maps

$$\Phi_{c_\Phi} = \Pi_Z \circ u_{c_\Phi} \circ \rho_\Phi.$$

Composition with the fixed map ρ_Φ changes the preceding C^0- and C^1-parameter bounds only by constants depending on $\|D\rho_\Phi\|_{C^0}$ and $\mathrm{Lip}(D\rho_\Phi)$, not on W. Moreover, $\rho_\Phi = \mathrm{Id}$ on a neighborhood of $H(S)$, so the value and derivative estimates entering the three integrand classes retain the displayed h_Φ^{-1} scaling. Here c_Φ and c_F denote the encoder and transition coefficient vectors.

We now verify the parameter-Lipschitz estimates for the three integrand classes.

Let $\theta := (c_\Phi, c_F)$ be the combined parameter vector. We define

$$g_\theta^M(s, a) := \|\psi_{c_\Phi}(G(s, a)) - F_{c_F}(\psi_{c_\Phi}(s), a)\|_Z^2$$

and

$$R_\theta(s, a) := \psi_{c_\Phi}(G(s, a)) - F_{c_F}(\psi_{c_\Phi}(s), a),$$

where $\psi_{c_\Phi} := \Phi_{c_\Phi} \circ H$.

Since the two terms in R_θ take values in K_Z, the residuals are uniformly bounded:

$$\|R_\theta(s, a)\|_Z \le D_Z := \mathrm{diam}(K_Z).$$

With $\theta' = (c'_\Phi, c'_F)$, we estimate

$$\left|g_\theta^M(s, a) - g_{\theta'}^M(s, a)\right| \le 2D_Z\|R_\theta(s, a) - R_{\theta'}(s, a)\|_Z.$$

We also estimate

$$\begin{aligned}\|R_\theta(s, a) - R_{\theta'}(s, a)\|_Z \le{} & \|\psi_{c_\Phi}(G(s, a)) - \psi_{c'_\Phi}(G(s, a))\|_Z \\ & + \|F_{c_F}(\psi_{c_\Phi}(s), a) - F_{c'_F}(\psi_{c'_\Phi}(s), a)\|_Z.\end{aligned}$$

For the second term, we add and subtract $F_{c_F}(\psi_{c'_\Phi}(s), a)$:

$$\begin{aligned}\|F_{c_F}(\psi_{c_\Phi}(s), a) - F_{c'_F}(\psi_{c'_\Phi}(s), a)\|_Z \le{} & \|F_{c_F}(\psi_{c_\Phi}(s), a) - F_{c_F}(\psi_{c'_\Phi}(s), a)\|_Z \\ & + \|F_{c_F}(\psi_{c'_\Phi}(s), a) - F_{c'_F}(\psi_{c'_\Phi}(s), a)\|_Z.\end{aligned}$$

The transition class has a uniform Lipschitz budget in z, say L_F^{bud}. Therefore

$$\|F_{c_F}(\psi_{c_\Phi}(s), a) - F_{c_F}(\psi_{c'_\Phi}(s), a)\|_Z \le L_F^{\mathrm{bud}}\|\psi_{c_\Phi} - \psi_{c'_\Phi}\|_{C^0(S)}.$$

The parameter-value estimate for F gives

$$\|F_{c_F}(\psi_{c'_\Phi}(s), a) - F_{c'_F}(\psi_{c'_\Phi}(s), a)\|_Z \le C\|c_F - c'_F\|_{\ell^\infty}.$$

Combining these estimates yields

$$\|R_\theta - R_{\theta'}\|_{C^0(M)} \le C\|\theta - \theta'\|_{\ell^\infty}.$$

Consequently,

$$\|g^M_\theta - g^M_{\theta'}\|_{C^0(M)} \le C\|\theta - \theta'\|_{\ell^\infty}.$$

Thus one may take

$$L_{\theta,M,W} \le C.$$

A similar computation for $\mathcal{G}_U(W)$ and $\mathcal{G}_P(W)$ yields

$$L_{\theta,U,W} \le Ch_\Phi^{-1} \quad \text{and} \quad L_{\theta,P,W} \le C.$$

We recall from the construction in Proposition S1.6 that

$$h_\Phi^{-1} \le C_\Phi W^{1/d_x}.$$

Since the coefficient radius $R_W = R$ is independent of W, it follows that

$$R_W L_{\theta,M,W}, \qquad R_W L_{\theta,U,W}, \qquad R_W L_{\theta,P,W}$$

all grow at most polynomially in W. ■

S3. Secondary deterministic-control diagnostics

The main article retains the deterministic simulation lemma and the modular planning corollary. This section records the horizon-amplification table, a concentrated-residual construction, the adversarial-planner diagnostic, and secondary consequences of metric non-collapse. Together these results show constructively why uniform residual control is the correct companion to metric faithfulness and how the constants produced by the representation theorem propagate to planning, Lyapunov, and reachability statements.

The rollout factor $C(T, L_F)$ is finite for every finite T and L_F. It remains uniformly bounded in T when $L_F < 1$, grows linearly when $L_F = 1$, and grows geometrically when $L_F > 1$. The accumulated stage-cost factor $D(T, L_F)$ has one additional order in the neutral case: $D(T, 1) = T(T-1)/2$. Thus a terminal-state estimate can be $O(\eta T)$ at $L_F = 1$, while an undiscounted running-cost estimate is generally $O(\eta T^2)$. Table S1 reports $\eta\, C(T, L_F)$ for $\eta = 10^{-2}$ and several values of L_F and T.

T	$L_F = 0.9$	$L_F = 1.0$	$L_F = 1.1$	$L_F = 1.5$
1	1.00×10^{-2}	1.00×10^{-2}	1.00×10^{-2}	1.00×10^{-2}
5	4.10×10^{-2}	5.00×10^{-2}	6.10×10^{-2}	1.32×10^{-1}
10	6.51×10^{-2}	1.00×10^{-1}	1.59×10^{-1}	1.13×10^{0}
25	9.28×10^{-2}	2.50×10^{-1}	9.83×10^{-1}	5.05×10^{2}
50	9.95×10^{-2}	5.00×10^{-1}	1.16×10^{1}	1.28×10^{7}

Table S1. Cumulative rollout factor $\eta\, C(T, L_F)$ for $\eta = 10^{-2}$. For $L_F < 1$ it is bounded by $\eta/(1 - L_F)$ uniformly in the horizon, for $L_F = 1$ it equals ηT, and for $L_F > 1$ it grows geometrically.

S3.1. Sharpness of the interpolation exponent.

The exponent $1/(q+2)$ supplied by Lipschitz–Ahlfors interpolation cannot be improved under those hypotheses alone.

Proposition S3.1 (Sharpness of the $1/(q+2)$ exponent)**.** *Let $q \in \mathbb{N}$, let*

$$Y = [0, 1]^q$$

with Lebesgue measure, and fix $L > 0$. The exponent $1/(q+2)$ in the estimate

$$\|u\|_{L^\infty(Y)} \lesssim \left(\int_Y u^2 \, \mathrm{d}y\right)^{1/(q+2)}$$

cannot be improved uniformly over all nonnegative L-Lipschitz functions $u : Y \to \mathbb{R}$.

More precisely, for arbitrarily small $h > 0$, there exists a nonnegative L-Lipschitz function $u_h : Y \to \mathbb{R}$ such that

$$\|u_h\|_{L^\infty(Y)} = h, \qquad \int_Y u_h^2 \, \mathrm{d}y \le C_q L^{-q} h^{q+2},$$

where $C_q > 0$ depends only on the dimension q. Consequently, for every

$$\alpha > \frac{1}{q+2},$$

there is no constant $C_\alpha < \infty$ such that

$$\|u\|_{L^\infty(Y)} \le C_\alpha \left(\int_Y u^2 \, \mathrm{d}y \right)^\alpha$$

holds for all nonnegative L-Lipschitz functions $u : Y \to \mathbb{R}$.

Proof. We set $y_0 := \left(\frac{1}{2}, \dots, \frac{1}{2}\right) \in (0,1)^q$ and choose $h \in (0, L/2)$. Then, we observe that

$$\frac{h}{L} < \frac{1}{2} = \operatorname{dist}(y_0, \partial Y),$$

and hence $B_{h/L}(y_0) \subset Y$. We define the map

$$u_h(y) := \big[h - L|y - y_0|\big]_+, \qquad y \in Y.$$

This is the truncated cone of height h, slope L, and base radius h/L. We first verify the basic properties of u_h. The map u_h is the composition of the 1-Lipschitz map $y \mapsto |y - y_0|$, the affine map $r \mapsto h - Lr$ on $\mathbb{R}$, and the 1-Lipschitz map $r \mapsto [r]_+$ on $\mathbb{R}$. Therefore, u_h is L-Lipschitz. It is also nonnegative by definition. Moreover, $u_h(y_0) = h$, and since $u_h \le h$ everywhere, we have

$$\|u_h\|_{L^\infty(Y)} = h.$$

We compute

$$\int_Y u_h(y)^2 \, \mathrm{d}y = \int_{B_{h/L}(y_0)} \big(h - L|y - y_0|\big)^2 \, \mathrm{d}y.$$

Using the change of variables

$$y = y_0 + \frac{h}{L}\xi, \qquad \mathrm{d}y = \left(\frac{h}{L}\right)^q \mathrm{d}\xi,$$

we obtain

$$\begin{aligned}
\int_Y u_h(y)^2 \, \mathrm{d}y &= \int_{B_1(0)} \big(h - L(h/L)|\xi|\big)^2 \left(\frac{h}{L}\right)^q \mathrm{d}\xi \\
&= h^2 \left(\frac{h}{L}\right)^q \int_{B_1(0)} (1 - |\xi|)^2 \, \mathrm{d}\xi \\
&= C_q \, L^{-q} \, h^{q+2},
\end{aligned}$$

where $C_q := \int_{B_1(0)} (1 - |\xi|)^2 \, \mathrm{d}\xi < \infty$.

We now establish sharpness by contradiction. Suppose that there exist $\alpha > 1/(q+2)$ and a constant $C_\alpha < \infty$ such that

$$\|u\|_{L^\infty(Y)} \le C_\alpha \left(\int_Y u^2 \, \mathrm{d}y \right)^\alpha$$

for every nonnegative L-Lipschitz function $u : Y \to \mathbb{R}$. Applying this estimate to u_h, we obtain

$$h \le C_\alpha \left(C_q L^{-q} h^{q+2}\right)^\alpha,$$

or equivalently,

$$1 \le C_\alpha \left(C_q L^{-q}\right)^\alpha h^{\alpha(q+2)-1}.$$

Since $\alpha(q+2) - 1 > 0$, the right-hand side tends to 0 as $h \downarrow 0$, which is a contradiction. Therefore, no uniform estimate with exponent $\alpha > 1/(q+2)$ can hold.

The family $\{u_h\}$ shows that an L^2-error of order h^{q+2} can conceal a pointwise error of order h. Hence the exponent $1/(q+2)$ in the Lipschitz–Ahlfors L^2-to-L^∞ interpolation estimate is optimal under a Lipschitz-only regularity assumption. ∎

S3.2. **Why an L^2-mean residual is not enough.** The natural training objective minimizes an empirical squared residual, while its population counterpart $\mathcal{E}_{\text{pred}}$ is an $L^2(\mu)$-average. One might hope that a small average is sufficient for downstream planning; in general, it is not. The obstruction is geometric: in the deterministic simulation lemma in the main article, η is a supremum over $(s, a) \in M$, whereas an $L^2(\mu)$ loss can be small even when the residual is large on a small region. An optimizing planner may then prefer trajectories that pass through precisely those regions where the surrogate is overly optimistic.

Spike example. Let $M = [0,1]^{d_m}$ carry normalized Lebesgue measure, and choose an interior center x_0 such that $B_r(x_0) \subset [0,1]^{d_m}$. Define

$$R_{h,r}(x) := h\left[1 - \frac{|x - x_0|}{r}\right]_+.$$

Then $\|R_{h,r}\|_{L^\infty} = h$, independently of r. A radial integration gives

$$\|R_{h,r}\|_{L^2} = h r^{d_m/2} \left[\frac{2\omega_{d_m}}{(d_m+1)(d_m+2)}\right]^{1/2},$$

where

$$\omega_{d_m} = \frac{\pi^{d_m/2}}{\Gamma(d_m/2+1)}$$

is the volume of the d_m-dimensional unit ball. Table S2 reports representative values.

d_m	r	$\|R_{h,r}\|_{L^\infty}$	$\|R_{h,r}\|_{L^2}$	$\|R_{h,r}\|_{L^2}/\|R_{h,r}\|_{L^\infty}$
2	0.50	1.00×10^{-1}	3.618×10^{-2}	3.618×10^{-1}
2	0.10	1.00×10^{-1}	7.236×10^{-3}	7.236×10^{-2}
2	0.01	1.00×10^{-1}	7.236×10^{-4}	7.236×10^{-3}
4	0.10	1.00×10^{-1}	5.736×10^{-4}	5.736×10^{-3}
8	0.10	1.00×10^{-1}	3.003×10^{-6}	3.003×10^{-5}

TABLE S2. Supremum norm and L^2-mean of the spike residual for $h = 10^{-1}$, with the support ball contained in the cube. The L^2-mean can be made arbitrarily small at fixed supremum norm. The factor $r^{d_m/2}$ makes the discrepancy especially pronounced in high dimension.

Adversarial-planner diagnostic. The spike construction proves the average-versus-supremum separation, but one may still ask whether a concentrated model error can materially affect planning. We illustrate this on a fully specified one-dimensional receding-horizon problem. This diagnostic isolates the average-versus-uniform planning mechanism in a transparent finite problem; the principal representation theorem supplies the corresponding smooth, finite-sample certification framework.

Let

$$S = [0, 1], \qquad A = \{-1, -0.5, 0, 0.5, 1\},$$

and define

$$G(s, a) := \mathrm{clip}_{[0,1]}(s + 0.05a).$$

The running cost is

$$\ell(s, a) := (s - \tau(s))^2, \qquad \tau(s) := \begin{cases} 0.7, & s < 0.5, \\ 0.3, & s \geq 0.5, \end{cases}$$

and the terminal cost is zero. Thus the preferred target switches across the decision boundary $s = 0.5$.

At each closed-loop step, an MPC controller enumerates all 5^4 action sequences $\mathbf{a} = (a_0, \ldots, a_3) \in A^4$, rolls them out for horizon $T = 4$ under its planning model $\widetilde{G}$, and minimizes

$$\sum_{t=0}^{3} \ell(z_t, a_t), \qquad z_{t+1} = \widetilde{G}(z_t, a_t).$$

It applies the first action of a minimizing sequence; ties are broken by the first sequence in the lexicographic order induced by $-1 < -0.5 < 0 < 0.5 < 1$.

We compare the two surrogate models

$$\widetilde{G}_A(s, a) := \mathrm{clip}_{[0,1]}\big(s + 0.05a + r_A(s)\big),$$

$$r_A(s) := 0.15 \exp\left(-\frac{(s - 0.48)^2}{2\sigma^2}\right), \qquad \sigma = 0.04,$$

and

$$\widetilde{G}_B(s, a) := \mathrm{clip}_{[0,1]}\big(s + 0.05a + r_B(s)\big), \qquad r_B(s) \equiv 0.04.$$

For $i \in \{A, B\}$, define the actual one-step model error

$$e_i(s, a) := |\widetilde{G}_i(s, a) - G(s, a)|.$$

We measure e_i in the product of normalized Lebesgue measure on S and the uniform counting measure on A. Composite trapezoidal quadrature on 2,000,001 equally spaced state points gives

$$\|e_A\|_{L^\infty} = 0.15, \qquad \|e_A\|_{L^2} \approx 0.03994,$$

$$\|e_B\|_{L^\infty} = 0.04, \qquad \|e_B\|_{L^2} \approx 0.03916.$$

Thus the two models have nearly equal L^2 one-step errors, while Model A has a much larger worst-case error concentrated near the decision boundary.

For each initial state in the uniform grid of 11 points in $[0.4, 0.6]$, we run the receding-horizon controller for 10 steps. At every step, the action selected using the relevant planning model is applied to the true dynamics G, and the true running cost is accumulated. Let $J_{10}^{A}(s_0)$, $J_{10}^{B}(s_0)$, and $J_{10}^{\mathrm{exact}}(s_0)$ denote the resulting true closed-loop costs when planning with $\widetilde{G}_A$, $\widetilde{G}_B$, and the exact model G, respectively. We report the excess costs

$$\Delta J_i(s_0) := J_{10}^{i}(s_0) - J_{10}^{\mathrm{exact}}(s_0), \qquad i \in \{A, B\}.$$

Model	$\|e\|_{L^\infty}$	$\|e\|_{L^2}$	mean excess cost	max. excess cost
A (spike near boundary)	**0.150**	0.0399	$\mathbf{2.10\times10^{-1}}$	2.82×10^{-1}
B (uniform offset)	0.040	0.0392	1.31×10^{-1}	2.03×10^{-1}

TABLE S3. Adversarial-planner diagnostic. The two models have nearly equal L^2 one-step errors, but Model A has a much larger supremum error and incurs approximately 60% more mean excess closed-loop cost relative to exact-model MPC. The observed ordering displays the additional information supplied by uniform-error control beyond the L^2-error ordering. The theorem places this mechanism in a complete quantitative envelope that also incorporates transition and cost Lipschitz constants and horizon factors.

For a planning state z, the MPC optimizer selects a sequence

$$\mathbf{a}^\star(z) \in \operatorname*{argmin}_{\mathbf{a}\in A^4} \sum_{t=0}^{3} \ell(z_t, a_t), \qquad z_{t+1} = \widetilde{G}(z_t, a_t),$$

and applies its first component. Optimization can therefore preferentially select trajectories passing through regions in which the surrogate is optimistic, even if those regions have small training-measure mass. A uniform semiconjugacy modulus supplies the natural guarantee over all initial states and action sequences, strengthening the average residual into exactly the control-relevant form required by the simulation argument.

S3.3. **Why metric non-collapse is the additional ingredient.** The discussion above explains why the *sup-norm* matters. The lower co-Lipschitz constant c_* of the principal representation theorem in the main article plays a separate but complementary role: even when the semiconjugacy modulus η vanishes, a folded encoder produces nonsensical plans. We explain this in two ways.

Cost and optimizer transfer. The lower co-Lipschitz estimate makes the original costs well-defined and Lipschitz on the encoded state set, with a quantitative loss proportional to c_*^{-1}; the latent-cost extension lemma in the main article extends these costs to all of K_Z. The deterministic planning-transfer corollary in the main article then compares the true and latent objectives uniformly over the common action space A^T. It applies to arbitrary optimizer selections, including nonunique optima, and gives

$$J_{\mathrm{true}}(s_0; \widehat{\mathbf{a}}(s_0)) - J^\star(s_0) \le 2\Delta_{T,L_F}(\eta) + \xi_{\mathrm{plan}}$$

for every ξ_{plan}-optimal latent action sequence. Thus c_* controls the regularity of the cost representation, while η controls the dynamical error accumulated over the planning horizon.

Practical Lyapunov transfer. Suppose $z^* = \psi(s^\dagger)$ for some target state $s^\dagger \in S$, and let $V_F : Z \to \mathbb{R}_+$ be L_{V_F}-Lipschitz on the relevant compact set. Assume

$$\beta_1 \|z - z^*\|_Z^2 \le V_F(z), \qquad V_F(F^\star(z, \pi_F(z))) \le \beta_2 V_F(z),$$

with $\beta_1 > 0$ and $0 < \beta_2 < 1$. Then the lower co-Lipschitz estimate and the one-step semiconjugacy error imply

$$\beta_1 c_*^2 d_S(s, s^\dagger)^2 \le V_F(\psi(s))$$

and

$$V_F(\psi(G(s, \pi_F(\psi(s))))) \le \beta_2 V_F(\psi(s)) + L_{V_F} \eta_{W,n,\delta}(\varepsilon_{\mathrm{train}}).$$

Thus $V_F \circ \psi$ is a practical Lyapunov function with an additive disturbance proportional to the semi-conjugacy modulus. In particular, iteration yields

$$V_F(\psi(s_t)) \le \beta_2^t V_F(\psi(s_0)) + \frac{L_{V_F}\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})}{1-\beta_2}.$$

The role of c_* is to convert latent closeness into state-space closeness; without a positive lower metric constant, small latent Lyapunov values need not localize the true state.

Reachability in latent space. Let $\mathrm{Reach}_t(S_0) \subset S$ be the set of states reachable in t steps from S_0, and let $\mathrm{Reach}_t^F(\psi(S_0)) \subset Z$ be the corresponding set generated by the learned latent transition under the same admissible action sequences. The deterministic simulation lemma in the main article gives the valid latent-space Hausdorff estimate

$$d_H\big(\psi(\mathrm{Reach}_t(S_0)),\, \mathrm{Reach}_t^F(\psi(S_0))\big) \le \eta C(t, L_F).$$

The theorem already yields the displayed latent-space Hausdorff estimate. When a decoder or a quantified projection onto $\psi(S)$ is available, the same argument extends this certificate to state-space reachable sets.

S3.4. **Comparison with stochastic bisimulation bounds.** The stochastic bisimulation-metric program of Ferns–Panangaden–Precup [3, 4], DeepMDP [5], and DBC [6] provides expected value-error or representation-error bounds under specified distributions, which need not coincide with the visitation distribution induced by a downstream planner. For a Δ-accurate stochastic transition model, Asadi–Misra–Littman [7] prove the multi-step Wasserstein estimate

$$W\Big(\widehat{T}^t(\cdot \mid \mu), T^t(\cdot \mid \mu)\Big) \le \Delta \sum_{j=0}^{t-1} \overline{K}^j, \qquad \overline{K} := \min\{K_{\widehat{T}}, K_T\}.$$

This is a transition-distribution bound; their separate discounted value-error result additionally uses Lipschitz reward assumptions. The geometric factor has the same recurrence form as $C(t, L_F)$ in the deterministic simulation lemma, but the driving discrepancy is Wasserstein-1 rather than a pointwise supremum. In the deterministic setting considered here, the transition kernel reduces to a point map and the uniform residual gives a worst-case guarantee over initial states and action sequences. The results are therefore complementary rather than ordered by a universal notion of sharpness.

S3.5. **Summary.** The constant c_* and modulus $\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})$ of the principal representation theorem are precisely the quantities that yield (i) a uniform finite-horizon cost-transfer estimate and a deterministic latent-optimizer suboptimality bound for compatible Lipschitz costs, (ii) the resolution of state recovery from latent codes (§S3.3), (iii) Lyapunov and practical-stability estimates after transfer, and (iv) latent-space Hausdorff approximations of forward-reachable sets; a state-space pullback would require an additional decoder or projection assumption. Under the stated assumptions, the local–global certificate strengthens average prediction and global-spread information into these uniform control conclusions. The constants c_* and $\eta_{W,n,\delta}(\varepsilon_{\mathrm{train}})$ are explicit, and the planning guarantee keeps the empirical-training tolerance $\varepsilon_{\mathrm{train}}$ separate from the latent-solver tolerance ξ_{plan}. The finite-sample planning corollary in the main article further quantifies learned-cost errors through ε_ℓ and ε_g. The guarantee holds whenever the regularization strength belongs to the admissible range of the principal representation theorem in the main article.

S4. Numerical protocols, summarized records, and supporting-material availability

The experiments below provide three complementary layers of evidence: exact geometric obstruction results for prediction and covariance-spread objectives, sampled geometric diagnostics in finite trained models, and a nonlinear benchmark linking restored metric resolution to improved control. The analytical layer establishes the obstruction exactly; the finite-dimensional layers quantify the mechanism, threshold ingredients, optimization behavior, and planning consequences on declared fixed evaluation sets. Geometry-aware objectives consistently restore nontrivial metric resolution and strongly improve control relative to pure prediction, while the local–global hinge retains the distinctive advantage of matching the sufficient mechanism proved in the main theorem. In our reports, the quantities in the principal representation theorem in the main article are computed using the learned representation and learned latent cost heads. For example, if $(\widehat{\Phi}, \widehat{F})$ is a learned representation, we compute

$$\widehat{\eta}_{\mathrm{eval}} := \max_{(s_i,a_i)\in D_M^{\mathrm{eval}}} \|\widehat{\Phi}(H(G(s_i,a_i))) - \widehat{F}(\widehat{\Phi}(H(s_i)), a_i)\|,$$

Furthermore, we denote the pairs consisting of minimum and maximum of the pairwise chord ratios and the norm of $D(\widehat{\Phi}\circ H)$ by $(\widehat{c}_{\min}, \widehat{c}_{\max})$, $(\widehat{m}_{\mathrm{loc}}, \widehat{M}_{\mathrm{loc}})$, respectively. We also denote by $\widehat{m}_{\mathrm{glob}}$ the minimum distance of $\widehat{\Phi}\circ H(s)$ and $\widehat{\Phi}\circ H(s')$ for all samples s, s' satisfying $|s - s'| \geq \rho$.

The analytic obstructions in Experiment A are exact, while the trained studies provide complementary finite-architecture and sampled-diagnostic evidence. In particular, the coefficient-box spline class in Experiment B is deliberately chosen as a tractable finite-capacity proxy for restart analysis and threshold decomposition, whereas the uniform-capacity $C^{1,1}$ class constructed in Proposition S1.6 supplies the theorem-level approximation architecture. This division combines a rigorous constructive class with a computationally transparent calibration study. Three design decisions distinguish the experiments below from a naive implementation.

(1) **A common compact latent range and explicit upper control.** The proposed hinge is one-sided. In an unconstrained network, multiplying the encoder output by a large constant improves every lower diagnostic without improving the representation at all. The theorem excludes this by assuming a compact latent range K_Z and a uniform $C^{1,1}$ budget. Accordingly every trained encoder in every experiment has the form

$$\psi_\theta(s) \;=\; R_Z \tanh\bigl(h_\theta(H(s))\bigr) \qquad \text{coordinatewise,} \tag{S4.1}$$

with the *same* fixed R_Z for every objective and seed, and we report the upper diagnostics alongside the lower ones. Resolution improvement is evaluated jointly with $\widehat{c}_{\max}$, $\widehat{M}_{\mathrm{loc}}$, and the latent norm, so the reported finite-set gain documents improved lower geometry rather than scale inflation. Continuum certification is reserved for the validated finite-net conditions of the main article.

(2) **A fixed empirical objective.** The principal representation theorem in the main article concerns three fixed independent samples D_M, D_U, D_P. In the theorem-facing experiments these arrays are drawn once per seed and frozen; minibatches are index draws into them, and no direction or separated pair is generated during optimization. A separate, clearly labeled ablation compares this with resampled training.

(3) **Restart-aware optimization accounting.** The theorem keeps the optimization tolerance $\varepsilon_{\mathrm{train}}$ explicit, so every run reports final objective components, sampled total-objective curves, and terminal gradient norms. In Experiment B, the observed restart gap is computed within the same fixed empirical objective, with seed, capacity, sample size, and regularization strength

experiment	R_Z	upper control	$n_M/n_U/n_P$	runs per objective	architecture	params	preprocessing time (s)
A3 folding	2.0	bounded range; upper hinge in objective (vi)	512/512/256	12 seeds × 2 regimes	MLP 1-128-128-1	–	–
B spline (main grid)	2.0	coefficient box ±3.0	$n/n/n$, $n \le 2048$	2 seeds × 3 restarts	cubic B-spline	–	–
B spline (capacity sweep)	2.0	coefficient box	512/512/512	1 seed(s)	cubic B-spline	–	–
B bounded MLP	2.0	approx. PI spectral target 1.6	$n/n/n$	2 seeds	MLP width 64	–	–
C pendulum + planning	2.0	approx. PI spectral target 1.5	2048/1024/512	3 seeds	MLP width 64	13896	20.64 [14.87, 50.40]

TABLE S4. Protocol alignment. Every trained encoder in a given experiment maps into the same compact latent box $K_Z = [-R_Z, R_Z]^{d_z}$, uses the same fixed D_M, D_U, D_P arrays, and receives the same number of updates. Run counts are read from the authors' retained result records, which will be provided upon request. The timing column is the median [min, max] over individual representation preprocessing runs, including representation training, cost-head fitting, and diagnostics but excluding MPC planning; it is not a total experiment wall-clock.

held fixed. Experiment A reports cross-seed objective distributions separately. This accounting makes the achieved optimization quality visible and connects the numerical study directly to the theorem's tolerance parameter.

Evidence hierarchy. We state in advance how each analytical and numerical layer contributes to the paper's conclusions.

(H1) *Analytically established.* Zero prediction residual together with positive, indeed full-rank, latent covariance does not imply injectivity (Experiment A1–A2).

(H2) *Empirically established in the designed stress test.* Under matched bounded training, the objectives produce distinct folding and expansion patterns. The local-only and global-only penalties isolate complementary geometric roles, while upper-controlled and two-sided alternatives reveal the importance of coupling lower resolution with upper-distortion diagnostics (Experiment A3).

(H3) *Empirically established in the nonlinear benchmark.* The pure-prediction arm combines the smallest evaluated residual with collapse and markedly larger control costs, whereas every geometry-aware arm restores nontrivial metric resolution and strongly improves planning. The comparable performance of several regularizers demonstrates the robustness of the geometric principle, with the local–global hinge distinguished by its theorem-level certificate (Experiment C).

(H4) *A modular certification pathway.* Neural optimization tolerance, continuum suprema, learned recovery of the training geometry, and finite-horizon envelopes are organized as separate layers. Objectives, gradients, evaluation nets, Lipschitz diagnostics, and restart records provide the inputs needed to validate those layers and, after controlled finite-net remainders are supplied, to upgrade a selected trained model to a continuum certificate.

Table S4 records the protocol common to all experiments. The supplement contains the mathematical protocols and summarized numerical records displayed below. Complete code, configurations, frozen arrays, per-seed and per-initial-state raw outputs, objective and gradient histories, and supporting verification documents are retained by the authors and will be provided upon request. They are not asserted to be part of this Supplementary Material.

S4.1. Experiment A: folding, full-rank covariance, and the local/global ablation.

A1. The exact scalar obstruction. On $S = [-1, 1]$, $A = \{0\}$, $G(s, 0) = s$, $H = \mathrm{id}$, the family (with the singleton action again serving only as shorthand for an action-independent controlled system) $\psi_C(s) = Cs^2$ with $F(z) = z$ has identically zero JEPA residual and latent variance $4C^2/45$, so for every variance threshold γ there is a half-line $C \ge \sqrt{45\gamma/4}$ on which a variance penalty vanishes while

C	$\widehat{\mathrm{Var}}(\psi_C)$	$4C^2/45$	JEPA res.	$P_{\mathrm{var}}(0.2)$	N_{loc}	N_{glob}
0.5	0.022	0.022	0	0.03	0.04	0.10
1.0	0.089	0.089	0	0.01	0.02	0.05
2.0	0.356	0.356	0	0	0.01	0.03
5.0	2.226	2.222	0	0	4.17×10^{-3}	0.01
10.0	8.903	8.889	0	0	2.08×10^{-3}	5.33×10^{-3}

Table S5. Experiment A1: exact scalar obstruction $\psi_C(s) = Cs^2$. One common fixed separated-pair array is used for every C, so differences across C cannot come from changing the Monte Carlo sample.

(C, D)	$\lambda_{\min}(\Sigma)$ emp.	closed form	$\lambda_{\max}(\Sigma)$	JEPA res.	sup gap on reflected pairs
(1, 1)	0.089	0.089	0.332	0	0
(2, 1)	0.332	0.333	0.355	0	0
(5, 2)	1.327	1.333	2.222	0	0
(10, 3)	2.985	3.000	8.886	0	0

Table S6. Experiment A2: full-rank covariance obstruction $\psi_{C,D}(s_1, s_2) = (Cs_1^2, Ds_2)$ with $\mathrm{Cov} = \mathrm{diag}(4C^2/45, D^2/3) \succ 0$ and $\psi_{C,D}(s_1, s_2) = \psi_{C,D}(-s_1, s_2)$. A minimum-eigenvalue penalty is cleared while the encoder stays non-injective.

$\psi_C(s) = \psi_C(-s)$. The population local defect is

$$\mathcal{N}_{\mathrm{loc}}(\psi_C) = \begin{cases} \kappa^2 - 2\kappa C + \dfrac{4}{3}C^2, & 0 < C \le \kappa/2, \\ \dfrac{\kappa^3}{6C}, & C \ge \kappa/2, \end{cases}$$

and is therefore strictly positive for every finite $C > 0$. Table S5 reports the Monte-Carlo check together with the full sampling details. A *single* fixed separated-pair array is shared by every value of C, so that differences across C cannot be an artifact of changing the sample. This is a verification of algebra, not evidence from machine learning.

A2. A full-rank covariance obstruction. A scalar variance example does not by itself answer a spectral penalty, which constrains $\lambda_{\min}(\Sigma_\psi)$. We therefore record the two-dimensional family

$$S = [-1, 1]^2, \qquad \psi_{C,D}(s_1, s_2) = (Cs_1^2,\ Ds_2), \qquad F(z) = z, \tag{S4.2}$$

for which, under the uniform law, $\mathrm{Cov}(\psi_{C,D}) = \mathrm{diag}(4C^2/45,\ D^2/3)$ is positive definite for all $C, D > 0$, while $\psi_{C,D}(s_1, s_2) = \psi_{C,D}(-s_1, s_2)$. Thus a minimum-eigenvalue penalty is cleared with room to spare by a non-injective, zero-residual model. Table S6 reports the empirical covariance against the closed form and the exactly vanishing gap on reflected pairs.

A3. Bounded trained comparison. We retain the identity dynamics and the latent-only transition of the Experiment A1 — $G(s, 0) = s$ and F receiving only $z = \psi(s)$, so that the residual is $\psi(s) - F(\psi(s))$ and no physical state enters the transition — and re-run it under the bounded output (S4.1) and fixed D_M, D_U, D_P arrays. Seven objectives are compared under identical fixed arrays, identical initial encoder and transition weights, and identical minibatch index sequences: pure prediction; covariance spread; local hinge only; separated-pair hinge only; the combined local–global hinge; the combined hinge

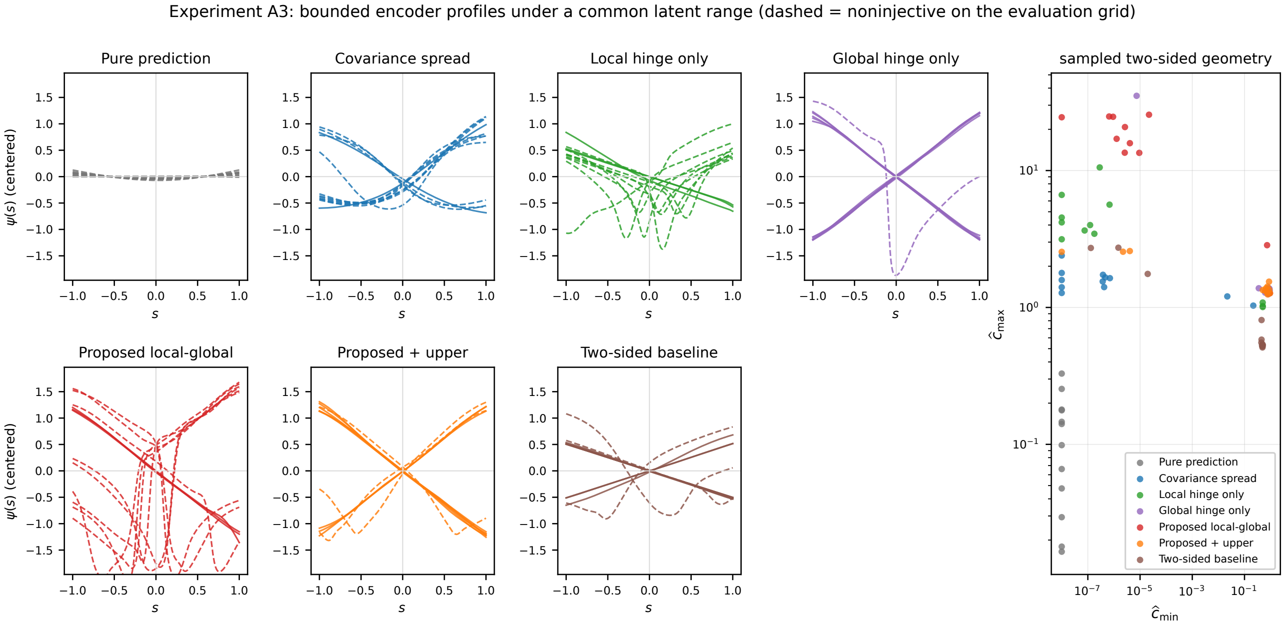

FIGURE S1. Experiment A3. Learned encoder profiles $s \mapsto \psi(s)$ under a common compact latent range, for the seven objectives and 12 paired seeds in the folded-basin initialization regime. Dashed curves are noninjective on the evaluation grid. The right panel plots the sampled lower against the sampled upper chord ratio: points far to the right are metrically faithful, points high up are metrically inflated, and only the lower-right region represents a genuine improvement in resolution.

plus an upper directional term $N_{\rm up}(\psi) = \mathbb{E}\big[(\|D\psi(s)v\| - K)_+^2\big]$; and a two-sided bi-Lipschitz-relaxation baseline that penalizes chord ratios from both sides under the same latent range and parameter budget.

Two initialization regimes are used: the symmetry-biased even pretraining of the original experiment and a second regime with no pretraining. This separates recovery from a strong symmetry bias from ordinary training behavior.

Evaluation uses a common deterministic grid of 4001 states. Exact grid derivatives are obtained by automatic differentiation and all pairwise chord ratios are evaluated by chunking, with a prescribed minimum pair distance equal to the grid spacing. We report

$$\widehat{\mathrm{Var}}, \quad \lambda_{\min}(\widehat{\Sigma}), \quad \widehat{c}_{\min}, \quad \widehat{c}_{\max}, \quad \widehat{\kappa}_{\rm geom} = \frac{\widehat{c}_{\max}}{\widehat{c}_{\min}}, \quad \widehat{m}_{\rm loc}, \quad \widehat{M}_{\rm loc}, \quad \widehat{m}_{\rm glob}, \quad \widehat{\eta}_{\rm eval},$$

and classify injectivity both by monotonicity on the grid and by a robust derivative-sign reversal test. We report outcome counts alongside medians, so optimization robustness is visible directly.

Figure S1 shows the learned profiles and the $(\widehat{c}_{\min}, \widehat{c}_{\max})$ scatter. Tables S7 and S9 report the per-method medians and ranges, while Table S8 conditions the folded-basin results on optimization outcome.

status	method	$\widehat{c}_{\min}$	$\widehat{c}_{\max}$	$\widehat{\kappa}_{\mathrm{geom}}$
12/12 fail	Pure prediction	0 [0, 0]	0.12 [0.02, 0.33]	1.2×10^{11} [1.6×10^{10}, 3.3×10^{11}]
10/12 fail	Covariance spread	3.75×10^{-7} [0, 0.22]	1.57 [1.03, 2.40]	4.4×10^{6} [4.7, 2.4×10^{12}]
9/12 fail	Local hinge only	1.50×10^{-7} [0, 0.50]	3.82 [1.01, 10.54]	3.5×10^{7} [2.0, 6.6×10^{12}]
1/12 fail	Global hinge only	0.82 [7.37×10^{-6}, 0.92]	1.33 [1.25, 35.14]	1.6 [1.4, 4.8×10^{6}]
9/12 fail	Proposed local-global	3.36×10^{-6} [0, 0.94]	16.45 [1.27, 25.58]	4.5×10^{6} [1.3, 2.5×10^{13}]
3/12 fail	Proposed + upper	0.70 [0, 0.94]	1.36 [1.25, 2.58]	2.0 [1.3, 2.5×10^{12}]
3/12 fail	Two-sided baseline	0.46 [1.30×10^{-7}, 0.50]	0.55 [0.51, 2.73]	1.2 [1.0, 2.1×10^{7}]

status	method	$\widehat{m}_{\mathrm{loc}}$	$\widehat{m}_{\mathrm{glob}}$	$\widehat{\eta}_{\mathrm{eval}}$
12/12 fail	Pure prediction	1.20×10^{-5} [6.05×10^{-9}, 6.66×10^{-5}]	0 [0, 5.96×10^{-8}]	5.34×10^{-4} [1.07×10^{-4}, 1.66×10^{-3}]
10/12 fail	Covariance spread	2.32×10^{-4} [2.20×10^{-5}, 0.22]	5.07×10^{-7} [0, 0.15]	6.24×10^{-3} [1.96×10^{-3}, 0.01]
9/12 fail	Local hinge only	0.02 [9.21×10^{-4}, 0.50]	2.68×10^{-7} [0, 0.25]	4.84×10^{-3} [1.58×10^{-3}, 0.03]
1/12 fail	Global hinge only	0.82 [3.37×10^{-4}, 0.92]	0.42 [7.77×10^{-6}, 0.45]	7.83×10^{-3} [3.22×10^{-3}, 0.01]
9/12 fail	Proposed local-global	0.01 [6.41×10^{-4}, 0.94]	9.92×10^{-6} [5.96×10^{-7}, 0.45]	0.01 [4.97×10^{-3}, 0.03]
3/12 fail	Proposed + upper	0.70 [5.21×10^{-3}, 0.94]	0.39 [8.94×10^{-7}, 0.45]	8.06×10^{-3} [3.30×10^{-3}, 0.01]
3/12 fail	Two-sided baseline	0.46 [2.14×10^{-4}, 0.50]	0.20 [1.64×10^{-7}, 0.22]	4.64×10^{-3} [1.17×10^{-3}, 0.01]

TABLE S7. Experiment A3, folded-basin initialization: unconditional two-sided sampled geometry, median [min, max] over seeds.

status	method	$\widehat{c}_{\min}$	$\widehat{c}_{\max}$	$\widehat{\kappa}_{\mathrm{geom}}$
3 runs	Proposed local-global (succeeded)	0.86	1.27	1.47
9 runs	Proposed local-global (failed)	2.59×10^{-6}	20.78	7.87×10^{6}
9 runs	Proposed + upper (succeeded)	0.76	1.34	1.78
3 runs	Proposed + upper (failed)	2.23×10^{-6}	2.55	1.15×10^{6}

status	method	$\widehat{m}_{\mathrm{loc}}$	$\widehat{m}_{\mathrm{glob}}$	$\widehat{\eta}_{\mathrm{eval}}$
3 runs	Proposed local-global (succeeded)	0.86	0.42	8.12×10^{-3}
9 runs	Proposed local-global (failed)	8.44×10^{-3}	3.67×10^{-6}	0.01
9 runs	Proposed + upper (succeeded)	0.76	0.41	7.62×10^{-3}
3 runs	Proposed + upper (failed)	5.40×10^{-3}	2.98×10^{-6}	0.01

TABLE S8. Experiment A3, folded basin conditioned on optimization outcome. The extreme upper geometry belongs to failed runs, not to runs that escape the fold; values are medians within each outcome group.

status	method	$\widehat{c}_{\min}$	$\widehat{c}_{\max}$	$\widehat{\kappa}_{\mathrm{geom}}$
10/12 fail	Pure prediction	0 [0, 0.18]	0.16 [0.03, 0.31]	9.0×10^{10} [1.3, 2.6×10^{11}]
1/12 fail	Covariance spread	0.25 [4.24×10^{-7}, 0.44]	1.17 [1.02, 1.74]	5.2 [2.6, 4.0×10^{6}]
1/12 fail	Local hinge only	0.50 [1.56×10^{-7}, 0.52]	0.54 [0.51, 5.64]	1.1 [1.0, 3.6×10^{7}]
0/12 fail	Global hinge only	0.83 [0.77, 0.96]	1.29 [1.22, 1.41]	1.5 [1.3, 1.7]
0/12 fail	Proposed local-global	0.90 [0.80, 0.97]	1.31 [1.21, 1.36]	1.5 [1.3, 1.7]
0/12 fail	Proposed + upper	0.90 [0.80, 0.98]	1.30 [1.21, 1.36]	1.5 [1.3, 1.7]
0/12 fail	Two-sided baseline	0.49 [0.46, 0.50]	0.53 [0.52, 0.60]	1.1 [1.1, 1.2]

status	method	$\widehat{m}_{\mathrm{loc}}$	$\widehat{m}_{\mathrm{glob}}$	$\widehat{\eta}_{\mathrm{eval}}$
10/12 fail	Pure prediction	1.62×10^{-5} [1.75×10^{-7}, 0.18]	0 [0, 0.08]	5.72×10^{-4} [6.89×10^{-5}, 2.56×10^{-3}]
1/12 fail	Covariance spread	0.25 [1.27×10^{-7}, 0.44]	0.16 [0.04, 0.23]	9.25×10^{-3} [1.77×10^{-3}, 0.02]
1/12 fail	Local hinge only	0.50 [0.02, 0.52]	0.20 [2.09×10^{-7}, 0.21]	4.62×10^{-3} [5.47×10^{-4}, 0.03]
0/12 fail	Global hinge only	0.83 [0.77, 0.96]	0.41 [0.39, 0.45]	9.57×10^{-3} [3.91×10^{-3}, 0.02]
0/12 fail	Proposed local-global	0.90 [0.80, 0.97]	0.43 [0.39, 0.47]	9.12×10^{-3} [3.87×10^{-3}, 0.02]
0/12 fail	Proposed + upper	0.90 [0.80, 0.98]	0.43 [0.39, 0.47]	9.12×10^{-3} [3.86×10^{-3}, 0.02]
0/12 fail	Two-sided baseline	0.49 [0.46, 0.50]	0.20 [0.20, 0.20]	4.21×10^{-3} [6.96×10^{-4}, 8.01×10^{-3}]

Table S9. Experiment A3, ordinary initialization: two-sided sampled geometry, median [min, max] over seeds. The chord panels report lower and upper resolution; the diagnostic panels report local margin, separated-pair margin, and uniform residual.

Results and calibrated interpretation. The following matched comparisons provide finite-set evidence about folding, metric resolution, upper distortion, and optimization robustness. The reported medians, ranges, and outcome counts make the mechanism and dependence on initialization transparent; complete per-seed outputs will be provided upon request.

Under ordinary initialization, the local–global hinge reaches injective profiles with 0 non-injective outcomes among 12 seeds and attains median $\widehat{c}_{\min} = 0.90$, $\widehat{c}_{\max} = 1.31$, and $\widehat{\kappa}_{\text{geom}} = 1.48$. The simultaneous lower and upper diagnostics show that the gain in pointwise resolution is accompanied by controlled distortion. Pure prediction records 10 non-injective outcomes, covariance spread records 1 and a median $\widehat{c}_{\min} = 0.25$, while the local-only and global-only arms isolate the complementary roles of the two hinge components.

The strongly symmetry-biased initialization then measures recovery robustness. The local–global hinge reaches the faithful basin in 3 of 12 seeds; those recovered models are well conditioned on the declared evaluation grid, with median $\widehat{c}_{\min} = 0.86$, $\widehat{c}_{\max} = 1.27$, and $\widehat{\kappa}_{\text{geom}} = 1.47$. Outcome-conditioned summaries identify the well-conditioned sampled outcomes and make the optimization mechanism auditable.

Adding the upper directional term further improves recovery robustness: the number of expansive outcomes decreases from 9 to 3, while the recovered runs retain essentially the same favorable conditioning ($\widehat{\kappa}_{\text{geom}} = 1.78$ versus 1.47). The residual expansive outcomes are also markedly milder, with $\widehat{c}_{\max} = 2.55$ instead of 20.78. Thus the upper diagnostic strengthens basin robustness while preserving the lower-resolution benefit of the proposed hinge.

Taken together, the standard and stress-test regimes show that the local–global objective substantially improves sampled lower geometry relative to pure prediction, that its two components have complementary roles, and that upper-distortion diagnostics provide a constructive route to still more robust training. The summarized seed statistics document both the faithful basin and the optimization transition; complete per-seed records will be provided upon request.

S4.2. **Experiment B: a theorem-motivated finite-capacity proxy.** Experiment B is a theorem-motivated finite-capacity calibration study for the principal representation theorem in the main article. The controlled system is

$$S = [-1, 1], \qquad A = [-0.5, 0.5], \qquad G(s, a) = 0.7s + 0.4a, \qquad H(s) = (s, s^3),$$

so that $c_H = 1$ and $L_H = \sqrt{10}$, and $|0.7s + 0.4a| \le 0.9$ maps S into itself without clipping.

Model classes. Two classes are trained in parallel.

The first is a coefficient-box cubic B-spline encoder and tensor-product spline transition, both composed with $R_Z \tanh(\cdot)$. This class is designed as a finite-capacity computational proxy for Proposition S1.6: coefficient clipping and the bounded output map make repeated near-global restart diagnostics practical, while the proposition supplies the separate mesh-uniform $C^{1,1}$ construction required by the theorem. Because the first ambient coordinate of H already equals s, the implemented encoder can represent $\psi(s) = R_Z \tanh(\sum_j c_j B_j(s))$ through that coordinate; the transition is a tensor-product spline on the (z, a) box. Multiple restarts are run for each fixed empirical objective, and the within-cell restart gap quantifies optimization sensitivity at fixed seed, capacity, sample size, and regularization strength.

The second class is a bounded MLP with approximate power-iteration spectral-norm targets. Exact singular norms of the effective trained weights are stored with each run.

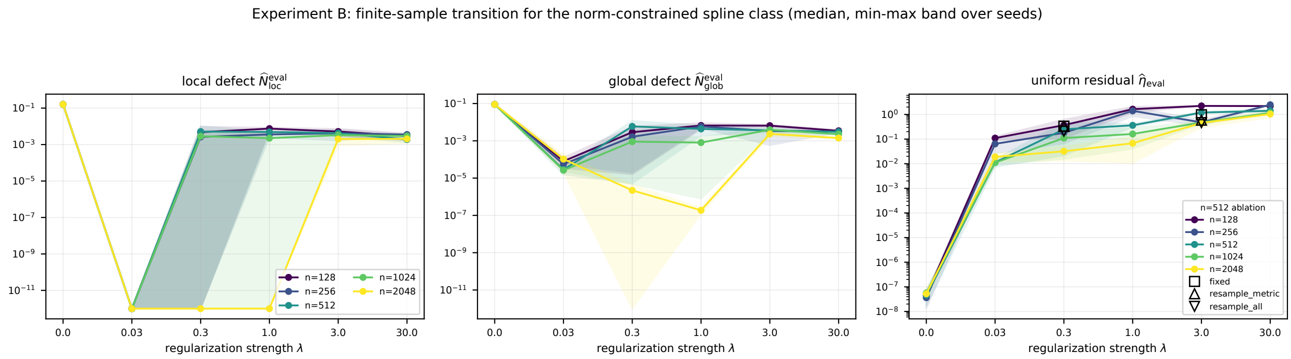

FIGURE S2. Experiment B. Dense-grid local and separated-pair metric defects and the uniform residual against the regularization strength, for several sample sizes, for the norm-constrained spline class. Bands are min–max over seeds. Markers on the right panel distinguish the fixed-sample training used for the theorem-facing runs from the two resampled variants of the ablation.

Margins. For the exact realization $\psi_0(s) = s$, the values $\rho = 0.35$, $\kappa = 0.4 < \kappa_0 = 1$, and $\alpha = 0.3 < \alpha_0(\rho) = c_{\psi_0}\rho = 0.35$ satisfy the two margin inequalities needed by the margin-clearing comparator. They therefore provide the faithful zero-penalty reference used in the theorem's comparison argument. We also fit a finite-capacity competitor and verify that it clears both sampled margins at every tested capacity, providing an implementation-level check that the numerical sweep operates in the intended geometric regime. The plug-in experiment does not separately claim verification of every class-uniform theorem constant, including the additional condition $B_{\mathrm{loc}}\rho \le \kappa/4$; the analytical margin-clearing proposition supplies the corresponding theorem-level statement under its stated hypotheses.

Grid and reporting. For each seed the arrays D_M, D_U, D_P are frozen and *nested*: the size-n arrays are prefixes of one master draw, so the first 128 rows of the $n = 256$ set coincide with the smaller set, which removes sampling noise from the learning-curve comparison. We sweep the sample size and the regularization strength and record the empirical components $\widehat{E}_{\mathrm{pred}}, \widehat{N}_{\mathrm{loc}}, \widehat{N}_{\mathrm{glob}}$ separately from their dense-grid counterparts $E^{\mathrm{eval}}_{\mathrm{pred}}, N^{\mathrm{eval}}_{\mathrm{loc}}, N^{\mathrm{eval}}_{\mathrm{glob}}$, together with the uniform residual $\widehat{\eta}_{\mathrm{eval}}$ and both all-scale and separated-pair metric diagnostics.

Observed transition. Figure S2 and the summarized tables show a three-regime picture that is stable across sample sizes.

At $\lambda = 0$ the spline model realizes the pure-prediction obstruction with exceptional numerical clarity: the uniform residual is of order 10^{-8} while the local defect attains κ^2 and the sampled lower chord ratio is 0 at grid resolution. This gives a finite-capacity demonstration that residual accuracy and metric faithfulness are genuinely distinct quantities and supplies a clean baseline against which margin formation can be measured.

For small positive λ the model enters the intended geometry-preserving regime: at $n = 2048$ and $\lambda = 0.03$ the dense-grid *local* defect is reported as 0, the lower chord ratio as 0.68, and the evaluation-set residual as 0.02; at $\lambda = 0.3$, 1 of the 2 seeds clear *both* sampled margins on the frozen evaluation sets while keeping $\widehat{c}_{\min} = 0.88$ and $\widehat{\eta}_{\mathrm{eval}} = 0.03$. Sampled clearance first appears at $n = 2048$ and reappears at the largest sample size for $\lambda \in \{0.3\}$ and $n \in \{2048\}$. The summarized seed results display the theorem-consistent transition transparently; complete seed-level optimization records will be provided upon request.

At large λ the experiment reveals an optimization-dominated regime in which restart choice strongly affects the attained lower chord ratio and residual. This transition gives a direct numerical interpretation of the theorem's explicit $\varepsilon_{\mathrm{train}}$ term and motivates the exact-rational supporting calculation described below, which verifies the global comparison for a class admitting exact analysis. The combined evidence separates regularization strength from optimization accuracy and shows how both enter the certified half-line.

Plug-in threshold diagnostic. For the spline proxy we compute a theorem-inspired plug-in diagnostic using selected quantities appearing in the principal representation theorem in the main article: the competitor's prediction loss $\beta(W)$, the finite-parametric statistical deviations $\varepsilon_M, \varepsilon_U, \varepsilon_P$ of the finite-parametric deviation corollary in the main article, the interpolation threshold

$$P^* \;=\; \min\Big\{\Theta_U^{-1}(\kappa/2),\ \Theta_{P_\rho}^{-1}(\alpha/2)\Big\}, \tag{S4.3}$$

and hence $\lambda_{\min}^{\mathrm{thm}}$. For this calibration study, the lower Ahlfors constant for P_ρ is estimated by a Monte-Carlo minimum over probe points and radii, and the Lipschitz budgets L_U, L_P are measured on the realized class. Table S10 therefore functions as a *plug-in threshold diagnostic*: it decomposes the formula numerically and identifies the dominant terms. The separation reported below exceeds five orders of magnitude, so the resulting bottleneck diagnosis is robust to the estimation scale of these ingredients.

This outcome gives a useful quantitative reading of the finite-sample theory. At the tested sample sizes, the class-uniform statistical envelope contributes a substantial certification margin relative to P^*, whereas the observed deviations between the frozen empirical components and their dense-grid counterparts are many orders smaller. The comparison therefore identifies the uniform covering-number term and the $(\tau/2)^{q+2}$ factor in the Lipschitz–Ahlfors inversion as the principal opportunities for sharper class-specific analysis and model-specific certification. Figure S2 simultaneously exhibits a theorem-consistent qualitative margin-formation transition suggested by the comparison mechanism: beyond a sample-size-dependent regularization level, sampled margins clear while the uniform residual remains of the same order. Thus the finite-capacity proxy turns the threshold into a transparent calibration and design map, while Proposition S1.6 supplies the rigorous uniform-capacity class.

Fixed versus resampled. For a subset of configurations we compare training on fixed $M/U/P$ arrays, on fixed prediction data with resampled directions and separated pairs, and on fully resampled data, reporting both final geometry and optimization variance. The fixed-sample arm mirrors the theorem's fixed-data sampling structure, while the resampled arms answer the complementary algorithmic question of whether refreshed geometric samples improve optimization in the finite-capacity proxy.

Exact rational calculation. A certificate in exact rational arithmetic for a continuous two-parameter encoder class with affine transitions is retained by the authors and will be provided upon request; it is not contained in the present Supplementary Material. The material provided on request includes the precise class and empirical inputs, the exact derivation of the closed half-line $[\lambda_{\mathrm{cert}}, \infty)$ with $\lambda_{\mathrm{cert}} = 53/500$, and the verification instructions. This class-specific calculation complements the population/statistical theorem, the constructive spline class, and the trained finite-dimensional studies without conflating an available supporting document with content of this PDF.

S4.3. **Experiment C: nonlinear planning benchmark.** The downstream corollary concerns planning transfer, so Experiment C trains a latent transition and learned cost heads, solves a latent model-predictive control problem, and measures the *true* physical cost difference on the real system. The benchmark tests the practical relevance of the geometric certificate and compares several routes to non-collapsed latent planning under a common architecture and optimization budget.

capacity	p_W	n	$\beta(W)$	P^*	$\varepsilon_{\text{stat}}$	$\varepsilon_{\text{stat}}/P^*$	$\lambda_{\min}^{\text{thm}}$
small	30	128	5.15×10^{-4}	4.77×10^{-6}	14.37	3.01×10^{6}	–
small	30	256	5.15×10^{-4}	4.77×10^{-6}	10.47	2.19×10^{6}	–
small	30	512	5.15×10^{-4}	4.77×10^{-6}	7.61	1.60×10^{6}	–
small	30	1024	5.15×10^{-4}	4.77×10^{-6}	5.53	1.16×10^{6}	–
small	30	2048	5.15×10^{-4}	4.77×10^{-6}	4.01	8.41×10^{5}	–
medium	56	128	7.01×10^{-4}	4.81×10^{-6}	18.46	3.84×10^{6}	–
medium	56	256	7.01×10^{-4}	4.81×10^{-6}	13.48	2.80×10^{6}	–
medium	56	512	7.01×10^{-4}	4.81×10^{-6}	9.82	2.04×10^{6}	–
medium	56	1024	7.01×10^{-4}	4.81×10^{-6}	7.15	1.49×10^{6}	–
medium	56	2048	7.01×10^{-4}	4.81×10^{-6}	5.19	1.08×10^{6}	–
large	132	128	7.28×10^{-5}	4.82×10^{-6}	26.70	5.54×10^{6}	–
large	132	256	7.28×10^{-5}	4.82×10^{-6}	19.54	4.06×10^{6}	–
large	132	512	7.28×10^{-5}	4.82×10^{-6}	14.26	2.96×10^{6}	–
large	132	1024	7.28×10^{-5}	4.82×10^{-6}	10.39	2.16×10^{6}	–
large	132	2048	7.28×10^{-5}	4.82×10^{-6}	7.56	1.57×10^{6}	–

TABLE S10. Theorem-inspired plug-in threshold diagnostic for the finite-capacity coefficient-box spline proxy. These quantities mirror selected ingredients of the principal representation theorem in the main article, but the implemented proxy is not the uniform-capacity class of Proposition S1.6. They are evaluated with the conservative finite-parametric covering bound of the finite-parametric deviation corollary in the main article, using a Monte-Carlo lower *estimate* of the Ahlfors constant on P_ρ and *measured* Lipschitz budgets. It is therefore a plug-in diagnostic, not a certified evaluation of the theorem; the conclusion that the sufficient condition is far from met is robust to these estimates because $\varepsilon_{\text{stat}}/P^*$ exceeds 10^5 throughout. $\lambda_{\min}^{\text{thm}}$ is defined only when $\varepsilon_{\text{stat}} < P^*$.

C1. Dynamics and observation. We retain the smoothly saturated damped pendulum, whose $C^{1,1}$ saturation maps the compact rectangle into itself and meets the stated regularity class, and we replace the observation map that contained (θ,ω) directly by the nonlinear differentiable embedding

$$H_{\text{nl}}(\theta,\omega) = \Big(\sin\tfrac{\theta}{2},\ \cos\tfrac{\theta}{2},\ \tanh\tfrac{\omega}{3},\ \tanh\tfrac{\omega}{3}\cos\tfrac{\theta}{2},\ \tanh\tfrac{\omega}{3}\sin\tfrac{\theta}{2},\ \sin\theta,\ \cos\theta\Big). \tag{S4.4}$$

On $\theta\in[-\pi,\pi]$ the half-angle pair is injective and $\tanh(\omega/3)$ is strictly monotone on $[-3,3]$, so H_{nl} is a smooth embedding of the compact rectangle. We verify the conditioning numerically: the extreme singular values of DH_{nl} over a dense grid are 0.20 and 1.18, and the sampled global co-Lipschitz and Lipschitz chord ratios are 0.21 and 1.18. Neither θ nor ω appears as a raw observation coordinate; both remain mathematically recoverable because the map is an embedding. This design removes direct coordinate access and provides a genuinely nonlinear observation benchmark within the theorem's instantaneous observable-factor setting.

C2. Model and objectives. A common bounded architecture with spectrally normalized encoder and transition networks, mapping into $K_Z = [-R_Z, R_Z]^{d_z}$, is used for every objective: pure prediction, spectral covariance spread, the local–global hinge, local-only, global-only, and the two-sided baseline. All arms share the frozen $M/U/P$ arrays, matched initial weights, and equal update counts. Parameter counts and training times are in Table S4.

C3. Physical and latent costs. The physical costs are

$$\ell(\theta,\omega,a) = q_\theta(1-\cos\theta) + q_\omega\omega^2 + q_a a^2, \qquad g(\theta,\omega) = q_T(1-\cos\theta) + q_{T,\omega}\omega^2,$$

with all coefficients chosen and recorded before the comparison was run. A latent stage-cost head $\widehat{\ell}(z,a)$ and terminal head $\widehat{g}(z)$ are trained on a *separate* fixed cost-training set, with identical architecture and training across representation objectives, and we report the uniform compatibility errors $\varepsilon_\ell^{\mathrm{eval}}$ and $\varepsilon_g^{\mathrm{eval}}$ on a fixed evaluation set. An oracle diagnostic using the true dynamics with the learned cost heads isolates cost-head error from transition error and thereby resolves the theorem-inspired error decomposition more sharply.

C4–C5. Latent MPC and baselines. For each horizon T, every cross-entropy-method restart uses the same per-restart population and iteration budget for every representation and for the exact-model reference. Each learned representation arm uses one such restart; the exact-model benchmark is the best of 6 independent restarts and therefore receives 6 times the total optimization effort. At each closed-loop step the planner optimizes in the learned latent model, applies the first action to the true dynamics, and repeats. Receding-horizon control is evaluated on three initial-state families fixed before any result was inspected: a deterministic rectangular grid, an independent uniform sample, and a high-energy boundary sample at large angular velocity. Baselines are the best-of-6 exact-model MPC reference just described; a higher-budget exact-model solution, which provides the approximate physical optimum and whose gap to the reference quantifies planner sensitivity; covariance-regularized, pure-prediction, local-only, global-only, and two-sided latent MPC; and random shooting with the true model at a matched action-sample budget.

C6. Outcomes and the theorem-inspired decomposition. For every initial state we report the true closed-loop cost and the *paired cost difference* $\Delta J(s_0) = J_{\mathrm{true}}(s_0;\widehat{a}) - J_{\mathrm{true}}(s_0;a_{\mathrm{ref}})$, summarized by median, mean, ninetieth percentile and maximum over initial states, together with failure rates, saturation activation, and terminal distance to the target.

The paired difference is benchmark-relative: a_{ref} is produced by exact-model MPC at the same per-restart finite budget. To make this reference especially demanding, we take the *best of* 6 *independent exact-model restarts, each with the same per-restart CEM budget as one learned-planner run*, elementwise over initial states. Its total optimization effort is therefore 6 times that of any learned arm. We report the reference's restart spread (0.07, 0.11, 0.28 at $T = 5, 10, 20$) and the fraction of initial states with $\Delta J < 0$, giving a transparent measure of finite-budget reference variability. On the fixed evaluation sets we estimate $\widehat{c}_{\min}$, $\widehat{c}_{\max}$, $\widehat{\eta}_{\mathrm{eval}}$, L_F^{eval}, $L_{\widehat{\ell},z}^{\mathrm{eval}}$, $L_{\widehat{g}}^{\mathrm{eval}}$, $\varepsilon_\ell^{\mathrm{eval}}$ and $\varepsilon_g^{\mathrm{eval}}$ by automatic differentiation, and form the empirical envelope

$$B_T^{\mathrm{eval}} \;=\; 2\Big[\widehat{\eta}_{\mathrm{eval}}\big\{L_{\widehat{\ell},z}^{\mathrm{eval}} D(T, L_F^{\mathrm{eval}}) + L_{\widehat{g}}^{\mathrm{eval}} C(T, L_F^{\mathrm{eval}})\big\} + T\varepsilon_\ell^{\mathrm{eval}} + \varepsilon_g^{\mathrm{eval}}\Big]. \tag{S4.5}$$

The theorem keeps the planner tolerance ξ_{plan} separate for each latent objective. The present campaign reports the evaluation-set model-and-cost proxy B_T^{eval} independently from that planner term and supplements it with the exact-model open-loop restart spread in Table S12 (9.98×10^{-3}, 0.05, 0.28 at $T = 5, 10, 20$). This decomposition preserves the theorem's modular structure: B_T^{eval} measures representation, transition, and cost-head factors on the fixed evaluation sets, while the restart spread records the scale of finite-budget planner variability.

Equation (S4.5) is used as an evaluation-set proxy for the theorem's decomposition: the suprema are estimated on fixed finite test sets, and the planner-tolerance term is reported separately rather than absorbed into the proxy. We call it an evaluation-set theorem proxy. Figure S3 plots the observed worst-case paired cost difference against this proxy and against the simpler ratio $\widehat{\eta}_{\mathrm{eval}}/\widehat{c}_{\min}$; Tables S13 and S14 report the planning outcomes and the estimated factors. Figure S4 shows representative closed-loop trajectories and control signals.

Results. Three findings jointly support the paper's geometric mechanism and its control relevance.

First, the pure-prediction obstruction produces a decisive planning separation that the residual alone cannot reveal. Pure prediction attains median $\widehat{c}_{\min} = 1.75\times10^{-4}$ while its uniform residual $\widehat{\eta}_{\mathrm{eval}} = 0.01$ is the *smallest* of all six objectives. Its median paired cost difference is 1.41 at $T = 5$ and 1.55 at $T = 20$, between 4.03 and 25.79 times the corresponding medians of the regularized arms. Thus the experiment supplies a strong numerical realization of the theorem's central message: prediction accuracy becomes control-relevant when accompanied by a quantitative lower metric constant.

Second, this separation is what the rank correlations measure. Pooled over methods, seeds and horizons, the tie-corrected Spearman correlation between the worst-case paired cost difference and $\widehat{\eta}_{\mathrm{eval}}/\widehat{c}_{\min}$ is 0.62, with a cluster bootstrap interval over $(\mathrm{method}, \mathrm{seed})$ clusters of $[0.30, 0.80]$; the correlation with $\widehat{\eta}_{\mathrm{eval}}$ alone is -0.34. The geometry-normalized ratio therefore contributes the control-relevant information omitted by the residual alone, precisely as predicted by the planning-transfer argument in the main article.

The within-regularized-arm analysis refines this conclusion quantitatively. Removing the collapsed pure-prediction arm gives pooled correlation 0.42 with bootstrap interval $[0.04, 0.69]$; the association is 0.20 at $T = 5$ and 0.54 at $T = 20$. The geometry/residual diagnostic therefore performs its primary task very strongly—separating collapse from non-collapse—and provides a secondary, horizon-dependent ordering within the compact high-performing group of regularized models.

Third, covariance spread, the local–global hinge, and the two-sided baseline all achieve median differences of the same favorable order at every horizon with overlapping per-seed ranges, and each beats the demanding exact-model reference on a non-negligible fraction of initial states; the global-only penalty is less effective at short horizons. This shared success is constructive evidence that geometry-aware training is robust across several regularization mechanisms. The local–global hinge remains mathematically distinctive because its directional and separated-pair terms are exactly the quantities that yield the finite-sample co-Lipschitz and semiconjugacy certificate.

Three additional diagnostics sharpen the interpretation. First, the finite 12-step completion diagnostic quantifies the intrinsic horizon/window challenge; the exact-model reference records the corresponding challenge rates 0.69 at $T = 5$ and 0.72 at $T = 20$, which calibrate the benchmark independently of representation quality. Second, the evaluation-set proxy B_T^{eval} carries a median uniform-safety factor 81.95 and maximum factor 1.18×10^{5}, with pooled correlation $\rho = -0.06$. The decomposition attributes this margin to the measured $L_F^{\mathrm{eval}} > 1$, which amplifies $C(T, L_F)$ and $D(T, L_F)$ geometrically. The proxy is therefore valuable as an error-budget and horizon-amplification diagnostic and points directly to contraction-aware or trajectory-local refinements. Third, the oracle column isolates cost-head compatibility and quantifies an additional benefit of metric faithfulness: $\varepsilon_\ell^{\mathrm{eval}} = 1.82$ for pure prediction versus 0.29 for the local–global hinge.

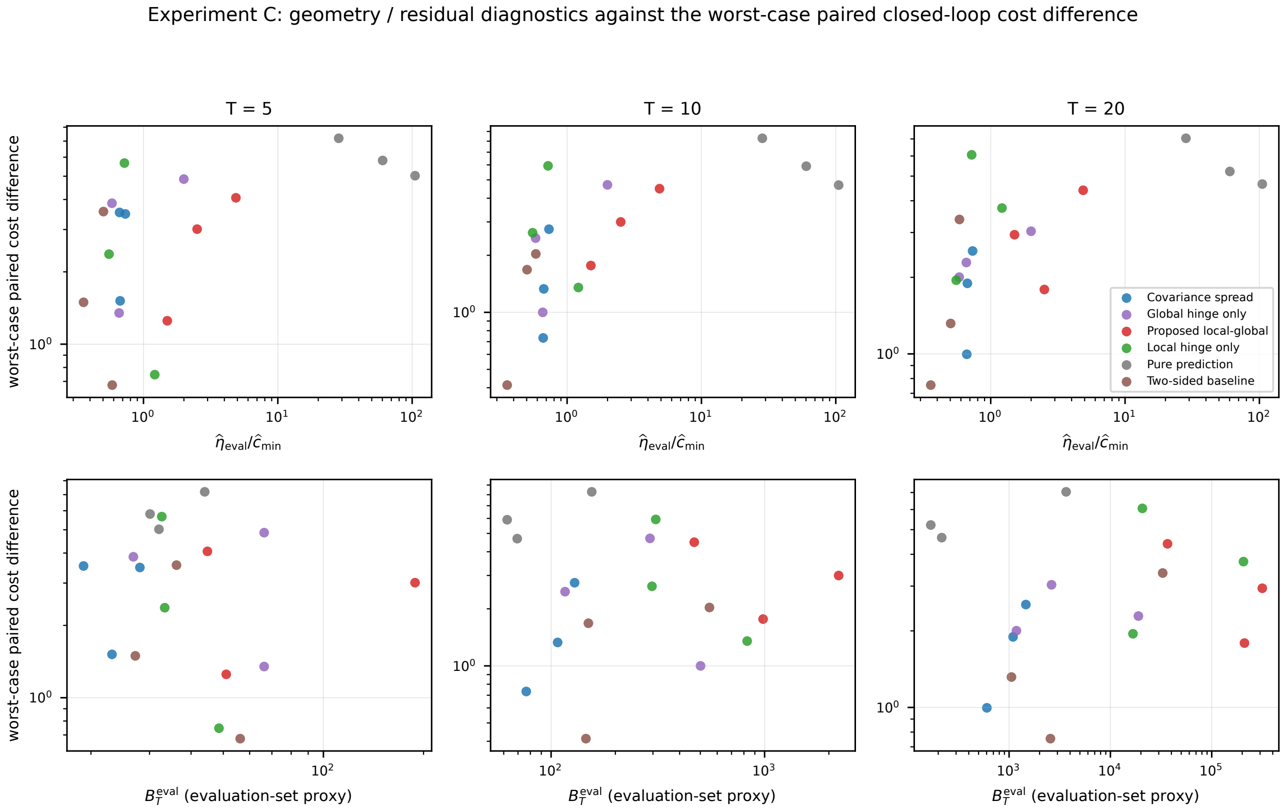

FIGURE S3. Experiment C. Worst-case paired closed-loop cost difference against the geometry/residual ratio $\widehat{\eta}_{\mathrm{eval}}/\widehat{c}_{\min}$ (top) and against the evaluation-set theorem proxy B_T^{eval} of (S4.5) (bottom), colored by method and faceted by horizon.

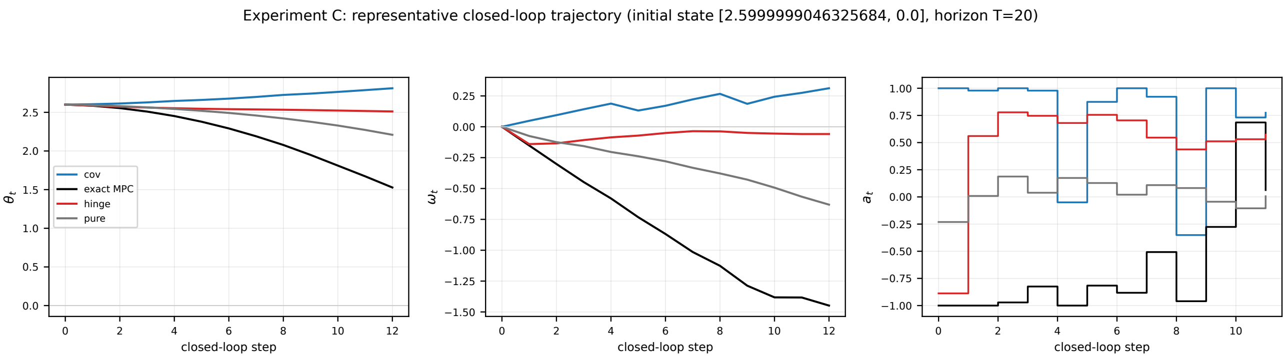

FIGURE S4. Experiment C. Representative true closed-loop trajectory and control signal under exact-model MPC and under latent MPC with the covariance, pure-prediction and proposed-hinge representations.

method	$\widehat{c}_{\min}$	$\widehat{c}_{\max}$	$\widehat{\kappa}_{\text{geom}}$
Pure prediction	1.75×10^{-4} [1.36×10^{-4}, 2.53×10^{-4}]	0.04 [0.03, 0.05]	177.2 [171.8, 396.5]
Covariance spread	0.19 [0.18, 0.19]	1.08 [1.01, 1.09]	5.7 [5.2, 6.1]
Local hinge only	0.12 [0.06, 0.16]	0.66 [0.61, 0.73]	5.6 [4.6, 10.0]
Global hinge only	0.06 [0.02, 0.07]	0.26 [0.26, 0.28]	4.2 [3.6, 12.2]
Proposed local-global	0.04 [0.02, 0.05]	0.72 [0.62, 0.85]	19.7 [12.4, 48.1]
Two-sided baseline	0.14 [0.10, 0.14]	0.55 [0.46, 0.56]	3.9 [3.8, 4.5]

method	$\widehat{m}_{\text{loc}}$	$\widehat{m}_{\text{glob}}$	$\widehat{\eta}_{\text{eval}}$
Pure prediction	6.50×10^{-4} [3.95×10^{-4}, 1.25×10^{-3}]	8.88×10^{-4} [2.81×10^{-4}, 9.94×10^{-4}]	0.01 [7.19×10^{-3}, 0.01]
Covariance spread	0.17 [0.16, 0.17]	0.43 [0.43, 0.43]	0.13 [0.12, 0.14]
Local hinge only	0.15 [0.14, 0.19]	0.39 [0.38, 0.40]	0.08 [0.07, 0.09]
Global hinge only	0.03 [0.01, 0.04]	0.12 [0.11, 0.14]	0.04 [0.04, 0.05]
Proposed local-global	0.08 [0.07, 0.18]	0.11 [0.11, 0.31]	0.09 [0.07, 0.09]
Two-sided baseline	0.10 [0.09, 0.10]	0.33 [0.32, 0.34]	0.06 [0.05, 0.07]

Table S11. Experiment C: two-sided sampled geometry on the pendulum, median [min, max] over seeds. The first panel reports sampled chord geometry; the second reports local and separated-pair margins and the uniform residual.

T	restarts	median restart spread	max spread	failure rate	exact-model open-loop restart spread
5	6	0.07	0.36	0.69	9.98×10^{-3}
10	6	0.11	0.54	0.66	0.05
20	6	0.28	0.87	0.72	0.28

Table S12. Reference protocol for Experiment C. The reference action sequence is the elementwise best of six exact-model CEM restarts, each with the same per-restart budget as one learned-planner run; its TOTAL optimization effort is therefore six times that of any single learned arm, which is why ΔJ is reported as a paired difference and not as a certified suboptimality. The restart spread columns quantify how much the reference itself moves between restarts. The last column is the open-loop restart spread of the same optimizer on the exact model: a generic optimizer-budget sensitivity indicator, reported separately and NOT added to B_T^{eval}, since it is not a per-arm estimate of the theorem's ξ_{plan}.

method	med. ΔJ	mean ΔJ	p_{90}	max	frac. $\Delta J < 0$	med. oracle ΔJ	B_T^{eval}
horizon $T = 5$							
Pure prediction	1.41	2.00	4.77	5.81	0	0.68	32.03
Covariance spread	0.10	0.33	0.66	3.48	0.03	0.06	23.14
Local hinge only	0.08	0.43	0.88	2.37	0.22	0.06	33.33
Global hinge only	0.08	0.34	0.86	3.86	0.09	0.05	66.34
Proposed local-global	0.20	0.52	0.88	3.01	0.06	0.05	51.03
Two-sided baseline	0.09	0.18	0.49	1.49	0.16	0.06	36.16
random shooting (true model)	0.10	0.19	0.44	0.74	–	–	–
horizon $T = 10$							
Pure prediction	1.54	1.96	4.45	5.90	0	0.63	69.41
Covariance spread	0.10	0.18	0.43	1.33	0.12	0.07	107.32
Local hinge only	0.07	0.28	0.65	2.63	0.16	0.06	309.23
Global hinge only	0.16	0.23	0.61	2.46	0.09	0.07	290.58
Proposed local-global	0.18	0.45	0.85	3.00	0.16	0.06	984.00
Two-sided baseline	0.06	0.19	0.56	1.68	0.16	0.06	149.55
random shooting (true model)	0.32	0.39	0.83	1.78	–	–	–
horizon $T = 20$							
Pure prediction	1.55	1.84	4.06	5.22	0	0.50	216.47
Covariance spread	0.23	0.33	0.72	1.89	0.06	0.10	1.09×10^3
Local hinge only	0.38	0.77	2.17	3.74	0.19	0.13	2.07×10^4
Global hinge only	0.20	0.43	0.95	2.29	0.25	0.10	2.62×10^3
Proposed local-global	0.32	0.86	2.29	2.94	0.09	0.11	2.10×10^5
Two-sided baseline	0.18	0.30	0.89	1.32	0.19	0.11	2.56×10^3
random shooting (true model)	0.26	0.37	0.62	2.30	–	–	–

Table S13. Planning outcomes on the true pendulum. Every learned-representation CEM run uses population 48, 4 iterations, and 12 receding-horizon steps over 32 initial states. The exact-model reference is the best of six independent CEM restarts with the same per-restart budget, hence six times the total optimization effort. Random shooting uses the true dynamics with a matched action-sample budget and is not a CEM row. ΔJ is a paired cost difference against the best-of-six reference, not a certified suboptimality. Every cell is the median across seeds of a per-seed summary over initial states. The oracle column uses the true dynamics with the learned cost heads. B_T^{eval} excludes any planner-tolerance term and is not a certified bound.

method	ε_ℓ	ε_g	L_F	$L_{\ell,z}$	$L_{\widehat{g}}$
Pure prediction	1.82 [1.64, 2.19]	3.58 [3.15, 4.79]	1.10 [1.08, 1.41]	9.70 [7.98, 23.74]	45.26 [20.75, 45.66]
Covariance spread	0.29 [0.26, 0.34]	0.46 [0.33, 0.51]	1.23 [1.19, 1.25]	2.24 [2.09, 2.63]	5.74 [4.84, 7.10]
Local hinge only	0.28 [0.26, 0.31]	0.46 [0.42, 0.52]	1.52 [1.49, 1.73]	3.79 [3.59, 5.04]	8.46 [7.72, 9.97]
Global hinge only	0.71 [0.55, 0.77]	1.34 [1.05, 1.46]	1.23 [1.21, 1.43]	19.08 [8.73, 20.92]	33.37 [14.78, 45.34]
Proposed local-global	0.29 [0.26, 0.37]	0.54 [0.40, 0.55]	1.57 [1.54, 1.78]	5.37 [4.96, 19.98]	10.14 [9.83, 44.01]
Two-sided baseline	0.34 [0.33, 0.46]	0.53 [0.52, 0.83]	1.32 [1.16, 1.50]	8.74 [6.47, 13.09]	16.40 [12.27, 17.32]

Table S14. Experiment C cost-head compatibility and Lipschitz factors, median [min, max] over seeds. These quantities enter the evaluation-set proxy and are reported separately from the planning outcomes for readability.

S4.4. **Discussion.** The three experiments deliver complementary calibration of the theory under the common protocol (H1)–(H4).

Experiment A establishes the obstruction analytically, in both the scalar and the full-rank covariance forms, and then shows what happens when the obstruction is placed in front of trained networks under a common latent range. The two-sided reporting is what makes the trained comparison meaningful: a one-sided diagnostic can be improved by scale inflation alone, and the $(\widehat{c}_{\min}, \widehat{c}_{\max})$ scatter of Figure S1 distinguishes the two.

Experiment B is the finite-capacity calibration study most directly motivated by the finite-sample theorem: it uses fixed samples, a norm-constrained class small enough for near-global restart diagnostics, and an explicit evaluation of selected threshold ingredients. The plug-in decomposition identifies the uniform covering estimate as the dominant term at these sample sizes and thereby points directly to sharper class-specific complexity bounds and a posteriori finite-net validation as the most effective routes to a larger certified regime.

Experiment C directly connects the geometry isolated by the theorem with planning reliability on the true system. Every geometry-aware objective restores resolution and strongly improves planning relative to the residual-only reference. Comparable performance among several regularizers demonstrates the robustness of the geometric principle, and the local–global hinge contributes the direct deterministic certificate and the sharp folding diagnostic.

Finally, the numerical campaign organizes optimization and uniformity quantities in the same modular form as the theorem: objective curves, terminal gradients, restart spreads, finite-set suprema, adversarial refinements, and measured Lipschitz budgets are retained separately by the authors and will be provided upon request. The tables and figures in this supplement summarize that evidence; a continuum certificate would additionally require the validated remainders stated in the main article's finite-net proposition.

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(Alain Bensoussan) Naveen Jindal School of Management, University of Texas at Dallas, Richardson, TX 75080, USA

Email address: alain.bensoussan@utdallas.edu

(Minh-Nhat Phung) Department of Mathematics, Texas A&M University, College Station, TX 77843, USA

Email address: pmnt1114@tamu.edu

(Minh-Binh Tran) Department of Mathematics, Texas A&M University, College Station, TX 77843, USA

Email address: minhbinh@tamu.edu